

Fundamentals of Quantitative Aptitude

YOUR IDEAL COMPANION FOR THE PREPARATION OF
APTITUDE BASED ENTRANCE TESTS

(Part – I)



**Career
Launcher**

BBA/HM

Preface

Dear Student

If 'Quantitative Aptitude' does give you jitters, then, first things first, get over this fear, before starting the journey. If 'quantitative aptitude' is what you enjoy the most, then get going. Here is a section where you can maximize your scores.

The quantitative aptitude section of BBA will test your understanding of basic fundamentals of mathematics. Like all other sections this one is also designed to see how fast you can work in a given time. So this is more of a game of speed and accuracy than one of difficulty. Performance in QA section is vital too; about one-fourth of the number of questions across various tests is from QA.

But remember speed can never become the first step. You can run faster only if you know how to run. So when it is important to increase your speed in calculations, the initial focus should be on mastering the fundamentals.

So work towards laying a sound foundation first. Take a look at your approach. See if it can be bettered. Look beyond the numbers and formulae. Visualize every problem, try to get the logic behind every formula. Come up with your own list of formulae. Then move on to increase your speed.

The focus of this booklet is to help you master arithmetic. This booklet will take you through fundamentals, solved examples and practice exercises of various topics.

To get maximum output from this book, you are advised to concentrate first on the fundamentals and solved examples before moving to unsolved exercises.

For more doubts, please send your queries at helpme@careerlauncher.com.

Happy reading!

All The best

Team CL

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SAMPLE
CL Media

Number System

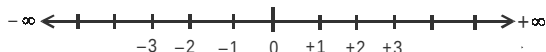
Real Numbers

The set of real numbers consists of all **rational** and **irrational numbers** i.e. all those numbers which can be represented on a number line are called **Real Numbers**.

e.g: -1.3 , $+8.7$, -56.69 , $\sqrt{2}$, $\sqrt{3}$, etc.

Number Line

It is a line on which all the positive and negative numbers can be marked in a sequence.



Rational Numbers

Numbers which can be expressed in the form $\frac{x}{y}$, where

x and y are integers and $y \neq 0$, are known as **rational numbers**. These have a terminating or recurring decimal representation. Terminating decimal numbers can be converted into rational numbers by putting as many zeros after 1 in the denominator as the number of digits after decimal.

e.g: $\frac{1}{2}$, $\frac{3}{7}$, $-\frac{4}{5}$, 1 , 0 , etc.

Irrational Numbers

Numbers which cannot be expressed in the form $\frac{x}{y}$,

where x and y are integers and $y \neq 0$, are known as **irrational numbers**. e.g: π , $\sqrt{2}$ are irrational

Integers

The rational numbers which do not have fraction or decimal part are called integers.

e.g: $-n, \dots, -3, -2, -1, 0, 1, 2, 3, \dots$,

Whole Numbers

The numbers $0, 1, 2, 3, 4, \dots, n$ are called **whole numbers**. The smallest whole number is 0 . Fractions and negative numbers are not whole numbers.

Natural Numbers

The numbers $1, 2, 3, 4, \dots, n$ are called **natural numbers**. 0 is not a natural number. The smallest natural number is 1 . Fractions and negative numbers are not natural numbers.

Even Numbers

The numbers which are divisible by 2 are called **even numbers**. e.g.: $2, 4, 6, 8, \dots$.

Odd Numbers

The numbers which are not divisible by 2 are called **odd numbers**. e.g. $1, 3, 5, 7, \dots$.

Prime Numbers

If a number is divisible by 1 and itself only, then it is called a **prime number**. e.g: $2, 3, 5, 7, \dots$.



Note:

1 is neither a prime number nor composite.

Composite Numbers

The numbers except 1 , which are not prime, are called **composite numbers**. e.g: $4, 6, 8, 9, \dots$.

Fractions

All rational numbers which are of $\frac{p}{q}$ form, where

p, q are integers and p is not a multiple of q .

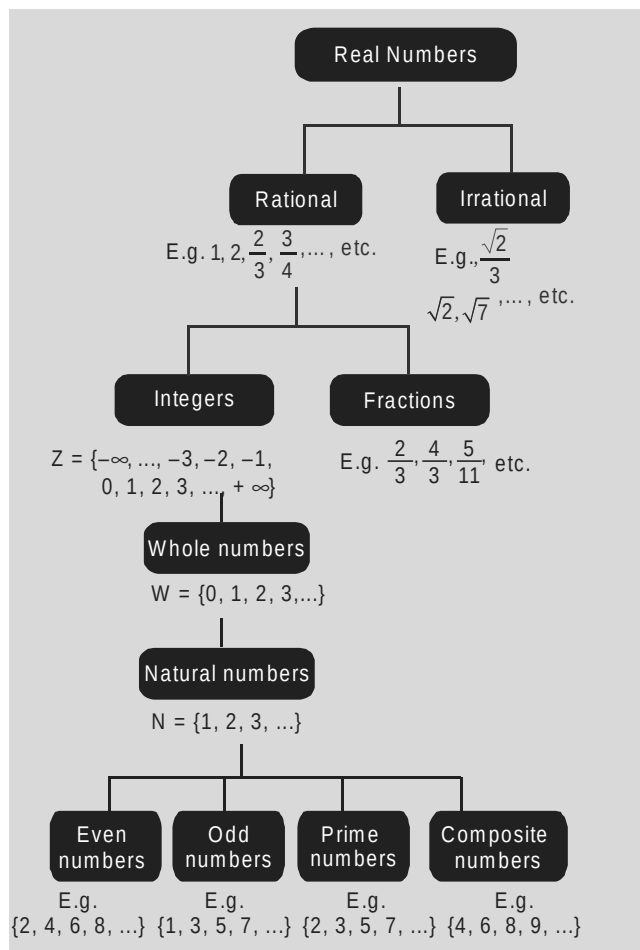
p is called numerator whereas q is known as denominator.

* Fractions are of the following types:

- **Proper:** $p < q$ e.g., $\frac{2}{7}$, $\frac{3}{8}$ etc.

- **Improper:** $p \geq q$ e.g., $\frac{6}{5}, \frac{5}{2}$ etc.
- **Mixed:** It is an integer plus a fraction
e.g., $3\frac{1}{5}, 7\frac{1}{3}$, etc.

The family of Numbers can easily be learned by learning the following Number tree



Complex Number

A number of the form $a + ib$, where $i = \sqrt{-1}$, is called a **complex number** and such number can't be represented on a number line.

We will restrict ourselves only to **Real Numbers**.

BODMAS

Order of simplification of expression of numbers

B: Bracket
D: Division
A: Addition
O: Of/Order
M: Multiplication
S: Subtraction

In a given expression of numbers, the above order of operations has to be strictly followed.

Facts about numbers

Odd + Odd = Even
Even + Even = Even
Odd × Even = Even
Odd + Even = Odd
Odd × Odd = Odd
Even × Even = Even

Then, 9 is the divisor, 57 is the dividend, 6 is the quotient, and 3 is the remainder.

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

Divisibility Tests

1. A number is divisible by 2 if its units digit is even or 0.
2. A number is divisible by 3 if the sum of its digits is divisible by 3.
3. A number is divisible by 4 if the number formed by the last two digits is divisible by 4 or the last two digits are 0.
4. A number is divisible by 5 if its units digit is 5 or 0.
5. A number is divisible by 6 if it is divisible by 2 and 3 both.
6. A number is divisible by 8 if the number formed by the last three digits is divisible by 8, or when the last three digits are 0.
7. A number is divisible by 9 if the sum of its digits is divisible by 9.
8. A number is divisible by 10 if its units digit is 0.
9. A number is divisible by 11 if the difference between the sum of the digits at the odd and the even places is 0 or a multiple of 11.
10. A number is divisible by 12 if it is divisible by 3 and 4 both.

Conversion of a non terminating recurring decimal

into $\frac{p}{q}$ form:

As seen from classification of numbers, non-terminating but recurring decimal numbers are rational numbers, they

can be expressed in the form of $\frac{p}{q}$.

Method 1:

? Example

Convert 0.4444 into the form of $\frac{p}{q}$

✓ Solution

Let $x = 0.4444$... (i)

multiply 10 on both sides.

$10x = 4.444$... (ii)

Subtract (i) from (ii)

$$9x = 4 \Rightarrow x = \frac{4}{9}.$$

? Example

Convert 0.434343 into the form of $\frac{p}{q}$.

✓ Solution

Let $x = 0.434343$... (i)

Multiply both sides by 100

$100x = 43.4343$... (ii)

Subtract (i) from (ii)

$$99x = 43 \Rightarrow x = \frac{43}{99}$$

Method 2:

The $\frac{p}{q}$ form of any recurring number

(The non-recurring and recurring part
written once) – (The non-recurring part)
= As many 9's as the number of digits in the
recurring part followed by as many 0's as
digits in the non-recurring part.

Factorial of a number:

The continued product of first 'n' natural numbers is called
'n factorial' of the number 'n' and is denoted by $n!$ or $\lfloor n$

$$n! = 1 \times 2 \times 3 \times \dots \times (n-1) \times n$$

$$\text{e.g: } 6! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$$

By definition $0! = 1$.

LCM and HCF

Factor

A number is said to be the factor of another,
if it divides the other number perfectly.

e.g: 7 and 9 are factors of 63.

Common factor

If one number perfectly divides two or more numbers,
then it is called a common factor of the numbers.

e.g: 5 is a common factor of 10, 15, 20 and 25.

Highest common factor (HCF)

HCF of two or more numbers is the greatest number that
perfectly divides each of them.

e.g: 5 is the HCF of 15 and 20.



Note:

HCF is also known as GCD (Greatest
Common Divisor)

? Example

Find the HCF of 24 and 72.

✓ Solution

$$24 = 2 \times 2 \times 2 \times 3$$

$$72 = 2 \times 2 \times 2 \times 3 \times 3$$

$$\text{HCF} = 2 \times 2 \times 2 \times 3 = 24.$$

Similarly, we can find the HCF of sets containing
more than two numbers.

Common Multiple:

A common multiple of two or more numbers is a number
which is perfectly divisible by each of the numbers.

e.g: 45 is a common multiple of 3, 5 and 15.

Lowest common multiple (LCM)

The LCM of two or more numbers is the least number
which is perfectly divisible by each of the numbers.

e.g: 57 is a least common multiple of 3 and 19.

Let's consider the LCM of 18, 27 and 30.

2	18, 27, 30
3	9, 27, 15
3	3, 9, 5
	1, 3, 5

$$\text{LCM} = 2 \times 3 \times 3 \times 3 \times 5 = 270.$$



Note:

$\text{LCM of two numbers} \times \text{HCF of two numbers}$
= Product of the two numbers

LCM & HFC of Fractions

$$\text{HFC}\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{HCF}(a, c, e)}{\text{LCM}(b, d, f)}$$

$$\text{LCM}\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM}(a, c, e)}{\text{HCF}(b, d, f)}$$

Cyclicity:

Take any two numbers – 46 and 57. If they are multiplied then the digit obtained at units place of the product will be the same as the units place digit of the product of 6 & 7.

$$6 \times 7 = 4 \boxed{2} \quad \& \quad 46 \times 57 = 262 \boxed{2}$$

If this concept is applied to exponent of numbers, an interesting pattern emerges when we look at them.

The last digits of the exponents of all the numbers have a cyclicity. If after every N^{th} power of the base have the same digit at units place as the first power of the base, then N is the cyclicity of that number.

For instance,

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 3 \boxed{2}$$

Since 2^5 has units place digit to be the same as 2^1 then it is said to have a cyclicity of 4. i.e $2^{1+4} = 32$

Similarly, we can find out cyclicity of other digits also.

You can check for yourself the cyclicity of all the digits.

Digit	Cyclicity
0, 1, 5 and 6	1
2, 3, 7 and 8	4
4 and 9	2

Some important facts about squares

- I. The square of an even number is always even.
- II. The square of an odd number is always odd.
- III. Square of an integer cannot end in 2, 3, 7, or 8.
- IV. The square of an integer (negative or positive) is always positive.
- V. The square of an integer is either a multiple of 3 or 4 or else exceeds a multiple of 3 or 4 by 1. It is of the form $3k \pm 1$ or $4k \pm 1$.
- VI. If a square ends in 9, the preceding digit is even.
- VII. The fifth power of any number has the same unit digit as the number itself.

Some important results

- I. $(x^n + y^n)$ is divisible by $(x + y)$ when n is odd.
- II. $(x^n - y^n)$ is divisible by $(x + y)$ when n is even and by $(x - y)$ always.
- III. $(x^n - x)$ is divisible by n (if n is prime).
- IV. $(p - 1)! + 1$ is divisible by p when p is prime.

Formulae

- I. $(a + b)^2 = a^2 + b^2 + 2ab$
- II. $(a - b)^2 = a^2 + b^2 - 2ab$
- III. $(a + b)^2 = (a - b)^2 + 4ab$
- IV. $(a - b)^2 = (a + b)^2 - 4ab$
- V. $a^2 - b^2 = (a - b)(a + b)$

$$\text{VI. } a^2 + b^2 = \frac{1}{2}[(a+b)^2 + (a-b)^2]$$

$$\text{VII. } a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$\text{VIII. } a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Rules on counting numbers

1. Sum of first n natural numbers = $\frac{n(n+1)}{2}$.
2. Sum of first n odd numbers = n^2 .
3. Sum of first n even numbers = $n(n+1)$.
4. Sum of the squares of first n natural numbers

$$= \frac{n(n+1)(2n+1)}{6}$$
5. Sum of the cubes of first n natural numbers

$$= \left[\frac{n(n+1)}{2} \right]^2$$

Solved Examples

? Example 1

$$\text{Simplify: } \frac{527 \times 527 \times 527 + 183 \times 183 \times 183}{527 \times 527 - 527 \times 183 + 183 \times 183}$$

✓ Solution

Expression

$$\frac{(527)^3 + (183)^3}{(527)^2 - 527 \times 183 + (183)^2} = \frac{a^3 + b^3}{a^2 - ab + b^2}$$

$$\frac{(a+b)(a^2 - ab + b^2)}{a^2 - ab + b^2} = a + b$$

(since, $a = 527$ and $b = 183$)
 $= 527 + 183 = 710$

? Example 2

Simplify:

$$\left(\frac{(614 + 168)^2 - (614 - 168)^2}{614 \times 168} \right)$$



✓Solution

Expression

$$\frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} = 4$$

?Example 3

Find the square of 1605.

✓Solution

$$\begin{aligned}(1605)^2 &= (1600 + 5)^2 \\ &= (1600)^2 + 2 \times 1600 \times 5 + (5)^2 \\ &= 2560000 + 16000 + 25 = 2576025\end{aligned}$$

?Example 4Find the value of $896 \times 896 - 204 \times 204$ **✓Solution**

$$\begin{aligned}a^2 - b^2 &= (a + b)(a - b), \\ \text{where } a &= 896 \text{ and } b = 204 \\ &= (896 + 204)(896 - 204) \\ &= 1100 \times 692 = 761200\end{aligned}$$

?Example 5Evaluate: $(57)^2 + (43)^2 + 2 \times 57 \times 43$ **✓Solution**

$$\begin{aligned}(a^2 + b^2 + 2ab) &= (a + b)^2 \\ &= (57 + 43)^2 = 100^2 = 10000\end{aligned}$$

?Example 6Simplify: $(81)^2 + (68)^2 - 2 \times 81 \times 68$ **✓Solution**

$$\begin{aligned}a^2 + b^2 - 2ab &= (a - b)^2 \\ &= (81 - 68)^2 = 13^2 = 169\end{aligned}$$

?Example 7Evaluate: $(313 \times 313 + 287 \times 287)$ **✓Solution**

$$a^2 + b^2 = \frac{1}{2} [(a+b)^2 + (a-b)^2],$$

where $a = 313$ and $b = 287$

$$\begin{aligned}&= \frac{1}{2} [(313 + 287)^2 + (313 - 287)^2] \\ &= \frac{1}{2} [(600)^2 + (26)^2] = 180338\end{aligned}$$

?Example 8

What is the divisor if dividend is 15968, quotient is 89, and remainder is 37?

✓Solution

$$\begin{aligned}\text{Divisor} &= \left(\frac{\text{Dividend} - \text{Remainder}}{\text{Quotient}} \right) \\ &= \left(\frac{15968 - 37}{89} \right) = 179\end{aligned}$$

?Example 9

Which least number must be subtracted from 2000 to get a number exactly divisible by 17?

✓Solution

On dividing 2000 by 17, we get 11 as remainder.
 $\therefore$ Required number to be subtracted = 11.

?Example 10

Which least number must be added to 3000 to obtain a number exactly divisible by 19?

✓Solution

On dividing 3000 by 19, we get 17 as remainder.
 $\therefore$ Number to be added = $(19 - 17) = 2$.

?Example 11

Find the number which is nearest to 3105 and exactly divisible by 21.

✓Solution

On dividing 3105 by 21, we get 18 as remainder.
 $\therefore$ Number to be added to 3105 is $(21 - 18) = 3$.
 $\therefore$ 3108 is the required number.

?Example 12

A number when divided by 342 gives a remainder 47. When the same number is divided by 19, what would be the remainder?

Solution

On dividing the given number by 342, let k be the quotient and 47 as remainder.

Then number = $342k + 47 = [(19 \times 18k) + (19 \times 2 + 9)] = [19(18k + 2) + 9]$

$\therefore$ The given number when divided by 19, gives $(18k + 2)$ as quotient and 9 as remainder.

Alternative method:

342 is a multiple of 19. Divide the remainder by the second dividend to get the remainder. 47, when divided by 19, gives 9 as remainder.

Example 13

The HCF of two numbers is 11 and their LCM is 693. If one number is 77, find the other.

Solution

The other number = $\frac{11 \times 693}{77} = 99$.

Example 14

Find the largest number that can exactly divide 513, 783 and 1107.

Solution

Required number = HCF of 513, 783 and 1107.

Now $513 = 3^3 \times 19$, $783 = 3^3 \times 29$,

$1107 = 3^3 \times 41$

$\therefore$ HCF = $3^3 = 27$.

Hence, the required number is 27.

Example 15

Find the least number exactly divisible by 12, 15, 20 and 27.

Solution

Required number = LCM of 12, 15, 20, 27.

$\therefore$ LCM = $3 \times 4 \times 5 \times 9 = 540$.

$\therefore$ Required number = 540.

Example 16

Find the least number, when divided by 6, 7, 8, 9 and 12, leaves remainder 1 in each case.

Solution

Required number = (LCM of 6, 7, 8, 9, 12) + 1

$\therefore$ LCM = $3 \times 2 \times 2 \times 7 \times 2 \times 3 = 504$.

Hence, required number = $(504 + 1) = 505$.

Example 17

Arrange the following rational numbers in ascending order.

$$-\frac{7}{10}, \frac{5}{-8}, \frac{2}{-3}$$

Solution

LCM of (10, 8, 3) = 120

$$\therefore \frac{-7}{10} = \frac{-84}{120}$$

$$\frac{5}{-8} = \frac{-75}{120}$$

$$\frac{2}{-3} = \frac{-80}{120}$$

as $-84 < -80 < -75$

$$\therefore \frac{-7}{10} < \frac{2}{-3} < \frac{5}{-8}$$

Alternate Method:

$$\frac{-7}{10} = -0.7, \frac{5}{-8} = -0.625 \text{ and } \frac{2}{-3} = -0.666$$

Clearly, $-0.7 < -0.666 < -0.625$.

$$\text{So } \frac{-7}{10} < \frac{2}{-3} < \frac{5}{-8}$$

Example 18

If square root of 15 = 3.88, find the value of square

root of $\frac{5}{3}$.

Solution

$$\sqrt{\frac{5}{3}} = \sqrt{\frac{5 \times 3}{3 \times 3}} = \frac{\sqrt{15}}{3} = \frac{3.88}{3} \approx 1.29$$

Example 19

A four-digit number that is divisible by 7 becomes divisible by 3 when 10 is added to it. Find the largest such number.

✓Solution

The largest four-digit number is 9999.
 On dividing 9999 by 7, we get 3 as remainder.
 The largest four-digit number divisible by 7 is 9996.
 Let $9996 - x + 10$ be divisible by 3.
 By trial and error, we find that $x = 7$. Hence, the required number = $(9996 - 7) = 9989$.

?Example 20

A three-digit number $4a3$ is added to another three-digit number 984 to give the four-digit number $13b7$ which is divisible by 11. Find the value of $(a + b)$.

$$\begin{array}{r} 4a3 \\ 984 \\ \hline 13b7 \end{array}$$

✓Solution

$a + 8 = b \Rightarrow b - a = 8$
 Also, $13b7$ is divisible by 11.
 So $(7 + 3) - (b + 1) = 0$ and $b = 9$.
 Now, $b - a = 8$ and $b = 9$.
 So, $a = 1$, and $a + b = (1 + 9) = 10$.

?Example 21

Express $0.\overline{643}$ into a fraction.

✓Solution

Let $x = 0.\overline{643}$. Thus, $1000x = 643.\overline{643}$

$$999x = 643 \text{ or } x = \frac{643}{999}$$

?Example 22

Of the three numbers, the sum of the first two is 45; the sum of the second and the third is 55; and the sum of the third and thrice the first is 90. Find the third number.

✓Solution

Let the numbers be x , y and z .
 Then $x + y = 45$, $y + z = 55$ and $3x + z = 90$.
 $y = 45 - x$, and $z = 55 - y = 55 - (45 - x)$
 $= 10 + x$
 $\therefore 3x + 10 + x = 90$ or $x = 20$
 $y = (45 - 20) = 25$, and $z = (10 + 20) = 30$.
 $\therefore$ The third number = 30.

?Example 23

The traffic lights at three different road-crossings change after every 24 sec, 72 sec and 120 sec respectively. If they all change simultaneously at 10.54.00 hr, then at what time will they change simultaneously for the second time?

✓Solution

Interval of change = LCM of (24, 72, 120) sec
 $= 360$ sec.
 The lights will change simultaneously after every 360 sec, i.e. 6 min.
 Next simultaneous change will take place at 11.00.00 hr.

?Example 24

The difference between a two-digit number and the number obtained by interchanging the digits is 72. What is the difference between the digits of the number?

✓Solution

Let tens digit be x and units digit be y .
 Then $(10x + y) - (10y + x) = 72$
 $\Rightarrow 9(x - y) = 72 \Rightarrow x - y = 8$.

?Example 25

How many three-digit numbers are divisible by 6 in all?

✓Solution

There are 16 numbers before 100, which are divisible by 6.
 There are 166 numbers before 999, which are divisible by 6.
 Total three-digit numbers divisible by 6
 $= (166 - 16) = 150$

?Example 26

If $(5^{2n} + 1)^2 \times 5^6 = 1$, then what is the value of n ?

✓Solution

$$\begin{aligned} (5^{2n} + 1)^2 \times 5^6 &= 1 \\ \Rightarrow 5^{4n+2} \times 5^6 &= 5^0 \Rightarrow 5^{4n+8} = 5^0 \\ \Rightarrow 4n + 8 &= 0 \Rightarrow n = -2. \end{aligned}$$



Practice Exercise

1. The LCM of two numbers is 4200 and their GCD is 20. If one number is 100, the other number is
(a) 840 (b) 420
(c) 210 (d) None of these
2. If $y = \sqrt{2} + 1$, then the value of $y + \frac{1}{y}$ is
(a) $\sqrt{\frac{3}{2}}$ (b) $\frac{\sqrt{3}}{2}$
(c) $2\sqrt{2}$ (d) $\frac{3}{2}$
3. The largest fraction among $\frac{2}{5}, \frac{3}{5}, \frac{1}{5}, \frac{7}{15}$ and $\frac{4}{5}$ is
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) $\frac{7}{15}$
4. The sum of the squares of first ten natural numbers is
(a) 281 (b) 402
(c) 385 (d) 502
5. The value of factorial zero (i.e. 0!) is
(a) 0 (b) 1
(c) 8 (d) $\frac{1}{2}$
6. Zero is counted as
(a) whole number (b) prime number
(c) natural number (d) All of these
7. Two rational numbers lying between $\frac{4}{5}$ and $\frac{6}{7}$ are
(a) $\frac{71}{84}, \frac{5}{6}$ (b) $\frac{29}{35}, \frac{5}{6}$
(c) $\frac{29}{35}, \frac{57}{70}$ (d) All of these
8. The lowest four-digit number which is exactly divisible by 2, 3, 4, 5, 6 and 7 is
(a) 1400 (b) 1300
(c) 1250 (d) 1260
9. The least number when divided by 35 leaves a remainder 25; when divided by 45 leaves a remainder 35; and when divided by 55 leaves the remainder 45. The number is
(a) 2515 (b) 3455
(c) 2875 (d) None of these
10. The sum of two positive numbers is twice their difference. If their product is 48, the numbers will be
(a) 3 and 16 (b) 6 and 8
(c) 4 and 12 (d) None of these
11. Solve: $3 + \frac{3}{3 + \frac{1}{3 + \frac{1}{3}}}$
(a) 1 (b) 3
(c) $\frac{43}{11}$ (d) $\frac{63}{19}$
12. Find the units digit in the product $254 \times 361 \times 159 \times 18$.
(a) 1 (b) 4
(c) 6 (d) 8
13. A heap of coconuts is divided into groups of 2, 3 and 5, and each time one coconut is left out. The least number of coconuts in the heap is
(a) 31 (b) 41
(c) 51 (d) 61
14. The ratio between a two-digit number and the sum of the digits of that number is 4 : 1. If the digit in the units place is 3 more than the digit in the tens place, what is that number?
(a) 69 (b) 25
(c) 36 (d) None of these
15. The smallest number among the following is
(a) $(7)^3$ (b) $(8.5)^3$
(c) $(4)^4$ (d) $(6^5)^{\frac{3}{5}}$
16. The largest fraction among the following is
(a) $\frac{17}{21}$ (b) $\frac{11}{14}$
(c) $\frac{12}{15}$ (d) $\frac{5}{6}$



17. If $x = (6 - \sqrt{35})$, then the reciprocal of x is
 (a) $\frac{1}{6 + \sqrt{35}}$ (b) $6 + \sqrt{35}$
 (c) 1 (d) 12
18. The sum of the first 50 even numbers is
 (a) 1275 (b) 2550
 (c) 5100 (d) 10100
19. If $x + 2y = 4$ and $\left(\frac{x}{y}\right) = 2$, then $y = ?$
 (a) 1 (b) 0.5
 (c) 2 (d) 8
20. If the product of three consecutive integers is 720, then their sum is
 (a) 27 (b) 45
 (c) 18 (d) 54
21. Solve: $18.18 \div 9 + 2.7$ of 3
 (a) 101.2 (b) 27.32
 (c) 10.12 (d) None of these
22. Solve: $8127 - 5422 + 1614 - 808$
 (a) 3580 (b) 3058
 (c) 3511 (d) 3088
23. Evaluate: $11^2 + 11^4 \div 11^3 - 11 + (0.5)11^2$
 (a) 302.5 (b) 181.5
 (c) 484.0 (d) 121
24. Which one of the following is incorrect?
 (a) Square root of 5184 is 72
 (b) Square root of 15625 is 125
 (c) Square root of 1444 is 38
 (d) Square root of 1296 is 34
25. The difference between two numbers is 1365. When the larger number is divided by the smaller one, the quotient is 6 and the remainder is 15. The smaller number is
 (a) 240 (b) 360
 (c) 270 (d) 295
26. 243 has been divided into three parts such that half of the first part, one-third of the second part and one-fourth of the third part are equal. The largest part is
 (a) 72 (b) 84
 (c) 92 (d) 108
27. How many numbers between 200 and 600 are divisible by 4, 5 and 6?
 (a) 5 (b) 6
 (c) 7 (d) 8
28. The units digit of the product $(247 \times 318 \times 577 \times 313)$ is
 (a) 2 (b) 3
 (c) 4 (d) 6
29. The sum of first 45 natural numbers is
 (a) 2070 (b) 1035
 (c) 1280 (d) 2140
30. Find the value of 'a' if the number '451a603' is exactly divisible by 9, such that 'a' is a single digit number.
 (a) 2 (b) 5
 (c) 8 (d) 7
31. Which least value must be assigned to 'a' so that the number 63576a2 is divisible by 8?
 (a) 1 (b) 2
 (c) 3 (d) 4
32. Which of the following is exactly divisible by 99?
 (a) 114345 (b) 135792
 (c) 3572404 (d) 913464
33. The number which is formed by writing any digit six times (e.g. 111111, 444444, etc.) is always divisible by
 (a) 7 (b) 11
 (c) 13 (d) All of these
34. The number nearest to 99547, which is exactly divisible by 687 is
 (a) 100166 (b) 98928
 (c) 99579 (d) 99615
35. Which largest number of five digits is divisible by 99?
 (a) 99999 (b) 99981
 (c) 99909 (d) 99990
36. Which smallest number of six digits is divisible by 111?
 (a) 111111 (b) 110011
 (c) 100011 (d) 111011
37. If n is any positive integer, then $(3^{4n} - 4^{3n})$ is always divisible by
 (a) 7 (b) 17
 (c) 112 (d) 145

38. Solve: 8756×99999
 (a) 815491244 (b) 796491244
 (c) 875591244 (d) None of these
39. Evaluate: 1399×1399
 (a) 1687401 (b) 1901541
 (c) 1943211 (d) 1957201
40. Find the value of $397 \times 397 + 104 \times 104 + 2 \times 397 \times 104$.
 (a) 250001 (b) 251001
 (c) 260101 (d) 261001
41. The expression

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)}$$
 for any natural number n, is
 (a) always greater than 1
 (b) always less than 1
 (c) always equal to 1
 (d) not definite
42. When 'n' is divided by 4, the remainder is 3. What is the remainder when '2n' is divided by 4?
 (a) 1 (b) 2
 (c) 3 (d) 6
43. Six bells commence tolling together and toll at intervals of 3 sec, 6 sec, 9 sec, 12 sec, 15 sec and 18 sec respectively. If all bells rang together at 9 o'clock, for how many more times will they ring together till 9:32?
 (a) 4 (b) 10
 (c) 11 (d) 15
44. The units digit of 3^{33} is
 (a) 2 (b) 3
 (c) 4 (d) 5
45. The units digit of $24^{37} \times 37^{24}$ is
 (a) 4 (b) 3
 (c) 2 (d) 6



My Doubts

SAMPLE
CL Media



Per cent means per 100, where 'cent' stands for 100. By a certain percent, we mean that many hundredth, thus x% means x divided by hundred.

$$\text{i.e. } x\% = \frac{x}{100}.$$

Percentage of a number

To find the percentage of a number, convert the percentage into fraction (by dividing by 100) and multiply the resulting fraction with the number.

$$\text{e.g. } 60\% \text{ of } 500 = \frac{60}{100} \times 500 = 300.$$

Conversion of a fraction or a decimal into percentage

A fraction or a decimal can be converted into a percentage by simply multiplying it by 100.

So, the fraction $\frac{1}{5}$ expressed as a percentage is

$$\frac{1}{5} \times 100 = 20\%.$$

And the decimal 0.05 expressed as a percentage is $0.05 \times 100 = 5\%$.

Converting a percentage into a fraction

A percentage when divided by 100 is converted into a

fraction. So, 20% as a fraction is $\frac{20}{100} = \frac{1}{5}$.

The '%' sign is dropped when we divide the percentage by 100.

Fraction of a fraction and Relative Percentage

To find the fraction of a fraction we multiply both the fractions.

e.g. $\frac{1}{4}$ of $\frac{1}{5}$ is nothing but $\frac{1}{4} \times \frac{1}{5} = \frac{1}{20}$, and

$$\frac{1}{3} \text{ of } \frac{3}{5} \text{ is } \frac{1}{5}, \text{ etc.}$$

Similarly, 25% of 20% is $25\% \times \frac{20}{100} = 5\%$,

$$\text{or } \frac{25}{100} \times 20\% = 5\%.$$

Successive percentage changes

If a number is changed (increased/decreased) by a% and in the second step, this changed number is again changed (increased/decreased) by b%, then

$$\text{net percentage change} = \left(a + b + \frac{a \times b}{100} \right) \%$$

If 'a' and 'b' show decrease, then put a -ve sign before 'a' and 'b', otherwise put +ve sign.

? Example

If the price of an item is increased by 20% and then a discount of 10% is given on the increased price, what will be the effective percentage change in the price of the item?

✓ Solution

Using percentage change

$$= a + b + \frac{a \times b}{100} \%$$

$$= 20 - 10 + \frac{20 \times 10}{100} = 8\% \text{ (increase)}$$



Notes:

The rule cannot be generalized. This can only be used for 2 values at a time.

Solved Examples

? Example 1

Express 40% in decimal terms.

You may encounter these types of questions in the examination

Solved Examples

Types of questions	Examples	Approach to the question
1. Convert percentage into fraction.	Express 12% in a fraction.	$x\% = \frac{x}{100} = \frac{12}{100} = \frac{3}{25}$
2. Convert fraction (or decimal) into percentage.	Express $\frac{5}{11}$ as percentage.	Multiply the fraction by 100. $= \frac{5}{11} \times 100 = 45.45\%$
3. If A is x% of B and B = P, then find A in term of P.	A's income is 40% of B's. If B's income is ₹10, 000, what is A's income?	$A = \frac{x}{100} \times P$ $= \frac{40}{100} \times 10,000 = ₹4,000$
4. If A is r% more than B, by how much percent is B less than A?	X's income is 25% more than Y's. By how much percent is Y's income less than X's?	Difference = $\frac{r}{100+r} \times 100$ $= \frac{25}{125} \times 100 = 20\%$
5. If A is r% less than B, by how much percent is B more than A?	X's income is 40% less than Y's. By how much per cent is Y's income more than X's?	Difference = $\frac{r}{100-r} \times 100$ $= \frac{40}{60} \times 100 = 66.67\%$
6. If the price of a commodity increases by r%, find the decrease in the consumption so as not to increase the expenditure. General Formula: % decrease/increase in consumption = $\frac{r}{100 \pm r} \times 100$ where r is % increase/decrease in the price of the commodity.	(a) If the price of potatoes is increased by 20%, by how much percent should the consumption be decreased so as to have no change in the expenditure? (b) If the price of potatoes is increased by 20% and the consumption is decreased by 10%, what will be the change in expenditure?	Expenditure = Price $\times$ Consumption (a) Decrease = $\frac{r}{100+r} \times 100$ $= \frac{20}{120} \times 100 = \frac{50}{3} = 16.67\%$ (b) Net change = $a + b + \frac{ab}{100}$ $= 20 - 10 - \frac{20 \times 10}{100} = 8\%$
7. The population of a town is N. It increases by x% during the first year and y% during the second year. Find the	The population of a town is 18000. It increases by 10% during first year and by 20% during the second year. What will be the population after 2 years?	$N_F = N \times \frac{100+x}{100} \times \frac{100+y}{100}$ $= 18000 \times \frac{110}{100} \times \frac{120}{100} = 23760$ Alternative method Successive increments of 10% and 20% = 32%. Then, population will increase by 32% of 18000 = 5760. Population after 2 years will be 18000 + 5760 = 23760.



✓Solution

$$40\% = \frac{40}{100} = 0.4$$

?Example 2

Express 30 as a percentage of 45.

✓Solution

$$\left(\frac{30}{45}\right) \times 100 = 66.67\%$$

?Example 3

Find 40% of 3340.

✓Solution

$$\left(\frac{40}{100}\right) \times 3340 = 1336$$

?Example 4

Express 50 g as a percentage of 4 kg.

✓Solution

$$\frac{50 \text{ g}}{4 \text{ kg}} \times 100 = \left(\frac{50 \text{ g}}{4000 \text{ g}}\right) \times 100 = 1.25\%$$

?Example 5

A's income is 70% of B's. B's income is 50% of C's. If C's income is ₹1,00,000 then A's income is

✓Solution

$$\text{B's income} = \left(\frac{50}{100}\right) \times ₹1,00,000 = ₹50,000$$

$$\text{A's income} = \left(\frac{70}{100}\right) \times ₹50,000 = ₹35,000$$

Alternative Method:

$$\text{B's income} = \frac{50}{100} \text{ of C's income}$$

$$\text{A's income} = \frac{70}{100} \text{ of B's income}$$

$$= \frac{70}{100} \times \frac{50}{100} \text{ of C's income}$$

$$\text{A's income} = \frac{35}{100} \times 1,00,000 = ₹35,000.$$

?Example 6

X's income is 50% more than Y's. By how much percentage is Y's income less than X's?

✓Solution

Using formula, $r = 50$

$$\Rightarrow \left[\frac{50}{(100 + 50)}\right] \times 100 = 33.33\%$$

?Example 7

In a market survey, 20% individuals opted for product B. The remaining individuals were uncertain. If the difference between those who opted for product B and those who were uncertain was 720. How many individuals were covered in the survey?

✓Solution

Clearly, 80% were uncertain. Let assume 'x' individuals were surveyed.

$$80\% \text{ of } x - 20\% \text{ of } x = 720$$

$$\Rightarrow 60\% \text{ of } x = 720 \Rightarrow \frac{60}{100} x = 720$$

$$\Rightarrow x = \frac{720 \times 100}{60} = 1200$$

?Example 8

Of the total amount received by Kiran, 20% was spent on purchases and 5% of the remaining on transportation. If he was left with ₹1,520, what was the initial amount?

✓Solution

Let 100 be the sum, 20% is spent on purchases. Hence, we are left with 80; 5% of 80 is 4. Hence, the remaining is 76. We are given that the remaining is 1520. Hence, 76 corresponds to 1520 and

$$\text{therefore } 100 \text{ corresponds to } \frac{100 \times 1520}{76} = ₹2000.$$

?Example 9

The length of a rectangle is increased by 10%. What will be the percentage decrease in its breadth, so as to have a constant area?

✓Solution

Let length and breadth of the rectangle are l and b respectively. Area = lb .

Increased length and corresponding breadth = l' and b' , area = $l'b'$

$$l' = \frac{110}{100}l = \frac{11}{10}l$$

$$\frac{11}{10}lb' = lb \Rightarrow b' = \frac{10}{11}b$$

$$\text{Decrease in breadth} = b - b' = b - \frac{10}{11}b = \frac{1}{11}b$$

Percentage decrease in breadth

$$\frac{\frac{1}{11}b}{b} \times 100 = \frac{100}{11} = 9\frac{1}{11}\%$$

Shortcut

Applying percentage change = $a + b + \frac{ab}{100}$.

Let decrease in breadth be $x\%$.

$$\text{Then } 0 = 10 - x - \frac{10 \times x}{100} \Rightarrow \frac{11x}{10} = 10$$

$$\text{or } x = \frac{100}{11} = 9\frac{1}{11}\%$$

?Example 10

A school has only three classes which contain 40, 50 and 60 students. The passing percentage of these classes are 10, 20 and 10 respectively. What is the percentage of the students who passed in the school?

✓Solution

Total number of students = $40 + 50 + 60 = 150$

Number of students passed

$$= \left(\frac{10}{100} \times 40 + \frac{20}{100} \times 50 + \frac{10}{100} \times 60 \right)$$

$$= (4 + 10 + 6) = 20$$

Percentage of students passed

$$= \frac{20}{(40 + 50 + 60)} \times 100 = 13\frac{1}{3}\%$$

?Example 11

Tax on water is increased by 20% but its consumption is decreased by 20%. Then, what is the percentage increase or decrease in the expenditure?

✓Solution

Let the original consumption = 100 units and tax = ₹100 per unit.

Then, original expenditure = ₹ (100×100) = ₹10,000.

New expenditure = $80 \times 120 = ₹9600$.

$$\text{Decrease in expenditure} = \left(\frac{400}{100 \times 100} \times 100 \right) = 4\%.$$

Shortcut

$$\text{Change in expenditure} = 20 - 20 - \frac{20 \times 20}{100}$$

$$= -4\%$$

Thus, expenditure decreases by 4%.

?Example 12

The number of seats in an auditorium is increased by 25%. The price of a ticket is also increased by 12%. What is the effect on the revenue collected?

✓Solution

Let the initial number of seats be 100 and price per ticket be ₹1.

Then, revenue = Number of seats $\times$ Price per ticket

$$\text{Increased number of seats} = \frac{125}{100} \times 100 = 125$$

$$\text{Increased price of a ticket} = \frac{112}{100} \times 1 = ₹1.12$$

$$\text{Increased revenue} = 125 \times 1.12 = ₹140$$

Percentage increase in revenue

$$= 140 - 100 = 40\%$$

Shortcut

Using successive percentage increase formula

$$= x + y + \frac{x \times y}{100}, \text{ where 'x' and 'y' are the percentage increase.}$$

$\therefore$ Percentage increase in revenue

$$= 25 + 12 + \frac{25 \times 12}{100} = 25 + 12 + 3 = 40\%.$$

?Example 13

A's 5% income is equal to 15% income of B, and 10% of income of B is equal to 20% income of C. If income of C is ₹2,000, then what is the total income of A, B and C?

Solution

$$\frac{5}{100}A = \frac{15}{100}B \text{ and } \frac{10}{100}B = \frac{20}{100}C$$

$$\therefore A = 3B \text{ and } B = 2C = 2 \times 2000 = ₹4,000$$

$$\therefore A = 3 \times 4000 = ₹12,000$$

$$\therefore A + B + C = (12,000 + 4,000 + 2,000) = ₹18,000$$

? Example 14

Arvind spends 75% of his income. His income is increased by 20% while his expenditure increased by 10%. By what per cent did Arvind's savings increase?

Solution

Let the income be 100. Expenditure = 75 and savings = 25. New income = 120,

$$\text{New expenditure} = \left(\frac{110}{100} \times 75 \right) = \frac{165}{2}$$

$$\text{New savings} = \left(120 - \frac{165}{2} \right) = \frac{75}{2}$$

$$\text{Increase in savings} = \left(\frac{75}{2} - 25 \right) = \frac{25}{2};$$

$$\text{Increase per cent} = \left(\frac{25}{2} \times \frac{1}{25} \times 100 \right) = 50\%$$

? Example 15

Two numbers are respectively 19% and 70% more than a third number. The first number as a percentage of the second number is:

Solution

Let the third number be 100.

Then, the first number is $100 + 19 = 119$ and the second number is 170.

$$\therefore \text{The first is } \frac{119}{170} \times 100 = 70\% \text{ of the second.}$$

Shortcut

$$\text{First number is } \frac{100+19}{100+70} \times 100 = 70\% \text{ of the second.}$$

? Example 16

Salaries of A, B and C are in the ratio 1 : 2 : 3. Salaries of B and C together is ₹6,000. By what percentage is the salary of C more than that of A?

Solution

Let $A = x$; $B = 2x$ and $C = 3x$.

Then $2x + 3x = 6000 \Rightarrow x = 1200$

$\therefore A = 1200$ and $C = 3600$

$$\text{Required percentage} = \left(\frac{2400}{1200} \times 100 \right) = 200\%$$

Shortcut

Salary of $A = x$ and $C = 3x$

Hence, salary of C is $2x$ more than that of A

$$\text{Required percentage} = \frac{2x}{x} \times 100 = 200\%$$

? Example 17

If the side of a square is increased by 25%, then by what percentage is its area increased?

Solution

Let the side be 10 cm. Then, the area will be 100 cm^2 .

New side = 125% of 10 = 12.5 cm;

$$\text{area} = (12.5)^2 = 156.25 \text{ cm}^2$$

$\therefore$ Percentage increase = 56.25%

Shortcut

If 'x' is the percentage increase in the side of a square, then increase in area is given by

$$x + x + \frac{x \times x}{100} = 2x + \frac{x^2}{100};$$

$$= 25 + 25 + \frac{25 \times 25}{100} = 56.25\%.$$

? Example 18

A cricket team won 40% of the total number of matches it played during a year. If it lost 50% of the matches played and 20 matches were drawn, what was the total number of matches played by the team during the year?

Solution

40% of $x + 50\%$ of $x + 20 = x$,

where x = Total number of matches

$$\Rightarrow \frac{40}{100}x + \frac{50}{100}x + 20 = x \text{ or } x = 200.$$



Practice Exercise

1. What is $18\frac{3}{4}\%$ of 2000?
(a) 300 (b) 400
(c) 390 (d) 375
2. What percentage of 48 is 26?
(a) 54.16% (b) 184.6%
(c) 56.33% (d) 57.16%
3. What is $33\frac{1}{3}\%$ of 972?
(a) 332 (b) 411
(c) 348 (d) 324
4. What percentage of 60 is 37?
(a) 60% (b) 61.66%
(c) 65.66% (d) 70%
5. The population of a town increases from 6,500 to 7,475. What is the percentage increase?
(a) 10% (b) 12%
(c) 15% (d) 20%
6. The population of a town increases by 20% annually. What is the population after 2 years, if present population is 2,500?
(a) 3,250 (b) 3,500
(c) 3,600 (d) 4,000
7. What is 90% of 90% of 100?
(a) 80 (b) 100
(c) 90 (d) 81
8. If 37% of a number is 990.86, then what will be (approximately) 19% of that number?
(a) 600 (b) 400
(c) 510 (d) 700
9. 50 min is what percentage of an hour?
(a) 83.33% (b) 50%
(c) 90% (d) 87.66%
10. I bought 20 kg mango, out of which 16 kg was fine and rest were rotten. What is my percentage loss, if I bought them for ₹30 per kilogram?
(a) 33% (b) 40%
(c) 15% (d) 20%
11. If A's income is 25% more than B's, then what is B's income as a percentage of A's income?
(a) 75% (b) 80%
(c) 90% (d) 125%
12. In an examination the passing percentage is 40. A obtained 72 out of 200. By what percentage of the total marks did he fail?
(a) 8% (b) 5%
(c) 4% (d) 16%
13. What is y as a percentage of x, if x is 120% of y?
(a) 80% (b) 83.33%
(c) 75% (d) 86.66%
14. If A is increased by 10%, by what per cent does A² increase?
(a) 20% (b) 21%
(c) 100% (d) 10%
15. A is what per cent of $\left(\frac{9}{15}\right)$ A?
(a) 60% (b) 100%
(c) 133.33% (d) 166.66%
16. If the length and breadth of a rectangle are decreased by 10%, then by what per cent does the area decrease?
(a) 19% (b) 25%
(c) 10% (d) 20%
17. The price of rice increased from ₹15 by 15% and then reduced by 30 paise. What was the net increase?
(a) 10% (b) 12%
(c) 13% (d) 28%
18. What is 30% of 55% of 100?
(a) 25 (b) 85
(c) 16.5 (d) 11.5
19. If 20% of a number exceeds 16% of the same number by 16, what is the number?
(a) 400 (b) 40
(c) 4000 (d) 160



20. Which is the largest?
 (a) $66\frac{1}{6}\%$ (b) $\frac{3}{5}$
 (c) 0.65 (d) $\frac{16}{25}$
21. In a town, there are 2,500 men and 2,500 women. If the number of men increased by 20% and women decreased by 20%, women as a percentage of men now is
 (a) 60% (b) 66.67%
 (c) 80% (d) 83.33%
22. In order to increase sales, price of a product was decreased by 20%. The net sales increased by 28%. What was the percentage increase in number of units sold?
 (a) 48% (b) 50%
 (c) 60% (d) 83%
23. In a class of 300 students, the number of boys is twice that of girls. If 50% of boys and 48% of girls appear in examinations, how many students did not appear?
 (a) 6 (b) 160
 (c) 152 (d) 144
24. Al Pacino invested 40% of his money in shares, 20% of rest in property and lost 25% of the remaining in a casino. What per cent does he have now?
 (a) 15% (b) 40%
 (c) 42% (d) 36%
25. How many votes did the loser get?
 (a) 200 (b) 150
 (c) 120 (d) 100
26. How many votes were cast?
 (a) 200 (b) 300
 (c) 400 (d) 500
27. What percentage of the total votes did the winner get?
 (a) 60% (b) 50%
 (c) 80% (d) 66.66%
28. If the numerator of a fraction is increased by 25% and the denominator is decreased by 20%, the new value is $\frac{5}{4}$. What was the original fraction?
 (a) $\frac{3}{5}$ (b) $\frac{4}{5}$
 (c) $\frac{7}{8}$ (d) $\frac{3}{7}$
29. If 28% of a number is less than 43% of the same number by 75. What is 30% of that number?
 (a) 120 (b) 150
 (c) 180 (d) 200
30. In an examination, it is required to get 45% marks to pass. A student got 138 marks and failed by 15%. What were the maximum marks?
 (a) 400 (b) 450
 (c) 460 (d) 600
31. A 'laddoo' is made of 70% flour, 20% sugar and rest is 'ghee'. What is the quantity of 'ghee' in 2 kg laddoos?
 (a) 200 g (b) 2 kg
 (c) 100 g (d) 400 g
32. Calculation shows that an angle is $37\frac{1}{2}^\circ$. The size obtained by drawing and measurement is 36° . The error percentage is
 (a) $1\frac{1}{2}\%$ (b) 3%
 (c) 4% (d) $4\frac{1}{6}\%$
33. Avinash spends 30% of his income on petrol for scooter, $\frac{1}{4}$ of the remaining on house rent and the balance on food. If he spends ₹300 on petrol, then what is the expenditure on house rent?
 (a) ₹525 (b) ₹1,000
 (c) ₹675 (d) ₹175

Directions for questions 25 to 27: Answer the questions based on the following information.

In an election, there were only 2 candidates. The losing candidate received $66\frac{2}{3}\%$ of the votes the winner got. The votes polled in favour of the loser were 60 less than that of the winner.

34. If x% of 'a' is the same as y% of 'b', then z% of 'b' is
 (a) $\frac{yz}{x}$ % of 'a' (b) $\frac{xy}{z}$ % of 'a'
 (c) $\frac{xz}{y}$ % of 'a' (d) None of these
35. In an examination, A got 10% marks less than B; B got 25% marks more than marks obtained by C; and C got 20% marks less than D. If A got 360 marks out of 500, the percentage of marks obtained by D was
 (a) 70% (b) 75%
 (c) 80% (d) 85%
36. p is six times as large as q. The percentage by which q is less than p is
 (a) 83.33% (b) 16.66%
 (c) 90% (d) 60%
37. In an election, involving two candidates, 68 votes were declared invalid. The winning candidate scores 52% of the valid votes and wins by 98 votes. The total number of votes polled is
 (a) 2,518 (b) 2,450
 (c) 2,382 (d) None of these
38. The price of sugar is increased by 20%. As a result, a family decreases its consumption by 25%. The expenditure of the family on sugar will be decreased by
 (a) 10% (b) 5%
 (c) 14% (d) 15%
39. The current birth rate per thousand is 32, whereas corresponding death rate is 11 per thousand. The net growth rate in terms of population is
 (a) 0.021% (b) 0.0021%
 (c) 21% (d) 2.1%
40. The length of a rectangle is increased by 60%. By what per cent the width must be decreased to maintain the same area?
 (a) 37.5% (b) 60%
 (c) 75% (d) None of these
41. A 50 L mixture of milk and water has 35% water. What is the quantity of milk in it?
 (a) 27.5 L (b) 15 L
 (c) 32.5 L (d) 65 L
42. Producer A gained 10% if the SP of the product is ₹121. What would the loss/profit be if the SP is reduced by ₹11?
 (a) Loss of 10% (b) Gain of 5%
 (c) Loss of 5% (d) No profit, no loss
43. A merchant makes a profit of 8% even after giving a 10% discount. What is the profit if he sells at marked price?
 (a) 10% (b) 18%
 (c) 26% (d) 20%
44. An increase of ₹60 in the monthly salary of Madan made it 50% of the monthly salary of Kamal. What is Madan's present monthly salary?
 (a) ₹180 (b) ₹240
 (c) ₹300 (d) Data inadequate
45. The salaries of A and B together amount to ₹2,000. A spends 95% and B spends 85% of their salaries. Now if their savings are the same, what is A's salary?
 (a) ₹1,500 (b) ₹1,250
 (c) ₹750 (d) ₹1,600
46. Raman's salary was decreased by 50% and subsequently increased by 50%. He has a loss of
 (a) 0% (b) 25%
 (c) 0.25% (d) 2.5%



My Doubts



Simple Interest & Compound Interest

Interest is the consideration (excess money) given by a borrower to the lender for the use of the money. The money which has been lent or borrowed is called the principal. The interest charged per ₹100 per year is known as the rate of interest per annum, and is always represented as a percentage.

Simple Interest

When the interest is payable on the principal alone, it is known as simple interest.

The amount of interest is always constant.

Basic formula for simple interest (SI) is $\frac{P \times R \times T}{100}$.

Where P = Principal or sum being borrowed or lent

R = Rate of interest per year

T = Time period for which the amount is borrowed

Amount (A) = Principal + Simple interest

$$A = P + \frac{P \times R \times T}{100}$$

? Example

What shall be the interest to be paid on a principal of ₹14,000 borrowed at a rate of 15% for a period of 3 years and 6 months?

✓ Solution

$$SI = \frac{P \times R \times T}{100}; P = 14,000, R = 15\% \text{ and}$$

$$T = 3.5 \text{ years}$$

$$\text{So, SI} = \frac{(14000 \times 15 \times 3.5)}{100} = ₹7,350$$

Compound Interest

In case of simple interest, the lender receives a fixed amount of money for every defined period of time. There is another case in which the interest amount received after every given period of time is reinvested at the same rate of interest as agreed upon initially.

In other words, the amount received at the end of year 1 becomes principal for period 2, and so on.

? Example

What shall be the amount for a sum of ₹1,000 at 10% per annum for 3 years compounded annually.

✓ Solution

Amount at the end of year 1 is

$$A_1 = \frac{(P \times R \times T)}{100} + P_1$$

$$= \frac{1000 \times 10 \times 1}{100} + 1000$$

$$= ₹1,100$$

This shall be the principal for year 2.

$$A_2 = \frac{(1100 \times 10 \times 1)}{100} + 1100 = ₹1,210$$

$$A_3 = \frac{(1210 \times 10 \times 1)}{100} + 1210 = ₹1,331$$

So, amount at the end of 3 years in case of compound interest (CI) is ₹1,331, while in the case of simple interest (SI), it shall be ₹1,300 (at 10%).

Alternative Method:

CI can be calculated using the following formula.

Principal + CI = Amount

$$A = P \left(1 + \frac{R}{100} \right)^n$$

$$CI = P \left(1 + \frac{R}{100} \right)^n - P$$

Where,

P = Principal or sum being borrowed

R = Rate of interest

n = Number of times the fixed interval is repeated.

**Note:**

If the interest is compounded semi-annually or half-yearly then the amount becomes

$$A = P \left(1 + \frac{r}{200} \right)^{2t}$$

If the interest is compounded quarterly then the amount becomes

$$A = P \left(1 + \frac{r}{400} \right)^{4t}$$

Important Note:

- I. If the word interest is given and nothing else is specified, the interest is considered as SI.
- II. If the interest is given by bank and nothing is specified, it is always CI.
- III. Population growth is always taken on compounding basis.
- IV. Difference between CI and SI for 2 years for the

same principle at the same interest is $P \left(\frac{r}{100} \right)^2$

Solved Examples**? Example 1**

If a certain sum amounts to ₹108 in 2 years, and ₹112 in 3 years, then find the principal and rate of interest (simple).

✓ Solution

Amount after 2 years = ₹108.

Amount after 3 years = ₹112.

In SI, interest amount remains the same for every year.

Therefore, interest for 1 year = ₹4.

Hence, the principal = $108 - 2(4) = ₹100$.

$$\text{Rate of interest} = \frac{4 \times 100}{100 \times 1} = 4\%$$

? Example 2

At what simple rate of interest shall a sum of money doubles itself in 4 years?

✓ Solution

Now important point to be noted is that the amount received by the lender is double the amount given,

that means interest = Principal.

So, if x is the Principal, then x is the simple interest.

$$\Rightarrow x = \frac{(x \times R \times 4)}{100}$$

$$\Rightarrow R = \frac{100}{4} = 25\%$$

? Example 3

Find the CI on ₹5,000 at 8% p.a. for 2 years, compounding being done annually.

✓ Solution

$P = ₹5000$, $R = 8\%$ and $n = 2$ years

$$5000 \left(1 + \frac{8}{100} \right)^2 = \text{Amount} = ₹5,832.$$

$$\begin{aligned} \text{CI} &= \text{Amount} - \text{Principal} \\ &= ₹(5832 - 5000) = ₹832 \end{aligned}$$

? Example 4

Find amount for ₹80,000 at 10% per annum, compounded bi-annually for 2 years.

✓ Solution

Here, $n = 2 \text{ years} \times 2 = 4$ periods

$$\text{Similarly, } R = \frac{10}{2} = 5\% \text{ (for half year);}$$

$P = ₹80,000$

$$A = 80000 \left(1 + \frac{5}{100} \right)^4 = ₹97,240.50$$

? Example 5

Find CI on ₹10,000 at 10% per annum for 9 months, compounded quarterly.

✓ Solution

$n = 3$ periods, $R = 2.5\%$ per period and

$P = ₹10,000$

$$\text{Amount} = 10000 \left(1 + \frac{2.5}{100} \right)^3 \approx ₹10,769 \text{ (approximately)}$$

$$\begin{aligned} \text{CI} &= \text{Amount} - \text{Principal} \\ &= 10769 - 10000 = ₹769 \end{aligned}$$



? Example 6

The difference between the CI and SI on a certain amount at 10% per annum for 2 years, compounded annually, is ₹372. Find the principal.

✓ Solution

Let the principal be 'a'.

$$SI = a \times \frac{10}{100} \times 2 = \frac{20}{100}a \text{ and}$$

$$CI = \text{Amount} - a$$

$$= a \left(1 + \frac{10}{100} \right)^2 - a = \frac{21}{100} \times a$$

$$CI - SI = ₹372$$

$$\frac{21}{100}a - \frac{20}{100}a = ₹372$$

$$\Rightarrow a = ₹37,200 = \text{Principal}$$

Alternative Method:

The above problem has an alternate. You need to understand the fact that for 1st period, $SI = CI$.

The difference between the values of CI and SI is because of accumulated interest building on interest which is reinvested. Therefore, for period 2, the difference between CI and SI is the interest for one period on the interest of period 1.

In the above example, the difference being 372 is the interest generated on interest for period 1 on the principal.

$$\text{Interest for period 1} = ₹ \left(372 \times \frac{100}{10} \right) = ₹3,720$$

$$\text{Therefore, principal} = ₹ \left(3720 \times \frac{100}{10} \right)$$

$$= ₹37,200$$

? Example 7

Ram invested a particular sum at 12% per annum with one of his friends Shyam and a similar amount in bank which pays interest at 12% p.a. compounded semi-annually. The difference between the amounts received after one year was ₹1,800. Find the total sum invested by Ram.

✓ Solution

Following the above illustration, we have the difference equal to the interest on the interest paid

on the principal for 6 months. (Rate of interest becomes $\frac{12}{2} = 6\%$).

Interest paid for 6 months = ₹ $\left(1800 \times \frac{100}{6} \right)$ on principal = ₹30,000.

Principal/sum invested in bank

$$= ₹ \left(30000 \times \frac{100}{6} \right) = ₹5,00,000.$$

$$\text{Total sum invested} = ₹(5,00,000 + 5,00,000) = ₹10,00,000.$$

? Example 8

If the CI on a certain sum for 3 years at 20% p.a. is ₹728, what is the sum invested?

✓ Solution

$$CI = 728 = P \left(1 + \frac{20}{100} \right)^3 - P$$

$$P \left(\frac{728}{1000} \right) = ₹728 \Rightarrow P = ₹1,000$$



Important Note:

When rates are different for different years, say r_1 , r_2 and r_3 for different years 1, 2 and 3 respectively, then, **Amount**

$$= \left(1 + \frac{r_1}{100} \right) \left(1 + \frac{r_2}{100} \right) \left(1 + \frac{r_3}{100} \right).$$

? Example 9

Find the amount after 3 years if the principal is ₹10,000 and rates are 10%, 8% and 12% in 1st, 2nd and 3rd year respectively.

✓ Solution

Amount

$$= ₹10,000 \left(1 + \frac{10}{100} \right) \left(1 + \frac{8}{100} \right) \left(1 + \frac{12}{100} \right)$$

$$= ₹10,000 \times 1.1 \times 1.08 \times 1.12$$

$$= ₹13,305.60$$

? Example 10

What is the CI received on a sum of ₹26,000 at rates 10%, 12% and 8% each, prevailing in that order for years 1, 2 and 3, interest being compounded semi-annually for the given three years?

✓ Solution

Amount

$$= 26000 \times \left(1 + \frac{5}{100}\right)^2 \left(1 + \frac{6}{100}\right)^2 \left(1 + \frac{4}{100}\right)^2$$

$$= ₹34,836.17$$

$$CI = ₹(34836.17 - 26000) = ₹8,836.17$$

? Example 11

What is the amount of interest on ₹1,000 compounded annually at the rate of 10% for 3 years.

✓ Solution

$$A = 1000 \left(1 + \frac{10}{100}\right)^3 = ₹1,331$$

$$\Rightarrow \text{Interest} = ₹331$$

? Example 12

If a sum of money doubles at 12.5% p.a. interest, what is the time period in case of (i) SI and (ii) CI?

✓ Solution

(i) $R = 12.5\%$, Amount = $2P$

$$\text{Time} = \frac{100}{12.5} = 8 \text{ years}$$

(ii) $A = P \left(1 + \frac{R}{100}\right)^t$

$$2P = P \left(1 + \frac{R}{100}\right)^t \Rightarrow 2 = \left(\frac{9}{8}\right)^t$$

$$\Rightarrow t = 5.88 \text{ years (approximately)}$$

? Example 13

For how many years should ₹600 be invested at 10% p.a. in order to earn the same simple interest as earned by investing ₹800 at 12% p.a. for 5 years?

✓ Solution

$$SI \text{ required} = ₹ \left(\frac{800 \times 12 \times 5}{100} \right) = ₹480$$

$$\text{Time} = \left(\frac{100 \times 480}{600 \times 10} \right) = 8 \text{ years}$$

? Example 14

Prabhat took a certain amount of loan from a bank at the rate of 8% p.a. S.I. and gave the same amount to Ashish as a loan at the rate of 12% p.a. If at the end of 12 years, he made a profit of ₹320 in the deal, what was the original amount?

✓ Solution

Let the original amount be ₹x. Then,

$$\frac{x \times 12 \times 12}{100} - \frac{x \times 8 \times 12}{100} = 320 \Rightarrow x = \frac{2000}{3}$$

$$= ₹666.67$$

? Example 15

Rahul borrowed ₹830 from Mr Lal at 12% p.a. SI for 3 years. He then added some money to the borrowed sum and lent it to Shobha for the same period at the rate of 14% p.a. interest. If Rahul gains ₹93.90 in the whole transaction, what amount did he add from his side?

✓ Solution

Let the amount added be ₹x.

$$\frac{(830 + x) \times 14 \times 3}{100} - \frac{830 \times 12 \times 3}{100} = 93.90$$

$$\Rightarrow 830 \times 42 + 42x - 830 \times 36 = 9390$$

$$\Rightarrow 42x + 830 \times (42 - 36) = 9390$$

$$\Rightarrow 42x = 9390 - 4980$$

$$\Rightarrow x = \frac{4410}{42} = 105$$

$$\therefore \text{Amount added} = ₹105.$$

? Example 16

If a sum of money at simple interest doubles in 6 years, in how many years will it become four times?

✓Solution

Let sum be x. Then SI = x

$$\therefore \text{Rate} = \left(\frac{100 \times x}{x \times 6} \right) \% = \frac{50}{3} \%$$

Now, sum is x and SI is 3x, rate = $\frac{50}{3} \%$.

$$\therefore \text{Time} = \frac{100 \times 3x}{x \times \frac{50}{3}} = 18 \text{ years}$$

Alternative method:

Let principal be P. Thus, amount is 2P after 6 years.

Interest in 6 years = 2P – P = P.

To become four times, interest = 4P – P = 3P.

When interest is P, time taken = 6 years.

When interest is 3P, time taken = 3 × 6 = 18 years.

?Example 17

The rate of interest on a sum of money is 4% p.a. for the first 2 years; 6% p.a. for the next 3 years; and 8% p.a. for the period beyond 5 years. If the simple interest collected by the sum for a total period of 8 years is ₹1,280, what is the sum?

✓Solution

Let the sum be ₹x. Then

$$\frac{x \times 4 \times 2}{100} + \frac{x \times 6 \times 3}{100} + \frac{x \times 8 \times 3}{100} = 1280$$

$$\Rightarrow 50x = 1280 \times 100$$

$$\therefore x = ₹2,560$$

?Example 18

Vinod Kumar invested ₹1,600 for 3 years and ₹1,100 for 4 years at the same rate of simple interest. If the total interest from these investments is ₹506, what was the rate of interest?

✓Solution

$$\frac{1600 \times 3 \times R}{100} + \frac{1100 \times 4 \times R}{100} = 506$$

$$\Rightarrow 92R = 506$$

$$\Rightarrow R = 5 \frac{1}{2} \%$$

?Example 19

A man invests an amount of ₹15,860 in the names of his three sons A, B and C in such a way that they get the same interest after 2, 3 and 4 years respectively. If the rate of interest is 5%, then what will be the ratio of amounts invested among A, B and C?

✓Solution

Let the amounts invested be x, y, z respectively.

$$\text{Then, } \frac{x \times 2 \times 5}{100} = \frac{y \times 3 \times 5}{100} = \frac{z \times 4 \times 5}{100} = k ;$$

$$\therefore x = 10k, y = \frac{20}{3}k \text{ and } z = 5k.$$

$$\text{So, } x : y : z = 10k : \frac{20k}{3} : 5k = 30 : 20 : 15$$

$$= 6 : 4 : 3.$$

?Example 20

The rates of simple interest in two banks A and B are in the ratio 5 : 4. A person wants to deposit his total savings in two banks in such a way that he receives equal half yearly interest from both. What should be the ratio of his savings in banks A and B?

✓Solution

Let the savings be X and Y and the rates of simple interest be 5x and 4x respectively. Then,

$$X \times 5x \times \frac{1}{2} \times \frac{1}{100} = Y \times 4x \times \frac{1}{2} \times \frac{1}{100}$$

$$\Rightarrow \frac{X}{Y} = \frac{4}{5} \text{ i.e. } X : Y = 4 : 5$$

?Example 21

The value of a machine depreciates at the rate of 10% every year. It was purchased 3 years ago. If its present value is ₹8,748, what was its purchase price?

✓Solution

Let the purchase price be P.

Rate of depreciation = 10%

$$P \left(1 - \frac{10}{100} \right)^3 = 8748 \Rightarrow P = \left(8748 \times \frac{10}{9} \times \frac{10}{9} \times \frac{10}{9} \right)$$

$$= ₹12,000.$$



Practice Exercise

1. Two investments of ₹500 and ₹1500 have a combined yearly return of 8.5% of the total of the two investments. If the ₹500 investment has a yearly return of 7%, what per cent yearly return does the ₹1,500 investment have?
(a) 9% (b) 10%
(c) $10\frac{5}{8}\%$ (d) 11%
 2. A family made a down payment of ₹75 and borrowed a set of encyclopaedias that cost ₹400. The balance with interest was paid in 23 monthly payments of ₹16 each and a final payment of ₹9. The amount of interest paid was what per cent of the amount borrowed?
(a) 6% (b) 12%
(c) 14% (d) 16%
 3. The difference between the interests received from two different banks on ₹500 for 2 years is ₹2.50. The difference between their rates is
(a) 1% (b) 0.5%
(c) 0.25% (d) 25%
 4. A sum of ₹3,500 is lent for 5 years at 5% p.a. SI. The interest accrued and the amount to be repaid are respectively
(a) ₹875 and ₹4,375
(b) ₹975 and ₹4,475
(c) ₹500 and ₹4,000
(d) ₹375 and ₹3,875
 5. In what time, a sum of money will triple itself at SI 20% p.a.?
(a) 5 years (b) 10 years
(c) 15 years (d) 20 years
 6. What will be the CI on ₹1,000 for 3 years at 10% p.a.?
(a) ₹300 (b) ₹330
(c) ₹331 (d) ₹361
 7. Find the difference between SI and CI on ₹700 at the rate of 10% for 3 years?
(a) ₹20.90 (b) ₹21
(c) ₹21.70 (d) ₹24
 8. SI on a sum of money is one-fourth of principal. The number of years is equal to the rate of interest. Find the rate of interest.
(a) 2.5% (b) 7.5%
(c) 6% (d) 5%
 9. If CI for a certain sum for 2 years at 2% p.a. be ₹1,010, what is the principal?
(a) ₹20,000 (b) ₹25,000
(c) ₹25,250 (d) ₹27,500
 10. If CI for a certain sum at 3% for 2 years is ₹203, what would be the SI for the same period and at the same rate?
(a) ₹200 (b) ₹207.50
(c) ₹213.33 (d) ₹190
 11. What is the sum which when lent at 5% SI for 2 years would yield interest of ₹154?
(a) ₹1,450 (b) ₹1,540
(c) ₹1,650 (d) ₹1,480
 12. I bought a watch for ₹400 and sell it for ₹460 at a credit of 8 months. What is my approx. gain percentage considering interest rate to be 15% p.a.?
(a) 4.55% (b) 6.66%
(c) 7.5% (d) None of these
 13. The simple interest on ₹1,820 from March 9, 1994 to May 21, 1994 at 7.5% rate will be
(a) ₹29 (b) ₹28.80
(c) ₹27.30 (d) ₹22.50
- Directions for questions 14 and 15:** Answer the questions based on the following information.
- 'A' borrowed ₹4,000 at 10% CI for 3 years with the condition that half of the amount at the end of first and second years will be paid at the end of those years.
14. What is the amount due at maturity?
(a) ₹2,200 (b) ₹2,324
(c) ₹1,331 (d) ₹1,242
 15. In the above question, how much did 'A' pay in total as interest?
(a) ₹1,200 (b) ₹944
(c) ₹881 (d) ₹741



16. At what rate per cent, the simple interest on ₹1,125 will be ₹225 in 4 years?
(a) 4% (b) 5%
(c) $6\frac{2}{3}\%$ (d) Can't be determined
17. If I lend ₹5,000 for 3 years in two schemes (i) 11% SI and (ii) 10% CI, which is the beneficial scheme and by what amount?
(a) (i), ₹150 (b) (i), ₹50
(c) (ii), ₹5 (d) (ii), ₹50
18. A sum was put at simple interest at a certain rate for 2 years. Had it been put at 3% higher rate, it would have fetched ₹72 more. The sum is
(a) ₹1,200 (b) ₹1,500
(c) ₹1,600 (d) ₹1,800
19. If a sum of money doubles itself in 8 years at simple interest, the rate percentage per annum is
(a) 11.5% (b) 12%
(c) 12.5% (d) 13%
20. If x, y and z are three sums of money such that 'y' is the simple interest on 'x'; 'z' is the simple interest on 'y' for the same time and at the same rate of interest, then we have
(a) $x^2 = yz$ (b) $y^2 = xz$
(c) $z^2 = xy$ (d) $xyz = 1$
21. A sum of ₹1,550 was partly lent at 5% and 8% p.a. simple interest. The total interest received after 3 years was ₹300. The ratio of the money lent at 5% to that lent at 8% is
(a) 8 : 5 (b) 5 : 8
(c) 31 : 6 (d) 16 : 15
22. A man invested one-third of his capital at 7%, one-fourth at 8% and the remainder at 10% simple interest respectively. If his annual income is ₹561, the capital is
(a) ₹5,400 (b) ₹6,000
(c) ₹6,600 (d) ₹7,200
23. A man lends ₹10,000 in four parts. If he gets 8% on ₹2,000; 7.5% on ₹4,000; 8.5% on ₹1,400, what per cent must he get for the remainder, if his average annual interest is 8.13%?
(a) 10.5% (b) 9.25%
(c) 9% (d) 7%
24. ₹2,189 is divided into three parts such that their interests after 1, 2 and 3 years respectively is equal. The rate of simple interest being 4% p.a. in all cases, the smallest part is
(a) ₹702 (b) ₹398
(c) ₹756 (d) ₹1,093
25. If the simple interest on a sum of money for 3 years at 5% p.a. is ₹1,200, the compound interest for the same period at the same rate is
(a) ₹1,260 (b) ₹1,261
(c) ₹1,264 (d) ₹1,265
26. If the difference between the compound interest, compounded half-yearly and the simple interest on a sum at 10% p.a. for one year is ₹25, the sum is
(a) ₹9,000 (b) ₹9,500
(c) ₹10,000 (d) ₹10,500
27. A man borrowed ₹800 at 10% p.a. SI and immediately lent the whole sum at 10% p.a. CI. What does he gain at the end of 2 years?
(a) ₹6 (b) ₹8
(c) ₹10 (d) ₹12
28. A sum of money amounts to ₹4,624 in 2 years and to ₹4,913 in 3 years at compound interest. The sum is
(a) ₹4,096 (b) ₹4,260
(c) ₹4,335 (d) ₹4,360
29. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10?
(a) $2\frac{1}{2}$ years (b) $1\frac{1}{2}$ years
(c) $2\frac{1}{3}$ years (d) $1\frac{1}{3}$ years
30. A sum of ₹12,000 deposited at compound interest becomes double after 5 years. After 20 years it will become
(a) ₹1,20,000 (b) ₹1,92,000
(c) ₹1,24,000 (d) ₹96,000
31. The least number of complete years in which a sum of money put at 20% compound interest will be more than double is
(a) 3 (b) 4
(c) 5 (d) 6



32. A tree increases annually by one-eighth of its height. What will be the height of it after 2 years, if it stands today 64 cm high?
 (a) 72 cm (b) 74 cm
 (c) 75 cm (d) 81 cm
33. A sum of money becomes ₹13,380 after 3 years and ₹20,070 after 6 years on compound interest. The sum is
 (a) ₹8,800 (b) ₹8,890
 (c) ₹8,920 (d) ₹9,040
34. The compound interest on ₹30,000 at 7% per annum for a certain period is ₹4,347. The period is
 (a) 2 years (b) 2.5 years
 (c) 3 years (d) 4 years
35. At what rate of compound interest per annum will a sum of ₹1,200 become ₹1,348.32 in 2 years?
 (a) 7% (b) 6%
 (c) 7.5% (d) 6.5%
36. A sum of money invested at compound interest amounts to ₹800 in 3 years and ₹840 in 4 years. What is the rate of interest per annum?
 (a) 2% (b) 4%
 (c) 5% (d) 10%
37. A sum of money becomes eight times of itself in 3 years at compound interest. The rate of interest is
 (a) 100% (b) 80%
 (c) 1% (d) Data inadequate
38. A father left a will of ₹85,000 to be divided between his two sons aged 10 years and 12 years such that they may get equal amount when each attains the age of 18 years. If the money is reckoned at 10% p.a., find how much the son of age 10 gets at the time of the will?
 (a) ₹32,000 (b) ₹36,000
 (c) ₹40,000 (d) ₹42,000
39. Simple interest on a certain sum is 9 over 25 of the sum. Find the rate per cent and time, if both are numerically equal.
 (a) 6%, 6 years
 (b) 7%, 7 years
 (c) 9%, 9 years
 (d) 8%, 8 years
40. Find the difference between SI and CI on ₹2,000 for 2 years at the rate of 10% per annum.
 (a) ₹20 (b) ₹22
 (c) ₹24 (d) ₹30
41. Some of the amount of ₹7,000 was lent at 6% p.a. and the remaining at 4% p.a. If the total simple interest from the fraction in 5 years was ₹1,600, what was the sum lent at 6% p.a.?
 (a) ₹2,000 (b) ₹2,500
 (c) ₹3,000 (d) ₹4,000
42. What annual payment will discharge a debt of ₹1,200 in 4 years at 4% p.a. simple interest?
 (a) ₹200 (b) ₹348
 (c) ₹400 (d) ₹500
43. In what time will ₹36 become ₹45 at 6.25% p.a. simple interest?
 (a) 2 years (b) 3 years
 (c) 4 years (d) 8 years
44. The simple interest on ₹400 for 8 months at 5 paise per rupee per month is
 (a) ₹120 (b) ₹160
 (c) ₹200 (d) ₹400
45. The simple interest on a sum of money for 3 years is ₹360, and the compound interest on the same sum at the same rate for 2 years is ₹270. The rate per cent per annum is
 (a) 25% (b) 50%
 (c) 60% (d) 80%
46. If ₹1 becomes ₹10 in 50 years at simple interest, the rate per cent per annum is
 (a) 16% (b) 18%
 (c) 20% (d) None of these
47. A man lent ₹400 and ₹600 for 3 years, at the same rate of simple interest and received only ₹90 as interest. What was the rate percentage per annum?
 (a) 1% (b) 2%
 (c) 3% (d) 4%



Profit, Loss & Discount

Cost Price (CP)

Cost price is the price at which an article is purchased. Loss or gain is reckoned on the cost price.

Selling Price (SP)

Selling price is the price at which an article is sold.

$$\text{Profit} = \text{SP} - \text{CP}$$

$$\text{Loss} = \text{CP} - \text{SP}$$

Commission

Commission is an incentive given by the parent or manufacturing company to the retailer based on the sales of product.

Formulae

- Profit percentage

$$= \frac{\text{Profit}}{\text{CP}} \times 100 = \left(\frac{\text{SP}}{\text{CP}} - 1 \right) \times 100$$
- Loss percentage

$$= \frac{\text{Loss}}{\text{CP}} \times 100 = \left(1 - \frac{\text{SP}}{\text{CP}} \right) \times 100$$
- $$\text{SP} = \frac{(100 + \text{Profit percentage}) \times \text{CP}}{100}$$

 or
$$\frac{(100 - \text{Loss percentage}) \times \text{CP}}{100}$$
- $$\text{CP} = \frac{100 \times \text{SP}}{(100 + \text{Profit percentage})}$$

 or
$$\frac{100 \times \text{SP}}{(100 - \text{Loss percentage})}$$
- If marked price be MP and discount percentage be 'd', then

$$\text{SP} = \frac{\text{MP}(100-d)}{100}; \text{MP} = \frac{100 \times \text{SP}}{(100-d)}$$

- M.P. $\xrightarrow{- \text{Discount}}$ S.P. $\xleftarrow{\text{Profit} +}$ C.P.
- If 2 items are sold, each at ₹X, one at a gain of P% and the other at a loss of P%, then overall

$$\text{loss percentage} = \frac{P^2}{100} \%$$
- Successive Discounts: If two discounts of a% and b% are given successively, then the net discount

$$\text{given in percentage is} = a + b + \frac{ab}{100}$$

 We have learnt this in the percentage chapter.

Solved Examples

? Example 1

A boy buys eggs at 10 for ₹1.80 and sells them at 11 for ₹2. What is his gain or loss percentage?

✓ Solution

To avoid fractions, let the number of eggs purchased be LCM (10, 11) = 110

$$\text{CP of 110 eggs} = \frac{110 \times 1.80}{10} = ₹19.80$$

$$\text{SP of 110 eggs} = \frac{110 \times 2.00}{11} = ₹20.$$

$$\text{Profit percentage} = \frac{0.20 \times 100}{19.80} = 1.01\%$$

? Example 2

A woman buys certain number of apples at 15 per rupee and the same number at 20 per rupee. She mixes and sells them at 35 for ₹2. What is her gain or loss percentage?

Types of question	Example	Approach to question
1. If a dealer sells a goods at cost price but uses a false weight, find his profit percentage.	A dishonest dealer professes to sell his goods at cost price, but he uses a weight of 960 gm for 1 kg. Find his profit %.	$\text{Profit \%} = \frac{x}{y} \times 100$ where x is the error and y is True value – x. $\therefore \frac{40}{1000 - 40} \times 100 = 4.16\%$
2. If A sells to B at a profit of x%, B sells to C at a profit of y% and C pays ₹P for it, find the cost for A.	A sells a cycle to B at a profit of 10%, B sells to C at a profit of 20%. If C pays ₹264 for it, what did A pay for it?	$C.P_c = \frac{100 + x}{100} \times \frac{100 + y}{100} \times P$ where x and y are the profit % for A and B, and P is the cost for A. $\therefore \frac{110}{100} \times \frac{120}{100} \times P = 264$ $P = ₹200$
3. If cost price of A articles is equal to the selling price of B articles, find the profit %.	The C.P. of 10 articles is equal to the S.P. of 9 articles. Find the profit %.	$\text{Profit \%} = \frac{A - B}{B} \times 100$ where A is the number of articles bought and B is the number of articles sold. $\therefore \frac{10 - 9}{9} \times 100 = 11.11\%$
4. The cost price of two articles is the same. If one is sold at a X% profit and the other at a loss of X%, find his profit or loss %.	Amit buys 2 cows for ₹200 each. He sells one at a profit of 10% and the other at a loss of 10%. Find his profit or loss %.	For the same cost price and equal profit and loss %, there is no profit and no loss. $\therefore \text{Profit or loss} = 0\%$
5. The selling price of two articles is the same. If one is sold at X% profit and the other at a loss of X%, find his profit or loss %.	Amit sells 2 cows for ₹200 each. On one he gets a profit of 10%, while loses 10% on the other. What is his overall profit or loss %?	$\text{Loss \%} = \frac{X^2}{100} \%$ $= \frac{10^2}{100} \% = 1\%$
6. Find the single rate of discount equal to two successive discounts of x % and y%.	What single rate of discount is equal to two successive discounts of 10% and 15%?	Using successive percentage change: $-10 - 15 + \frac{10 \times 15}{100}$ $-25 + 1.5 = -23.5\%$ Hence, discount = 23.5%
7. If x% discount on an article is given on cash payment, find he % that should be marked above the cost price so as to make a profit of y%.	A dealer allows a discount of 7% for cash payment. How much % above the cost price should he mark his goods to make a profit of 10%?	$M.P = \frac{100 + y}{100 - x} \times 100$ where x% is the discount and y% is the profit. $\therefore 110/93 \times 100 = 118.28$ Hence, $118.28 - 100 = 18.28\%$

✓Solution

Suppose the woman buys (LCM of 15, 20 and 35) 420 apples.

Cost at the rate of 15 per rupee = ₹28.

Cost at the rate of 20 per rupee = ₹21.

Total cost for 840 apples = ₹49.

$$\text{SP for 840 apples} = \frac{840 \times 2}{35} = ₹48;$$

$$\text{Loss percentage} = \frac{1 \times 100}{49} = 2.04\%$$

?Example 3

A man bought 80 kg rice for ₹88 and sold it at a loss of as much money as he received for 20 kg. At what price did he sell it?

✓Solution

CP of 80 kg – SP of 80 kg = SP of 20 kg

SP of 100 kg = CP of 80 kg = ₹88

SP of 1 kg = 88 paise;

He sold it at 88 paise per kilogram.

?Example 4

Goods are purchased for ₹450 and one-third is sold at a loss of 10%. At what profit per cent should the remainder be sold so as to gain 20% on the whole transaction?

✓Solution

Total cost price of goods = ₹450

$$\text{SP of total goods} = 450 \times \frac{120}{100} = ₹540$$

$$\text{SP of one-third goods} = \frac{90}{100} \times \frac{450}{3} = ₹135$$

$$\text{SP of the remaining goods} = 540 - 135 = ₹405$$

$$\text{CP of the remaining (two-thirds) goods} = ₹300$$

$$\text{Hence, profit percentage} = \frac{105}{300} \times 100 = 35\%$$

Alternative method:

Applying weighted average, one-third of quantity there is a loss of 10% (or a profit of –10%) and balance two-thirds gives a profit of x%.

Hence, overall profit is given by $\frac{1}{3}(-10\%)$ of CP

$$+ \frac{2}{3}(x\%) \text{ of CP} = 20\% \text{ of CP.}$$

Thus, $x = 35\%$.

?Example 5

A reduction of 10% in the price of sugar enables a man to buy 25 kg more for ₹225. What is the original price of sugar (per kilogram)?

✓Solution

Let the original price be x.

$$\text{Original quantity} = \frac{225}{x}$$

$$\text{New price} = 0.9x$$

$$\text{New quantity} = \frac{225}{0.9x}$$

$$\Rightarrow \frac{225}{0.9x} - \frac{225}{x} = 25$$

$$\Rightarrow x = ₹1/\text{kg}$$

Alternative method:

$$\text{CP of 25 kg} = \frac{10}{100} \times 225 = ₹22.5;$$

$$\text{Reduced CP of 1 kg} = \frac{22.5}{25} = ₹0.90$$

$$\begin{aligned} \text{Original price of sugar (per kilogram)} \\ = \frac{0.90}{90} \times 100 = ₹1 \end{aligned}$$

?Example 6

A man sold an article at a profit of 25%. If he had bought it at 20% less and sold it for ₹10.50 less, he would have gained 30%. Find the CP of the article.

✓Solution

Let CP = ₹x; SP = 1.25x

New CP = 0.8x; new SP = 1.25x – 10.50

But new SP = 130% of new CP = 1.3 × 0.8x

Therefore, $1.3 \times 0.8x = 1.25x - 10.50$

$$\Rightarrow x = ₹50.$$

? Example 7

A vendor bought bananas at 6 for ₹5 and sold at 4 for ₹3. Find his gain or loss percentage.

✓ Solution

Let number of bananas be 24. (A multiple of 4 and 6)

$$\text{Cost price} = \frac{24}{6} \times 5 = ₹20$$

$$\text{Selling price} = \frac{24}{4} \times 3 = ₹18;$$

$$\therefore \text{Loss percentage} = \frac{2}{20} \times 100 = 10\%$$

? Example 8

If a commission of 10% is given on the marked price of an article, the gain is 25%. Find the gain percentage, if the commission is increased to 20%.

✓ Solution

Let marked price = ₹100

Commission = ₹10

SP after 10% commission = 90

$$\text{CP} = \frac{90}{125} \times 100 = ₹72$$

New commission = ₹20

New SP = ₹80, Gain = 80 – 72 = 8

$$\text{Gain percentage} = \frac{8 \times 100}{72} = 11.1\%$$

? Example 9

Peanuts are sold at 60 per rupee. If the vendor decides to hike SP by 20%, how many peanuts can be bought per rupee?

✓ Solution

$$\text{SP of one peanut} = ₹ \frac{1}{60}.$$

$$\text{New SP} = \frac{1.2}{60} = ₹ \frac{1}{50}$$

Therefore, 50 peanuts can be bought per rupee.

? Example 10

Sumit buys 9 books for ₹100 but sells 8 for ₹100. What is the net profit percentage?

✓ Solution

SP of 8 books = ₹100

$$\therefore \text{SP of one book} = \frac{100}{8} = ₹12.50$$

$$\therefore \text{SP of 9 books} = 12.50 \times 9 = ₹112.50$$

$$\therefore \text{Profit percentage} = 12.5\%$$

Alternative method:

CP of 9 books = SP of 8 books

CP of 8 books + CP of 1 book = SP of 8 books

CP of 1 book = SP of 8 books – CP of 8 books

Profit = CP of 1 book

$$\text{Profit percentage} = \frac{\text{CP of 1 book}}{\text{CP of 8 books}} \times 100$$

$$= 12.5\%.$$

? Example 11

If by selling an article for ₹100, a man gains ₹15, then what is his gain percentage?

✓ Solution

SP = ₹100, gain = ₹15. So, CP = SP – Gain

$$\therefore \text{Gain percentage} = \left(\frac{15}{85} \times 100 \right) \%$$

$$= 17 \frac{11}{17} \%$$

? Example 12

A grain dealer gains to the extent of 10% while buying as well as selling by using false weights. What is his total gain?

✓ Solution

Rule: Gain percentage

$$= \frac{(100 + \text{Common gain percentage})^2}{100} - 100$$

Gain percentage

$$= \left[\frac{(100 + 10)^2}{100} - 100 \right] \% = \left(\frac{12100 - 10000}{100} \right) \%$$

$$= 21\%$$

Alternate Method:

This question can also be done by using the approach

$$a + b + \frac{ab}{100} \Rightarrow 10 + 10 + \frac{10 \times 10}{100} = 21\%$$

? Example 13

A person bought 20 L milk at the rate of ₹8 per litre. He got it churned after spending ₹10 and got 5 kg cream and 20 L toned milk. If he sold the cream at ₹30 per kilogram and toned milk at ₹4 per litre, what was his profit in the transaction?

✓ Solution

Investment = ₹(20 × 8 + 10) = ₹170;

Receipt = ₹(30 × 5 + 20 × 4) = ₹230

$$\therefore \text{Gain percentage} = \left(\frac{60}{170} \times 100 \right)$$

$$= 35.29\% \approx 35.3\%$$

? Example 14

A person earns 15% on an investment but loses 10% on another investment. If the ratio of the two investments is 3 : 5, what is the gain or loss on the two investments taken together?

✓ Solution

Let the investments be 3x and 5x. Then, the total investment = 8x.

Total receipt = (115% of 3x + 90% of 5x)

$$= (3.45x + 4.5x) = 7.95x$$

$$\text{So loss} = 8x - 7.95x = 0.05x$$

$$\therefore \text{Loss percentage} = \left(\frac{0.05x}{8x} \times 100 \right) \%$$

$$= 0.625\%$$

? Example 15

Vivek purchased 120 tables at a price of ₹110 per table. He sold 30 tables at a profit of ₹12 per table and 75 tables at a profit of ₹14 per table. The remaining tables were sold at a loss of ₹7 per table. What is the average profit per table?

✓ Solution

$$\text{Total CP} = ₹(120 \times 110) = ₹13,200$$

$$\text{Total SP} = (30 \times 110 + 30 \times 12) + (75 \times 110 + 75 \times 14) + (15 \times 110 - 15 \times 7) = ₹14,505$$

$$\therefore \text{Average profit} = ₹ \left(\frac{14505 - 13200}{120} \right)$$

$$= ₹ \frac{1305}{120} = ₹10.875$$



My Doubts



Practice Exercise

1. If books bought at prices ranging from ₹200 to ₹350 are sold at prices ranging from ₹300 to ₹425, what can be the greatest possible profit?
(a) ₹400
(b) ₹600
(c) Cannot be determined
(d) None of these
2. The cost price of 20 articles is same as the selling price of 15 articles. The profit percentage is
(a) 25%
(b) 30%
(c) 33.33%
(d) 50%
3. If the selling price of an article is $\frac{4}{3}$ times of its cost price, the profit percentage is
(a) $33\frac{1}{3}\%$
(b) $25\frac{1}{4}\%$
(c) $20\frac{1}{2}\%$
(d) $20\frac{1}{3}\%$
4. If the cost price of 12 books is same as the selling price of 16 books, the loss percentage is
(a) 15%
(b) 20%
(c) 25%
(d) 30%
5. A man loses the selling price of 4 apples on selling 36 apples. His loss percentage is
(a) 12.5%
(b) 11.11%
(c) 10%
(d) None of these
6. By selling a table, Aditya earned a profit equal to one-fourth of the price for which he bought it. If he sold it for ₹375, what was the cost price?
(a) ₹281.75
(b) ₹300
(c) ₹312.50
(d) ₹350
7. A man bought a number of bananas at 3 for a rupee and an equal number at 2 for a rupee. At what price per dozen should he sell them to make a profit of 20%?
(a) ₹4
(b) ₹5
(c) ₹6
(d) ₹7
8. A man bought oranges at ₹5 a dozen and an equal number at ₹2 a dozen. He sold them at ₹5.50 a dozen and made a profit of ₹50. How many oranges (in dozens) did he buy?
(a) 25
(b) 40
(c) 50
(d) 60
9. A producer of tea blends two varieties costing ₹18 per kilogram and another ₹20 per kilogram in the ratio 5 : 3. If he sells the blended variety at ₹21 per kilogram, what is his gain percentage?
(a) 10%
(b) 12%
(c) 19%
(d) 22%
10. Ram purchased 35 kg rice at ₹9.50 per kilogram and 30 kg at ₹10.50 per kilogram, and mixed them. At what price (per kilogram) should he sell the mixture to gain 35%?
(a) ₹12
(b) ₹12.50
(c) ₹13
(d) ₹13.50
11. Oil costs ₹100 per liter. After adulterating it with another oil that costs ₹50 per liter, Ram sells the mixture at ₹96 per liter making a profit of 20%. In what ratio does he mix the two?
(a) 1 : 2
(b) 3 : 2
(c) 3 : 1
(d) None of these
12. Two varieties of rice are mixed in the ratio 2 : 3 and sold at ₹22 per kilogram, resulting in a profit of 10%. If the cost of the first variety rice be ₹14 per kilogram, the cost per kilogram of the second variety rice will be
(a) ₹23
(b) ₹24
(c) ₹25
(d) None of these
13. A dealer who professes to sell his goods at cost price uses a 900 g weight for a kilogram. His gain percentage is
(a) 9%
(b) 10%
(c) 11%
(d) 11.11%
14. A dealer professes to sell his goods at cost price, but he uses a false weight and gains $6\frac{18}{47}\%$. What is the weight used per kilogram of goods sold by him?
(a) 953 g
(b) 940 g
(c) 960 g
(d) 947 g



15. By selling toffees at 20 for a rupee, a man loses 4%. To gain 20% for a rupee he must sell
(a) 16 toffees
(b) 20 toffees
(c) 24 toffees
(d) 25 toffees
16. A man gains 10% by selling an article for a certain price. If he sells it at double the price, the profit is
(a) 20% (b) 120%
(c) 100% (d) 140%
17. 'A' bought a cycle and spent ₹110 on its repairs. He then sold it to 'B' at a profit of 20%. 'B' sold it to 'C' at a loss of 10%. 'C' sold it at a profit of 10% for ₹1,188. How much did 'A' buy it for?
(a) ₹850 (b) ₹890
(c) ₹930 (d) ₹950
18. If the manufacturer gains 10%, the wholesaler gains 15%, and the retailer gains 25%, what is the cost of production of the goods if the retail price is ₹1,265?
(a) ₹632.50 (b) ₹800
(c) ₹814 (d) ₹834.34
19. A man sells 2 cows for ₹4,000 each, neither gaining nor losing in the deal. If he sold one cow at a gain of 25%, then the other cow is sold at a loss of
(a) 16.66% (b) 18.22%
(c) 25% (d) None of these
20. Two horses were sold for ₹12,000 each, one at a loss of 20% and the other at a gain of 20%. The entire transaction resulted in
(a) no loss, no gain
(b) loss of ₹1,000
(c) gain of ₹1,000
(d) gain of ₹2,000
21. A vendor has 24 kg apples. He sells part of these at 20% gain and the balance at 5% loss. If on the whole he earns a profit of 10%, the part of apples sold at a loss is
(a) 6 kg (b) 4.6 kg
(c) 9.6 kg (d) 11.4 kg
22. The cost price of an article is 40% of the selling price. The percentage that selling price is of the cost price is
(a) 250% (b) 240%
(c) 60% (d) 40%
23. By selling an article, there is a loss of 2.5%. By selling it at ₹6 more, there is a gain of 5%. The cost price of the article is
(a) ₹78 (b) ₹80
(c) ₹82 (d) ₹84
24. A man sold an article for a gain of 5%. If he had bought it for 5% less and sold it for ₹1 less, he would have made a profit of 10%. The cost price of the article is
(a) ₹100 (b) ₹150
(c) ₹200 (d) ₹500
25. Profit after selling goods for ₹425 is the same as the loss after selling it at ₹355. What is its cost price?
(a) ₹385 (b) ₹390
(c) ₹395 (d) ₹400
26. The profit earned by selling a table for ₹900 is double the loss incurred when it is sold for ₹450. At what price should it be sold to make 25% profit?
(a) ₹600 (b) ₹750
(c) ₹800 (d) Data inadequate
27. Successive discounts of 30%, 20% and 10% is equivalent to a single discount of
(a) 50% (b) 49.6%
(c) 39.4% (d) 51%
28. The difference between the discount of 40% on ₹500 and two successive discounts of 36% and 4% on the same price is
(a) nil (b) ₹2
(c) ₹7.20 (d) ₹1.93
29. At what percentage above the cost price must an article be marked so as to gain 33% after allowing a discount of 5%?
(a) 38% (b) 40%
(c) 43% (d) 48%
30. A trader allows two successive discounts of 20% and 10%. If he sells the article for ₹108, then the marked price of the article is
(a) ₹150 (b) ₹148
(c) ₹142 (d) ₹140
31. A merchant intends to offer a discount of 10% but would like to maintain the current prices. By what percentage should he increase the list price?
(a) 10% (b) 9.09%
(c) 11.11% (d) 12.5%

32. A hotel offers 10% discount on food purchased during happy hours and 5% overall discount on all purchases exceeding ₹150. What is the net percentage discount offered to a customer who purchased food worth ₹190 during the happy hours?
 (a) 14.75% (b) 15%
 (c) 14% (d) 14.5%
33. A trader quotes ₹45 for an article whose cost price is ₹30. The customer pays him a fifty-rupee note. The trader does not have the change to return ₹5 to the customer. He thus goes to a neighbouring shop to get change for ₹50. The customer collects his balance of ₹5. The next day the neighbouring shop owner realizes that the fifty-rupee note was fake and demanded ₹50 back from the trader. What is the total loss to the trader?
 (a) ₹80 (b) ₹85
 (c) ₹35 (d) ₹30
34. A merchant sells rice and makes a profit of 6%. His cost price increases by 10% and thus he increases his selling price also by 10%. What profit percentage does he earn now?
 (a) 6%
 (b) 6.6%
 (c) 10%
 (d) None of these
35. A trader buys 78 kg of wheat for ₹492. He sells 40% of this at a loss of 20%. What should be the percentage mark up on the remaining so as to gain an overall 25%?
 (a) 40% (b) 55%
 (c) 28% (d) 45%
36. Sneha buys X eggs to resell them at a profit of 10% but loses 10% of the eggs. By how much should she mark up the selling price in order to retain 10% profit?
 (a) 30% (b) 40%
 (c) 33.33% (d) 22.22%
37. A merchant gives a discount of 10% on tea, but uses a weight of 900 gm per kilogram. Find his net profit/loss percentage.
 (a) 3.33% (b) 2.05%
 (c) 4.67% (d) No profit no loss
38. By selling 25 L of milk at ₹50 per litre, a merchant earns a profit equivalent to the cost price of 5 L. Find the profit percentage.
 (a) 15% (b) 25%
 (c) 20% (d) 18%
39. A man bought 100 kg of rice for ₹1,100 and sold it at a loss of as much money as he received for 20 kg of rice. At what price approximately did he sell the rice?
 (a) ₹9.17 (b) ₹10.50
 (c) ₹10.14 (d) ₹9.50



My Doubts



Ratio & Proportion, Mixture Alligation, Partnership & Average

Ratio of two quantities is the fraction that one quantity is relative to the other. Ratio of **a** to **b** is $\frac{a}{b}$ or **a** : **b**. **a**, the first term is called **antecedent** and the second term **b** is called **consequent**. It should be noted that both **a** and **b** are of the same units.



Important Note:

If any ratio $\frac{a}{b}$ is given such that $\frac{a}{b} > 1$ or $a > b$, and there is an integer x such that x is +ve, then

$$\frac{a+x}{b+x} < \frac{a}{b} \quad \text{and} \quad \frac{a-x}{b-x} > \frac{a}{b}$$

If any ratio $\frac{a}{b}$ is given such that $\frac{a}{b} < 1$ or $a < b$, and there is an integer x such that x is +ve, then

$$\frac{a+x}{b+x} > \frac{a}{b} \quad \text{and} \quad \frac{a-x}{b-x} < \frac{a}{b}$$

Proportion

A proportion is a statement of equality that exists between two ratios.

Example: $1 : 4 :: 2 : 8$ is a proportion consisting of four terms. The first and the last terms are called extremes, and the second and the third terms are called the means.

Direct Proportion

Two variables are directly proportional if their corresponding values have constant ratio, if one quantity is multiplied or divided by the same number, the ratio of the variables remains unchanged.

Suppose 'x' & 'y' are two variable under Direct proportion,

then their ratio remains constant. i.e. $\frac{x}{y} = \text{constant}$.

Hence, for two iterations in which the values of x are x_1 and x_2 , and the corresponding values of y are y_1 & y_2

$$\text{Then, } \frac{x_1}{y_1} = \frac{x_2}{y_2}$$

By **componendo** and **dividendo** property

$$\text{If } \frac{x_1}{y_1} = \frac{x_2}{y_2}, \text{ then } \frac{x_1 + y_1}{x_1 - y_1} = \frac{x_2 + y_2}{x_2 - y_2}$$

? Example

If one cup of oil can be obtained by crushing 50 groundnuts, then how many cups of oil can be obtained by crushing 150 groundnuts?

✓ Solution

	No. of Groundnut	No. of cups
Case - I	50	1
Case - II	150	y

Since no. of Groundnuts is more, then obviously we'll have more no. of cups of oil. So it is a clear case of Direct proportion.

$$\frac{50}{1} = \frac{150}{y}$$

$$\therefore y = 3$$

$$\therefore \text{No. of cups obtained} = 3.$$

Inverse Proportion

Two variables are inversely proportional, if they are so related that an increase in one variable results in corresponding decrease in the other, and a decrease in one variable results in a corresponding increase in the other. In this case product of the two variable remains same. Hence, for the case or iteration for two values of x to be x_1 and x_2 and the corresponding values of y to be y_1 and y_2 ,

$$\text{then, } x_1 y_1 = x_2 y_2$$

? Example

If one man can build a shed in 12 days, then how long will 4 men take to build the shed?

✓ Solution

	No. of men	No. of days
Case – I	1	12
Case – II	4	y

Since no. of men are increasing, hence work will be divided and it will take lesser no. of days. This is a case of inverse proportion. Hence, the product will remain constant.

$$4 \times y = 1 \times 12$$

$$\therefore y = 3$$

So, it will take 3 days for 4 men to build the shed.

Partnership

When two or more persons invest their money in a joint business, they are called partners and their association is known as partnership.

Types of Partnership

1. Simple partnership

If the money of the partners are invested for equal interval of time, the partnership is called **simple**.

? Example

A, B and C invested ₹1,000, ₹600 and ₹400 respectively to start a business. The profit is ₹200 which is to be divided among A, B and C in the ratio of their capital invested. How much share does each get?

✓ Solution

Ratios of investment of A, B and C
= 1000 : 600 : 400 = 5 : 3 : 2

$$\text{A's share of profit} = \frac{5}{10} \times 200 = ₹100$$

$$\text{B's share of profit} = \frac{3}{10} \times 200 = ₹60$$

$$\text{C's share of profit} = \frac{2}{10} \times 200 = ₹40$$

2. Compound partnership

If the money of the partners are invested for different intervals of time, the partnership is called **compound**.

? Example

A, B and C enter into a partnership with an amount of ₹10,000 each. After 4 months, A invests an additional ₹2,000. Three months later, B invests ₹4,000, and C at the same time withdraws ₹2,000. The profit at the end of the year was ₹2,17,000. What are their respective shares if C is allowed ₹2,000 as salary per month from profits at the end?

✓ Solution

A's capital (monthly)
= 10000 × 4 + 12000 × 8 = ₹1,36,000

B's capital (monthly)
= 10000 × 7 + 14000 × 5 = ₹1,40,000

C's capital (monthly)
= 10000 × 7 + 8000 × 5 = ₹1,10,000

Ratio in which profits are to be shared
= 68 : 70 : 55

Salary of C = 2000 × 12 = ₹24,000

$$\text{A's share} = \frac{193000 \times 68}{193} = ₹68,000$$

B's share = ₹70,000

C's share = ₹55,000



Important Note:

In the problems related to partnership, we always divide the profits in the proportion of the capital invested by the partners unless some special conditions are given.

Alligation

The word alligation means 'linking' and states that when different quantities of same or different ingredients of different value are mixed together to produce a mixture of a mean value, the ratio of their quantities are inversely proportional to the differences in their value from the mean value.

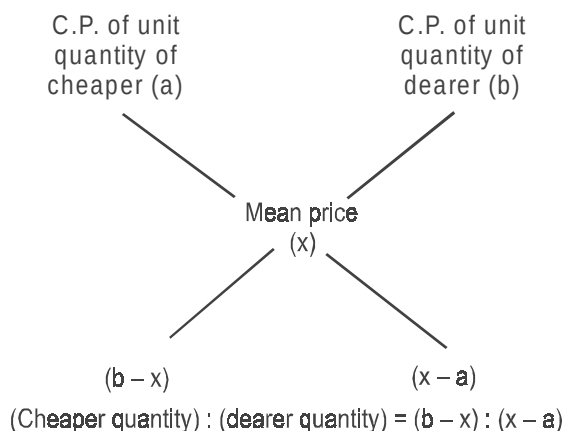
This rule is also applicable in case of their cost prices.

Rule of Alligation

$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{\text{CP of dearer} - \text{Mean Price}}{\text{Mean price} - \text{CP of cheaper}}$$



It can also be represented as:



Removal by equal amount

Consider a vessel containing only ingredient 'x'. From this 'a' units of ingredient is taken out and replaced by an equal amount of ingredient 'y'. This process is repeated 'n' times, then after 'n' operations,

$$\frac{\text{Amount of x left}}{\text{Amount of x originally present}} = \left(\frac{x-a}{x}\right)^n$$

? Example

A container contained 100 kg of milk. From this container 10 kg of milk was taken out and replaced by water. This process was further repeated three times. How much milk is now contained by the container?

✓ Solution

Amount of liquid left after 'n' operations, when the container originally contains 'a' units of liquid from which 'b' units is taken out each time, is

$$a\left(\frac{a-b}{a}\right)^n \text{ units} = 100\left(\frac{100-10}{100}\right)^4 \text{ kg}$$

$$= 100 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} = 65.61 \text{ kg}$$

Average

An average or an arithmetic mean is the sum of all observations divided by the total number of observations.

$$\text{Average} = \frac{\text{Sum of all observations}}{\text{Total number of observations}}$$

? Example

If the ages of five citizens in a club are 40, 45, 55, 60 and 40 years respectively, then

$$\text{Average age} = \frac{40 + 45 + 55 + 60 + 40}{5}$$

$$= \frac{240}{5} = 48 \text{ years.}$$

But, suppose in the same club, 2 citizens are 40-year-old, 3 of them are 45-year-old, 4 are 50-year-old and 5 are 55-year-old, then their average is

$$\frac{\{(2 \times 40) + (3 \times 45) + (4 \times 50) + (5 \times 55)\}}{(2 + 3 + 4 + 5)}$$

$$= \frac{690}{14} \text{ years.}$$

Here, $\left(\frac{2}{14}\right)$, $\left(\frac{3}{14}\right)$, $\left(\frac{4}{14}\right)$ and $\left(\frac{5}{14}\right)$ are called the weights of each category of citizens.

Weights are the fraction of the numbers in that category with respect to the total citizens in that club. This average is also called the **weighted average** of the club.

Solved Examples

? Example 1

Express the following in ratios.

- A and B are two students who got 175 and 225 marks respectively. Find the ratio of their marks.
- X got 105 marks out of 150, and Y got 175 marks out of 200. Express the ratio of marks they got to their respective total marks in ratio.

✓ Solution

$$(i) \quad \frac{175}{225} = \frac{7}{9} = 7 : 9$$

$$(ii) \quad \frac{105}{150} : \frac{175}{200} = 4 : 5$$

? Example 2

If 5 kg of wheat flour is mixed with 500 g of sugar extract, what is the ratio of sugar extract to the rest of the mixture, after adding 1.5 kg of water?

✓Solution

We first need to convert all figures into one single unit.

Wheat flour = 5 kg; Water = 1.5 kg

Sugar extract = 500 g = 0.5 kg

Total weight of the mixture = 7 kg

Ratio of sugar extract to the rest of mixture

$$= \frac{0.5}{6.5} = 1 : 13$$

?Example 3

Ratio of a : b is same as that of b : c, all three are positive integers. If a = 10 and c = 40, find b.

✓Solution

a : b = b : c (Same proportion)

$$b \times b = a \times c$$

$$b \times b = 400$$

$$b = \sqrt{400} = \pm 20. \text{ As it is +ve, so, } b = 20.$$

?Example 4

What must be subtracted from the numerator and denominator of fraction $\frac{6}{7}$ to give a fraction equal

to $\frac{16}{21}$?

✓Solution

Let the number to be subtracted be x.

$$\frac{6-x}{7-x} = \frac{16}{21};$$

On solving, we get x = 2.8

?Example 5

Pompo and Rompo are two countries engaged in a war. Pompo possesses 8 tanks and Rompo possesses 11 tanks. They get external support from neighbouring countries, of two tanks each. Which of the two countries is supposed to have a relatively greater increase in strength?

✓Solution

Pompo's strength : Rompo's strength (initial) = 8 : 11.

Pompo's strength : Rompo's strength (after aid received) = 10 : 13.

$$\frac{10}{13} > \frac{8}{11}$$

Therefore, Pompo's strength increased in a greater proportion as compared to Rompo's strength.

?Example 6

Divide ₹1,000 between A and B in the ratio 7 : 3.

✓Solution

7 : 3 implies $\frac{7}{10}$ and $\frac{3}{10}$ of ₹1,000.

Therefore, A gets ₹700 and B gets ₹300

?Example 7

₹1,150 is to be divided between A, B and C such that the ratio of share of A to that of B is equal to 3 : 2 and the share of B to share of C is equal to 3 : 4. Find their individual share?

✓Solution

Here A : B = 3 : 2 = $\frac{3}{2}$ and B : C = 3 : 4 = $\frac{3}{4}$

[We take the LCM of the values of the common term {B over here}]

e.g. LCM of 2 and 3 is 6

So A : B = 3 : 2 = 9 : 6 and

B : C = 3 : 4 = 6 : 8

Therefore, A : B : C = 9 : 6 : 8

1150 can be divided between them as follows,

$$\text{A's share} = \frac{1150 \times 9}{23} = ₹450$$

$$\text{B's share} = \frac{1150 \times 6}{23} = ₹300;$$

$$\text{C's share} = \frac{1150 \times 8}{23} = ₹400$$

?Example 8

Which is the greatest among the following fraction?

$$\frac{13}{11}, \frac{15}{13}, \frac{11}{9} \text{ and } \frac{12}{10}$$

✓Solution

Difference in the numerator and denominator is 2 in each case. Also, the fractions are all more

than 1. Therefore, the greatest fraction is $\frac{11}{9}$.



? Example 9

Which of the following is the smallest among following fraction?

$$\frac{14}{25}, \frac{57}{100}, \frac{49}{86}, \frac{3}{5}$$

✓ Solution

One way of solving such a problem is to convert each into decimals and then compare.

Another way is to either equalize the numerator or denominator of every two fractions that are being compared.

Suppose we have $\frac{a}{b}$ and $\frac{c}{d}$ to be compared.

If $a = c$, then the fraction with the higher value of denominator is smaller.

If $b = d$, then the fraction with the lower value of numerator is smaller.

Comparing $\frac{14}{25}$ with $\frac{57}{100}$, $\frac{14}{25} = \frac{56}{100} < \frac{57}{100}$.

So, $\frac{57}{100}$ is eliminated.

Now comparing $\frac{14}{25}$ with $\frac{49}{86}$, $\frac{14}{25} = \frac{98}{175}$ and

$\frac{49}{86} = \frac{98}{172}$. So, $\frac{49}{86}$ is eliminated. Comparing $\frac{14}{25}$

with $\frac{3}{5}$, $\frac{3}{5} = \frac{15}{25} > \frac{14}{25}$.

So, $\frac{14}{25}$ is the smallest.

? Example 10

A, B and C are to be given monthly allowance in year 1996 in the ratio 5 : 3 : 1. If C's share of such allowance is ₹1,000, how much did A get for the year ended in December 31, 1996.

✓ Solution

C's share = ₹1,000

A's share : C's share = 5 : 1

Therefore, A's share (monthly) = ₹5,000

A's share for whole year = $5000 \times 12 = ₹60,000$

? Example 11

Ram, Sham and Pran share profits in the ratio 12 : 1 : 5. If Pran's share is ₹12,500, what is the profit of the firm?

✓ Solution

Pran's share = 12500 = $\frac{5}{18}$ of total profit.

$\therefore$ Total profit = $\frac{12500 \times 18}{5} = ₹45,000$

? Example 12

₹1,59,000 is to be divided in the ratio 5 : 3 : 3 among A, B and C such that a certain sum of ₹5,000 from the total is divided in the ratio 2 : 3 between A and B before such division is made. What are the shares of A and B?

✓ Solution

₹5,000 is to be divided between A and B in ratio 2 : 3.

A's share = ₹2,000, and B's share = ₹3,000.

Amount left is ₹1,54,000 and to be divided in the ratio 5 : 3 : 3.

A's share = $\frac{5}{11} \times 1,54,000 = ₹70,000$

B's share = $\frac{3}{11} \times 1,54,000 = ₹42,000$

C's share = $\frac{3}{11} \times 1,54,000 = ₹42,000$

A's total share = ₹72,000 and B's total share = ₹45,000.

? Example 13

Speed of car 'A' is twice that of truck 'B'. Both start from a point A to point Y. If 'B' took 3 hr more than 'A', what is the time taken by 'B'?

✓ Solution

Since distance is constant, speed and time are inversely proportional.

Therefore, if ratio of speeds of 'A' and 'B' = 2 : 1, ratio of time taken by 'A' and 'B' = 1 : 2.

Let's assume the time taken by 'A' be X and by 'B' be 2X.

$$2X - X = 3$$

$$\Rightarrow X = 3$$

$\therefore$ Time taken by B = 6 hr.

? Example 14

Divide 156 in four parts in continuous proportion such that the ratio of sum of first and third part to the sum of second and fourth part is 1 : 5.

✓ Solution

Continuous proportion means that ratio is in a geometric progression, i.e. a ratio which is like $a : ad : ad^2 : ad^3 \dots$ is a continuous proportion.

$$\text{If } (a + ad^2) : (ad + ad^3) = 1 : 5$$

$$\Rightarrow a(1 + d^2) : ad(1 + d^2) = 1 : 5$$

$$\Rightarrow 1 : d = 1 : 5$$

$$\Rightarrow d = 5. \text{Ratio}$$

$$\Rightarrow a : 5a : 25a : 125a$$

$$\Rightarrow a + 5a + 25a + 125a = 156$$

$$\Rightarrow 156a = 156$$

$$\Rightarrow a = 1$$

Therefore, four parts are 1, 5, 25 and 125.

? Example 15

Two solutions contain alcohol and water in the ratio 3 : 2 and 5 : 3. If the two solutions are mixed in equal quantities, what is the ratio of alcohol to water in the resulting solution?

✓ Solution

Required ratio of alcohol to water

$$= \frac{\left(\frac{3}{5} + \frac{5}{8}\right)}{\left(\frac{2}{5} + \frac{3}{8}\right)} = \frac{49}{31}$$

? Example 16

B's income is $\frac{3}{2}$ of A's income and 75% of C's income. What is C's income as a percentage of A's income?

✓ Solution

$$\text{B's income} = \frac{3}{2} \text{ of A's income.}$$

$$\text{C's income} = \frac{4}{3} \text{ of B's income}$$

$$= \frac{4}{3} \times \frac{3}{2} \text{ of A's income} = 2(\text{A's income}).$$

$\therefore$ Required percentage = 200%.

? Example 17

In an election there are only 2 candidates, A and B. A got 59% and won by a majority of 144 votes. What is the total number of votes?

✓ Solution

A got 59%. Therefore, B got 41%.
Difference = 59 - 41 = 18%
18% = 144 votes

$$\text{Therefore, total votes} = \frac{144}{18} \times 100$$

$$= 800 \text{ votes.}$$

? Example 18

Out of an inheritance of ₹80,000, Anil invested ₹26,000 in shares, 30% of total in land and deposited the rest in 10% fixed deposit for 2 years. What is the gain from the fixed deposit after 2 years?

✓ Solution

$$\text{Amount in fixed deposit}$$

$$= 80000 - 26000 - \frac{30}{100} \times 80000 = ₹30,000$$

$$\text{Amount after 2 years} = 30000 \times \left(1 + \frac{10}{100}\right)^2$$

$$= ₹36,300$$

$$\text{Interest} = 36,300 - 30,000 = ₹6,300$$

? Example 19

A man has certain number of potatoes out of which 13% were bad, 75% of the rest were sold and he was left with 261. How many potatoes did he have initially?

✓ Solution

Let's assume that he had P potatoes initially.
Then Number of fresh potatoes = 100 - 13
= 87% and
Number of unsold potatoes = 25% of 87% of initial number potatoes

$$\left[\frac{87}{100} \times \frac{25}{100} \right] \times P = 261$$

$$\Rightarrow P = 1200$$

? Example 20

A man lost 12.5% of his money. After spending 70% of the remainder, he has ₹210 left with him. How much did he have at the beginning?

✓ Solution

Let 'x' be the actual amount of money.

$$\text{Now, } 12.5\% = \frac{1}{8}$$

So, money left with him, after losing = $\frac{7}{8}$ of x and

Money left after spending = 30% of $\frac{7}{8}$ of x

$$\Rightarrow \frac{30}{100} \times \frac{7}{8} \times x = 210 \Rightarrow \frac{3 \times 7}{80} \times x = 210$$

$$\Rightarrow \frac{21}{80} \times x = 210 \Rightarrow x = ₹800$$



My Doubts

SAMPLE

CL Media



Practice Exercise

1. If $A : B = 3 : 7$ and the sum of A and B is 45. Find the value of B.
(a) 28 (b) 33.5
(c) 31.5 (d) 36
2. A fraction bears the same ratio to $\frac{3}{7}$ as $\frac{1}{27}$ to $\frac{1}{35}$. The fraction is
(a) $\frac{4}{9}$ (b) $\frac{1}{3}$
(c) $\frac{3}{5}$ (d) $\frac{5}{9}$
3. Mean proportion between 8 and 72 is
(a) 24 (b) 40
(c) 16 (d) 32
4. Fourth proportional to 3, 15 and 27 is
(a) 39 (b) 45
(c) 81 (d) 135
5. Third proportional to 20 and 30 is
(a) 40 (b) 45
(c) 60 (d) 50
6. A, B and C join a partnership contributing ₹2,000, ₹1,500 and ₹1,250 respectively. What is A's share in a total profit of ₹3,610?
(a) ₹1,500 (b) ₹1,520
(c) ₹1,870 (d) ₹2,290
7. A starts a business with ₹4,000. B joins him after 3 months with ₹8,000. C puts a sum of ₹12,000 in the business for 2 months only. At the end of the year the business gave a profit of ₹5,200. Find the share of B.
(a) ₹1,500 (b) ₹2,000
(c) ₹2,600 (d) ₹4,000
8. If ratio of shares of X and Y is $2 : 7$, what is the ratio of X's share to the difference between Y's and X's shares?
(a) $2 : 7$ (b) $4 : 12$
(c) $2 : 5$ (d) $2 : 9$
9. ₹3,960 is divided among A, B and C such that half of A's part, one-third of B's part and one-sixth of C's part are equal. Then B's part is
(a) ₹1,080 (b) ₹960
(c) ₹1,720 (d) ₹1,540
10. In the above question, what is the ratio of A's share to the difference of B and C's share?
(a) $2 : 9$ (b) $2 : 5$
(c) $4 : 5$ (d) $2 : 3$
11. Two numbers are in the ratio $5 : 3$. If 9 is subtracted from both of them, they will be in the ratio $23 : 12$. The first number is
(a) 45 (b) 65
(c) 55 (d) None of these
12. If ratio of A's money to B's is $4 : 5$ and B's money to C's is $2 : 3$, and A has ₹800, then the total money of A, B, and C is
(a) ₹2,790 (b) ₹3,000
(c) ₹3,300 (d) ₹3,620
13. If $4x = 3y = 2z$, then $x : y : z$ is
(a) $4 : 3 : 2$ (b) $2 : 3 : 4$
(c) $3 : 4 : 2$ (d) $3 : 4 : 6$
14. If 68 is divided into two parts such that 7th part of the first is equal to 10th part of the second. The first part is
(a) 7 (b) 22
(c) 28 (d) 32
15. ₹6,800 is to be divided among A, B and C such that A gets two-thirds of what B gets, and B gets one-fourth of what C gets. Find B's share?
(a) ₹1,200 (b) ₹800
(c) ₹1,000 (d) ₹1,600
16. ₹9,700 has been divided among X, Y and Z such that if their shares are reduced by ₹30, ₹20 and ₹50, the balance is in the ratio $3 : 4 : 5$. What is Y's share?
(a) ₹3,180 (b) ₹3,220
(c) ₹3,253.33 (d) ₹3,200
17. The sum of ₹530 is divided among A, B and C such that A gets ₹70 more than B, and B gets ₹80 more than C. The ratio of A and C is
(a) $25 : 18$ (b) $18 : 10$
(c) $5 : 9$ (d) $5 : 2$



18. In a ratio equal to 4 : 9, the antecedent is 36. The consequent is
(a) 81 (b) 16
(c) 72 (d) None of these
19. An amount is distributed amongst A, B and C such that A gets half that of B and B gets twice that of C. What is the ratio of B to that of the sum of A and B.
(a) 2 : 5 (b) 2 : 3
(c) 3 : 2 (d) 4 : 3
20. A bag contains ₹600 in the form of one-rupee, 50-paise and 25-paise coins in the ratio 3 : 4 : 12. The number of 25-paise coins is
(a) 600 (b) 900
(c) 1,200 (d) 1,376
21. In a mixture of 180 L, the ratio of milk and water is 2 : 1. If the ratio of milk and water is to be 1 : 2, the water to be added is (in L)
(a) 80 L (b) 90 L
(c) 120 L (d) 180 L
22. In a mixture of 100 L, the ratio of milk and water is 3 : 1. If 200 L of water is added in the mixture, what is the new ratio of milk and water?
(a) 1 : 3 (b) 3 : 1
(c) 2 : 5 (d) None of these
23. Out of a class of 500, boys and girls are equal. In new batch, one-fifth of the girls left the class and 25 boys joined in. What is the ratio of boys and girls now?
(a) 3 : 2 (b) 12 : 7
(c) 11 : 8 (d) 9 : 8
24. The population of a country increased by one-tenth and then decreased by one-tenth in two successive years. If the original population was 15,000, what would be the population after 2 years?
(a) 14,550 (b) 15,000
(c) 14,850 (d) 16,270
25. The ages of a man and his son are in the ratio 7 : 2. After 15 years, they would be in the ratio 2 : 1. The father's age, when the son was born, was
(a) 25 years (b) 30 years
(c) 35 years (d) 42 years
26. Four years ago, father's age was six times that of his son. 12 years from now, father's age will be twice that of the son. The present ratio of ages of father and son is
(a) 6 : 1 (b) 7 : 1
(c) 8 : 2 (d) 7 : 2
27. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{7}$, then $\frac{(a+b+c)}{c}$ is
(a) 7 (b) 2
(c) $\frac{1}{2}$ (d) $\frac{1}{7}$
28. Two whole numbers whose sum is 84 cannot be in the ratio
(a) 9 : 3 (b) 3 : 5
(c) 19 : 2 (d) 5 : 7
29. If 0.35 of a number is equal to 0.07 of another number, the ratio of the number is
(a) 1 : 2 (b) 2 : 1
(c) 1 : 5 (d) 1 : 4
30. Two alloys containing copper and iron in the ratios 5 : 8 and 5 : 3 are melted in equal quantities. The ratio of iron to copper in the resulting alloy is
(a) 103 : 105 (b) 15 : 24
(c) 3 : 8 (d) 105 : 103
31. 80 L of a mixture of milk and water is in the ratio 5 : 3. If 16 L of this mixture is replaced by 16 L of milk, ratio of milk and water becomes
(a) 2 : 1 (b) 6 : 3
(c) 7 : 3 (d) 8 : 3
32. The ratio of copper and zinc in a brass piece is 13 : 7. How much copper will be there in 500 kg of such alloy?
(a) 300 kg (b) 325 kg
(c) 175 kg (d) 150 kg
33. A, B and C join a partnership. A invested ₹16,000 for 6 months, B invested ₹12,000 for $\frac{2}{3}$ year and C invested ₹1,000 for 12 months. Their profit sharing ratio is
(a) 8 : 7 : 10 (b) 10 : 8 : 7
(c) 6 : 8 : 12 (d) None of these

34. A sum of money is divided among A, B and C such that for each rupee that A gets, B gets 65 paise and C gets 35 paise. If C's share is ₹560, the sum is
(a) ₹2,400 (b) ₹2,800
(c) ₹3,200 (d) ₹3,600
35. A ratio of two numbers is 2 : 3. What is the ratio of their squares, if the two numbers have a difference of 12?
(a) 2 : 3 (b) 12 : 24
(c) 9 : 16 (d) 4 : 9
36. The cost of providing a product are labour, material and overheads in the ratio 5 : 7 : 3. If the profit is calculated as 20% on costs, the ratio of material costs to profit is
(a) 5 : 2 (b) 7 : 3
(c) 3 : 5 (d) 11 : 9
37. ₹11,250 is divided among A, B and C so that A receives half as much as B and C together receive; B receives one-fourth of what A and C receive together. The share of A is more than that of B by
(a) ₹1,500 (b) ₹3,000
(c) ₹1,550 (d) ₹3,200
38. The ratio of boys and girls in a class of 72 is 7 : 5. How many more girls should be admitted to make equal number of boys and girls?
(a) 9 (b) 12
(c) 220 (d) 240
39. The biggest of three positive numbers that are in the ratio 2 : 3 : 5 and the sum of their squares is 608, is
(a) 8 (b) 12
(c) 16 (d) 20
40. In a mixture of 28 L, the ratio of milk and water is 2 : 5. If 2 L of milk is added and 5 L of water is removed, the new ratio is
(a) 9 : 20 (b) 1 : 4
(c) 3 : 2 (d) 2 : 3
41. The incomes of A and B are in the ratio 3 : 2 and their expenses are in the ratio 5 : 3. If each one saves ₹3,000, then B's income is
(a) ₹12,000 (b) ₹6,000
(c) ₹9,000 (d) ₹15,000
42. A, B and C invested in a partnership with a ratio of 8 : 7 : 5. A withdraws half of her money after 5 months. If the profit was ₹26,500 for the year, then B's share is
(a) ₹9,800 (b) ₹10,200
(c) ₹10,500 (d) ₹12,600
43. The prices of a bicycle and a rickshaw are in the ratio 5 : 4. If the bicycle costs ₹800 more than the rickshaw, the price of the rickshaw is
(a) ₹800 (b) ₹3,200
(c) ₹4,000 (d) ₹2,400
44. If $A : B :: 5 : 2$, the value of $8A + 9B : 8A + 2B$ is
(a) 26 : 16 (b) 61 : 26
(c) 29 : 22 (d) 22 : 29
45. If 1 L water is added to 5 L of a 20% solution of sugar in water, what is the strength of the solution now?
(a) 16.67% (b) 10%
(c) 8.33% (d) 12.67%
46. The price of oil is increased by 25%. If the expenditure is not allowed to increase, the ratio of reduction in consumption to regular consumption is
(a) 1 : 3 (b) 1 : 4
(c) 1 : 5 (d) 1 : 6
47. Three year ago the average age of Abhishek and Aishwarya was 25 year. Now they have a one year old child. The average age of the family at present is:
(a) 25 years (b) 17 years
(c) 19 years (d) 28 years
48. In an ODI cricket match the run rate was only 4.2 in the first 20 overs. What should be the run rate in the remaining 30 overs to reach the target of 312 runs?
(a) 7.6 (b) 7.2
(c) 8 (d) 7.5
49. A family consists of six members A, B, C, D, E and F. The average age of A and B is 60 years, that of C and D is 38 years and that of E and F is 10 years. What is the average age of the family?
(a) 38 years (b) 42 years
(c) 32 years (d) 36 years



50. A shopkeeper has made a sale of ₹6,135, ₹7,227, ₹6,755, ₹7,430 and ₹6,462 for 5 consecutive days. How much sale must he have to make on the sixth day in order to manage an average sales of ₹6600 over the given six days.
 (a) ₹4991 (b) ₹6054
 (c) ₹5591 (d) None of these
51. The average of 50 numbers is zero. How many of them, at the most, can be greater than zero?
 (a) 49 (b) 25
 (c) 0 (d) 24
52. The average weight of 10 persons increases by 2 kg when a new person comes in place of one of them of weight 56 kg. The weight of the new person is
 (a) 66 kg (b) 76 kg
 (c) 72 kg (d) 36 kg
53. The average monthly income of Ram and Rahim is ₹4,500. The average monthly income of Rahim and Salim is ₹5,500 and the average monthly income of Ram and Salim is ₹5,100. The monthly income of Ram is
 (a) ₹4,100 (b) ₹4,000
 (c) ₹4,500 (d) ₹5,100
54. A library has an average of 720 visitors on Sundays and 450 on each of other days of the week. The average number of visitors per day in a month of 30 days beginning with Sunday is:
 (a) 500 (b) 495
 (c) 590 (d) 490
55. In a cricket match the run rate was only 4.5 in the first 24 overs. What must be the run scored in the remaining 30 overs to reach the overall run rate of 6?
 (a) 182 (b) 218
 (c) 212 (d) 192
56. A student's marks were wrongly entered as 83 instead of 38. Due to that the average marks for the class got increased by 0.3. The number of students in the class is:
 (a) 100 (b) 90
 (c) 150 (d) 200



My Doubts

CL Media

Time, Speed, Distance and Time & Work

Time, Speed and Distance

This section consists of problems on calculating speed, distance, time, head start and relative speed, etc.

Concepts

1. Distance = Speed $\times$ Time
2. $1 \text{ km/hr} = \frac{5}{18} \text{ m/sec}$
3. If the ratio of speed is $a : b : c$, then the ratio of time taken is $\frac{1}{a} : \frac{1}{b} : \frac{1}{c}$.

Speed is a relation between time and distance.

$S \propto D$, i.e. if speed is doubled, distance covered in a given time also gets doubled and $S \propto \frac{1}{T}$, i.e. if speed is doubled, time taken to cover the distance will be half.

Average speed

Average speed is defined as $\frac{\text{Total distance travelled}}{\text{Total time taken}}$.

Suppose, a man covers a distance d_1 kms at s_1 km/hr and a distance d_2 kms at s_2 km/hr, then

$$\text{Average speed of the entire journey} = \frac{d_1 + d_2}{\frac{d_1}{s_1} + \frac{d_2}{s_2}} \text{ km/hr.}$$

If the distances are equal, then

$$\text{Average speed} = \frac{d + d}{\frac{d}{s_1} + \frac{d}{s_2}} = \frac{2s_1s_2}{s_1 + s_2} \text{ km/hr.}$$

Relative speed

If two bodies are moving (in the same direction or in the opposite direction), then the speed of one body with respect to the other is called its relative speed.

Relative speed is a phenomenon that we observe everyday. Suppose you are travelling in a train and there

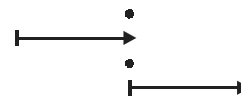
is a second train coming from the opposite direction on the parallel track, then it seems that the second train is moving much faster than actual. If both the trains are moving in the same direction on parallel tracks at same speeds, they seem to be stationary if seen from one of these trains, even though they might actually be at a speed of 100 km/hr each. So what you actually observe is your speed relative to the other.

Concepts

1. If two objects are moving in opposite directions towards each other or away from each other on a straight-line at speeds u and v , then they seem to be moving towards each other or away from each other at a relative speed = Speed of first + Speed of second = $u + v$.
This is also the speed at which they are moving towards each other or the speed at which they may be moving away from each other.
2. If the two objects move in the same direction with speeds u and v , then
relative speed = difference of their speeds = $u - v$.
This is also the speed at which the faster object is either moving closer to the slower object or moving away from the slower object as the case may be.
3. If two objects start from A and B with speeds u and v respectively, and after crossing each other, take a and b hours to reach B and A respectively, then $u : v = \sqrt{\frac{b}{a}}$.

Trains

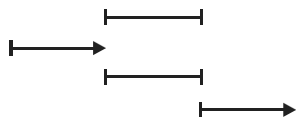
Train crossing a pole or a man :



Time taken by a train to cross a pole or a man is given by the ratio of length of the train to its speed.

$$\therefore \text{Time taken} = \frac{\text{Length of the train}}{\text{Speed of the train}}$$

Train crossing a bridge or a platform:



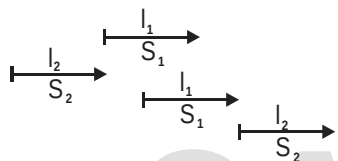
The distance covered in the time taken to cross a bridge or a platform will be equal to the sum of the length of the train and the length of the bridge or a platform being crossed.

∴ Time taken to cross a bridge or a platform

$$= \frac{\text{Length of the train} + \text{length of the bridge or a platform}}{\text{Speed of the train}}$$

Train crossing another train:

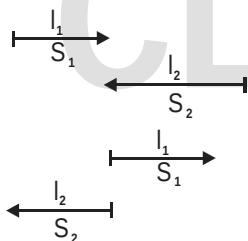
In the same direction :



∴ Time taken = $\frac{\text{Sum of lengths of the trains}}{\text{Relative speed of the trains}}$

$$= \frac{l_1 + l_2}{S_2 - S_1}$$

In the opposite direction :



∴ Time taken = $\frac{\text{Sum of lengths of the trains}}{\text{Sum of the speeds of the trains}}$

$$= \frac{l_1 + l_2}{S_1 + S_2}$$

Boats and Streams

Downstream and Upstream Motion

Downstream motion of a boat is its motion in the same direction as the flow of the river.

Upstream motion is exactly the opposite.

There are two parameters in these problems :

1. Speed of the river (R): This is the speed with which the river flows.
2. Speed of the boat in still water (B): If the river is still, this is the speed at which the boat would be moving.

The effective speed of a boat in upstream

$$= B - R$$

The effective speed of a boat in downstream

$$= B + R$$

3. The speed of the boat in still water is given as B

$$= \frac{1}{2}(d + u), \text{ and the speed of the river}$$

$$R = \frac{1}{2}(d - u),$$

where d and u are the downstream and upstream speeds, respectively.

Circular motion

The problems on circular motion deal with races on a circular track to calculate the time of meeting at the starting point and anywhere on the track.

Concepts

1. If two people A and B start from the same point, at the same time and move in the same direction along a circular track and take x minutes and y minutes respectively to come back to the starting point, then they would meet for the first time at the starting point according to the formula given below:
First time meeting of A and B at the starting point = (LCM of x and y)



Notes:

This formula would remain the same even if they move in the opposite directions.

2. If two people A and B start from the same point with speeds m km/hr and n km/hr respectively, at the same time and move in the same direction along a circular track, then the two would meet for the first time by the formula given below:

Time of the first meeting

$$= \frac{\text{Circumference of the track}}{\text{Relative speed}}$$

Time and Work

The Time and work topic involves problems on time taken to do a certain job by certain number of workers; the

change in the number of hours required to do a job if the number of workers is changed; the number of hours required to do the same job by different number of workers if their speeds are different, etc.

$$w = n \times h \times d = \text{constant}$$

where, w : Work

n : Number of men

h : Number of hours

d : Number of days

In general, we can say that,

If ' w_1 ' work is done by ' m_1 ' men working ' h_1 ' hours per day in ' d_1 ' days and ' w_2 ' work is done by ' m_2 ' men working ' h_2 ' hours per day in ' d_2 ' days, then

$$\frac{m_1 d_1 h_1}{w_1} = \frac{m_2 d_2 h_2}{w_2}$$

Generally, the following types of questions are asked in the examination.

Type of question	Example	Approach to question
1. Calculate the time taken by two persons working together to finish a work.	A can complete a piece of work in 10 days which B alone can complete in 12 days. In how many days can both complete it if they work together?	$T = \frac{X \times Y}{X + Y}$, where X and Y are the time taken by A and B individually. $T = \frac{10 \times 12}{10 + 12} = \frac{120}{22} = 5 \frac{5}{11}$ days
2. If A and B can finish a work in X days and A alone can complete it in Y days, find the time taken by B alone.	Two persons A and B working together can dig a trench in 8 hrs while. A alone can dig it in 12 hrs. In how many hours can B alone dig the trench?	$T = \frac{X \times Y}{Y - X} = \frac{12 \times 8}{12 - 8} = \frac{96}{4} = 24$ hrs
3. If one person is m times as good as another worker and takes n day less than the other person, in how many days can they together finish the work?	A is thrice as good as B and takes 60 days less than B for completing a job. Find the time taken by them if they work together.	$T = \frac{m \times l}{m^2 - 1} = \frac{3 \times 60}{9 - 1} = \frac{180}{8} = \frac{45}{2}$ days = 22.5 days
4. One pipe can fill in a cistern in T_1 min and another pipe in T_2 min. If both the pipes are opened together, find the time taken by them to fill in the cistern.	A tap can fill in a cistern in 30 min and another can fill it in 40 min. If both the taps are opened simultaneously, find the time taken to fill the cistern completely.	$T = \frac{T_1 \times T_2}{T_1 + T_2}$, where T_1 and T_2 are the time taken by each pipe individually. $= \frac{30 \times 40}{30 + 40} = \frac{1200}{70} = 17 \frac{1}{7}$ min
5. One pipe can fill in a cistern in T_1 min and another pipe can empty it in T_2 min. If both the pipes are opened together, find the time taken to fill in the cistern.	A cistern is filled in by a pipe A in 10 hr and emptied by a pipe B in 12 hr. If both the pipes are opened together, in how much time will the cistern be full?	$T = \frac{T_1 \times T_2}{T_2 - T_1}$, where T_1 and T_2 are the time taken by each pipe individually to fill and empty the cistern respectively. $= \frac{10 \times 12}{12 - 10} = \frac{120}{2} = 60$ hrs
6. A cistern takes T_1 min to be filled by the filling pipes, but takes x extra min to fill in due to a leakage in the cistern. Find out the time, in which the leak will empty the cistern.	Two pipes can fill in a cistern in 14 hrs and 16 hrs respectively. The pipes are opened simultaneously. Due to a leakage in the bottom of the cistern, 32 min extra are required to fill the cistern. If the cistern is full, in what time will the leak empty it?	$T = T_1 \times \left(1 + \frac{T_1}{x}\right)$, where T_1 is the time taken to fill in the cistern and x is the extra time taken due to a leakage. T is equal to $\frac{112}{15} \left(1 + \frac{112 \times 60}{15 \times 32}\right)$ = 112 hrs.



Concepts

1. If A can do a piece of work in 10 days, then in 1 day, A will do $\frac{1}{10}$ of the total work.
2. If A is thrice as good as B, then
 - (a) In a given amount of time, A will be able to do 3 times the work B does.
Ratio of work done by A and B (in the same time) = 3 : 1.
 - (b) For the same amount of work, B will take thrice the time as much as A takes.
Ratio of time taken by A and B (same work done) = 1 : 3.
3. Efficiency is directly proportional to the work done and inversely proportional to the time taken.

Pipes and Cisterns

This consists of problems on how long it will take for different pipes of different diameters to fill a cistern; the time taken to fill a cistern when one pipe is filling it while another empties it, etc.

Concepts

1. If a pipe can fill a tank in x hours and another pipe can fill it in y hours, then the fraction of tank filled by both pipes together in 1 hour = $\frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}$.
Or, time required to fill the tank by both the pipes = $\frac{xy}{x+y}$.
2. If a pipe can fill a tank in x hours and another pipe can empty it in y hours, then the fraction of tank filled by both the pipes together in 1 hour = $\frac{1}{x} - \frac{1}{y} = \frac{y-x}{xy}$.

Solved Examples

? Example

Three men can complete a piece of work in 6 days. Two days after they start, 3 more men join them. How many days will they take to complete the remaining work?

✓ Solution

$$\text{Work done by 3 men in 2 days} = \frac{1}{6} \times 2 = \frac{1}{3}$$

$$\text{Remaining work} = \left(1 - \frac{1}{3}\right) = \frac{2}{3}$$

$$3 \text{ men's 1 day's work} = \frac{1}{6}$$

$$6 \text{ men's 1 day's work} = \frac{1}{3}$$

$$\frac{2}{3} \text{ work is done by them in } \left(\frac{3 \times 2}{3}\right) = 2 \text{ days.}$$

Alternative method:

After two days, only four days' work remains. If 3 more men join the existing 3 men, work will be completed in half the time. Therefore, the work is completed in 2 days.

? Example

A and B undertake a piece of work for ₹600. A alone can complete it in 6 days, while B alone can complete it in 8 days. With the help of C, they finish it in 3 days. Find B's share.

✓ Solution

$$C's \text{ 1 day's work} = \frac{1}{3} - \left(\frac{1}{6} + \frac{1}{8}\right) = \frac{1}{24}$$

Ratio of work done in 1 day for A : B : C

$$= \frac{1}{6} : \frac{1}{8} : \frac{1}{24} = 4 : 3 : 1$$

$$B's \text{ share} = ₹ \left(\frac{600 \times 3}{8}\right) = ₹ 225$$

Alternative method:

Let the total work be LCM (6, 8, 3) = 24 units.

They finished the total work in 3 days.

Therefore, (A + B + C)'s 1 day's work = 8 units.

A's 1 day's work = 4 units.

B's 1 day's work = 3 units.

Hence, C's 1 day's work = 8 - 7 = 1 unit.

Ratio of work done in 1 day for A : B : C = 4 : 3 : 1

$$B's \text{ share} = ₹ \left(\frac{600 \times 3}{8}\right) = ₹ 225$$

? Example

Two men and 3 boys can complete a piece of work in 10 days while 3 men and 2 boys can complete the same work in 8 days. In how many days can 2 men and 1 boy complete the same work?

Solution

Let 1 man's 1 day's work be x .

Let 1 boy's 1 day's work be y .

$$2x + 3y = \frac{1}{10} \text{ and } 3x + 2y = \frac{1}{8}$$

On solving, we get $x = \frac{7}{200}$ and $y = \frac{1}{100}$

(2 men + 1 boy's) 1 day's work

$$= 2 \times \frac{7}{200} + 1 \times \frac{1}{100} = \frac{16}{200} = \frac{2}{25}$$

Thus, 2 men and 1 boy can finish the work in $\frac{25}{2}$ days.

Or

Alternative method 1:

Two men and three boys can complete a piece of work in 10 days.

Hence, in one day, the number of men and boys required = $(2m + 3b) \times 10$.

Similarly, if three men and two boys can complete a piece of work in 8 days, then in one day the number of men and boys required = $(3m + 2b) \times 8$.

Hence, equating both, we get

$$(2m + 3b) \times 10 = (3m + 2b) \times 8$$

$$\Rightarrow 20m + 30b = 24m + 16b$$

$$\Rightarrow 4m = 14b$$

$$\Rightarrow m = \frac{7}{2}b$$

$$\text{Now, } 2m + 3b = 2 \times \frac{7}{2}b + 3b = 10b.$$

$$\text{New requirement is } 2m + 1b = 2 \times \frac{7}{2}b + 1b = 8b$$

By unitary method,

$$10b \longrightarrow 10 \text{ days}$$

$$8b \longrightarrow ?$$

$$10 \times \frac{10}{8} = 12.5 \text{ days.}$$

Alternative method 2

Let the total work be LCM (10, 8) = 40 units.

Let 1 man's 1 day's work be x .

Let 1 boy's 1 day's work be y .

$$2x + 3y = 4 \text{ and } 3x + 2y = 5$$

On solving, we get $x = \frac{7}{5}$ and $y = \frac{2}{5}$

(2 men + 1 boy's) 1 day's work

$$= \left(2 \times \frac{7}{5} + 1 \times \frac{2}{5} \right) = \frac{16}{5} \text{ units.}$$

Number of days taken to complete the work

$$= 40 \times \frac{5}{16} = \frac{25}{2} \text{ days} = 12.5 \text{ days.}$$

Solved Examples**Example 1**

Three men can complete a piece of work in 6 days.

Two days after they start, 3 more men joined them.

How many days will they take to complete the remaining work?

Solution

$$\text{Work done by 3 men in 2 days} = \frac{1}{6} \times 2 = \frac{1}{3}$$

$$\text{Remaining work} = \left(1 - \frac{1}{3} \right) = \frac{2}{3}$$

$$3 \text{ men's 1 day work} = \frac{1}{6}.$$

$$6 \text{ men's 1 day work} = \frac{1}{3}.$$

$$\frac{2}{3} \text{ work is done by them in } \left(\frac{3 \times 2}{3} \right) = 2 \text{ days.}$$

Alternative method:

After 2 days, only 4 days' work remains. If 3 more men join the existing 3 men, work will be completed in half of the time. Therefore, the work is completed in 2 days.

Example 2

A and B undertake to do a piece of work for ₹600.

A alone can do it in 6 days, while B alone can do it in 8 days. With the help of C, they finish in 3 days. Find B's share.

Solution

$$C's \text{ 1 day work} = \frac{1}{3} - \left(\frac{1}{6} + \frac{1}{8} \right) = \frac{1}{24}.$$

Ratio of work done in 1 day for A, B and C

$$= \frac{1}{6} : \frac{1}{8} : \frac{1}{24} = 4 : 3 : 1$$



$$\text{B's share} = ₹ \left(\frac{600 \times 3}{8} \right) = ₹225$$

Alternative method:

Let the total work be LCM (6, 8, 3) = 24 units.

They finished the total work in 3 days.

Therefore, (A + B + C)'s 1 day work = 8 units.

A's 1 day work = 4 units.

B's 1 day work = 3 units.

Hence, C's 1 day work = 8 - 7 = 1 unit.

$$\text{B's share} = \frac{3}{8} \times 600 = ₹225$$

? Example 3

Two men and 3 boys can do a piece of work in 10 days, while 3 men and 2 boys can do the same work in 8 days. In how many days can 2 men and 1 boy do the work?

✓ Solution

Let 1 man's 1 day work = x.

Let 1 boy's 1 day work = y.

$$2x + 3y = \frac{1}{10} \text{ and } 3x + 2y = \frac{1}{8}$$

On solving we get, $x = \frac{7}{200}$ and $y = \frac{1}{100}$.

(2 men + 1 boy)'s 1 day work

$$= 2 \times \frac{7}{200} + 1 \times \frac{1}{100} = \frac{16}{200} = \frac{2}{25}$$

Thus, 2 men and 1 boy can finish the work in

$$\frac{25}{2} \text{ days.}$$

Alternative method:

Let the total work be LCM (10, 8) = 40 units.

Let 1 man's 1 day work = x.

Let 1 boy's 1 day work = y.

$$2x + 3y = 4 \text{ and } 3x + 2y = 5$$

On solving, we get $x = \frac{7}{5}$ and $y = \frac{2}{5}$.

(2 men + 1 boy)'s 1 day work

$$= \left(2 \times \frac{7}{5} + 1 \times \frac{2}{5} \right) = \frac{16}{5} \text{ units.}$$

Number of days taken to do the work

$$= 40 \times \frac{5}{16} = \frac{25}{2} \text{ days.}$$

? Example 4

A fort of 60 men has food for 28 days. 8 days later reinforcements arrive, leaving the number of days the food would last for 15 days. What was the strength of the reinforcement?

✓ Solution

Originally, the food would have lasted for 28 days.

After 8 days the food would have lasted for 20 days.

Let the reinforcement number be x.

The food that would have been consumed by 60 men in 20 days, was consumed by

(60 + x) men in 15 days.

$$60 \times 20 = (60 + x)15$$

$$\Rightarrow x = 20.$$

$\therefore$ The strength of the reinforcement was 20 men.

? Example 5

Two pipes A and B can fill a tank in 24 min and 32 min respectively. If both pipes are opened simultaneously, after how much time should pipe B be closed such that the tank is full in 18 min?

✓ Solution

Let B be closed after x minutes.

Part filled by (A + B) in x min + Part filled by A in (18 - x) min = 1

$$\text{Thus, } x \times \left(\frac{1}{24} + \frac{1}{32} \right) + (18 - x) \frac{1}{24} = 1$$

Solving for x, $x = 8$.

Hence, B must be closed after 8 min.

Alternative method:

Let B be closed after x minutes.

Let the total capacity be LCM (24, 32) = 96 units.

In 1 min, A can fill 4 units and B can fill 3 units.

$$7x + 4(18 - x) = 96. \text{ Solving for } x, x = 8 \text{ min.}$$

Hence, B must be closed after 8 min.

? Example 6

Two pipes A and B can fill a tank in 36 min and 45 min respectively. Pipe C can empty it in 30 min. A and B are opened and after 7 min C is also opened. In how much time will the tank be full?

✓ Solution

$$\text{Part filled in 7 min} = 7 \times \left[\frac{1}{36} + \frac{1}{45} \right] = \frac{7}{20}$$

$$\text{Remaining part} = 1 - \frac{7}{20} = \frac{13}{20}$$

Net part filled in 1 min when A, B and C are opened

$$= \frac{1}{36} + \frac{1}{45} - \frac{1}{30} = \frac{1}{60}$$

$$\frac{13}{20} \text{ part is filled in } 60 \times \frac{13}{20} = 39 \text{ min.}$$

Total time taken to fill the tank = (39 + 7) = 46 min.

Alternative method:

Let the capacity be LCM (36, 45, 30) = 180 units.
In 1 min, A fills 5 units, B fills 4 units, C empties 6 units.

In 7 min, A and B fill $(5 + 4) \times 7 = 63$ units.

After 7 min, A and B fill 9 units and C empties 6 units.

Effectively, only 3 units are retained in 1 min.

Tank to be filled = $180 - 63 = 117$ units.

$$\text{Time taken} = \frac{117}{3} = 39 \text{ min.}$$

Total time taken = (7 + 39) = 46 min.

Example 7

A scooterist covers a certain distance at 36 km/hr.
How many metres does he cover in 3 min?

Solution

$$\text{Speed} = 36 \text{ km/hr} = 36 \times \frac{5}{18} \times \text{m/s} = 10 \text{ m/s.}$$

$$\text{Thus, distance covered in 3 min} \\ = (10 \times 3 \times 60) = 1,800 \text{ m.}$$

Example 8

Walking at $\frac{3}{4}$ of his usual speed a man is $1\frac{1}{2}$ hr late. Find his usual travel time.

Solution

Let the usual time be t hours.

$$\frac{4}{3} \times t = t + \frac{3}{2} \Rightarrow t = 4.5 \text{ hr}$$

Alternative method:

$$st = \frac{3s}{4} \left(t + \frac{3}{2} \right) \Rightarrow 4st = 3st + 4.5s$$

$$t = 4.5 \text{ hr}$$

Example 9

A man starts from L to M, another from M to L at the same time. After passing each other,

they complete their journey in $3\frac{1}{3}$ hr and $4\frac{4}{5}$ hr

respectively. Find the speed of the second man if the speed of the first is 24 km/hr.

Solution

Speed of the first man : Speed of the second man

$$\sqrt{\frac{b}{a}} = \sqrt{\frac{24}{5} \times \frac{3}{10}} = \sqrt{\frac{36}{25}} = \frac{6}{5}$$

where 'a' and 'b' are the time taken by the first and the second man after passing each other.

$$\text{Thus, the second man's speed} = \frac{5}{6} \times 24 \\ = 20 \text{ km/hr.}$$

Example 10

Two cyclists cover the same distance with speeds 15 km/hr and 16 km/hr respectively. Find the distance travelled by each, if one takes 16 min longer than the other takes.

Solution

Let the required distance be x kilometres.

$$\frac{x}{15} - \frac{x}{16} = \frac{16}{60} \\ \Rightarrow 16x - 15x = 64 \Rightarrow x = 64.$$

Hence, the required distance = 64 km.

Example 11

A thief is spotted by a policeman at a distance of 200 m. If the speed of the thief be 10 km/hr and that of the policeman be 12 km/hr, at what distance will the policeman catch the thief?

Solution

Relative speed of the policeman = 2 km/hr.

Time taken by the policeman to cover 200 m

$$= \frac{200}{1000} \times \frac{1}{2} \text{ hr} = \frac{1}{2} \text{ hr.}$$

In this time the thief covers a distance

$$= 10 \times \frac{1}{10} = 1 \text{ km.}$$

The policeman catch the thief at a distance of (1 km + 0.2 km) 1.2 km

? Example 12

A train running at 54 km/hr takes 20 sec to cross a platform and 12 sec to pass a man walking in the same direction at a speed of 6 km/hr. Find the lengths of the train and the platform?

✓ Solution

Let the length of the train = x metres and the length of the platform = y metres.

Speed of the train relative to the man

$$= 48 \text{ km/hr} = \frac{40}{3} \text{ m/s.}$$

In passing the man, the train covers its own length with relative speed.

$$\text{Length of the train} = \frac{40}{3} \times 12 = 160 \text{ m.}$$

Since speed of the train = 54 km/hr = 15 m/s,

$$\frac{x+y}{15} = 20 \Rightarrow x+y = 300 \Rightarrow y = 140$$

Length of the platform = 140 m.

? Example 13

A can beat B by 20 m in a 200 m race. B can beat C by 10 m in a 250 m race. By how many metres can A beat C in a 100 m race?

✓ Solution

A runs 200 m when B runs 180 m.

B runs 250 m when C runs 240 m.

LCM of 180 and 250 = 4500.

Thus, when A runs 5000 m, B runs 4,500 m.

When B runs 4,500 m, C runs 4,320 m.

A beats C by 680 m in a race of 5,000 m.

$$\text{Hence, A beats C by } \frac{680}{5000} \times 100$$

$$= 13.6 \text{ m in a 100 m race.}$$

? Example 14

A hare makes 9 leaps in the same time as a dog

makes 4. But the dog's leap is $2\frac{1}{3}$ m while hare's

is only 1 m. How many leaps will the dog have to make before catching up with the hare if the hare has a start of 16 m?

✓ Solution

Distance covered by the dog in one minute

$$(4 \text{ leaps}) = 4 \times \frac{7}{3} = \frac{28}{3} \text{ m.}$$

Distance covered by the hare in one minute (9 leaps) = $9 \times 1 = 9 \text{ m.}$

Distance gained by the dog in one minute (4 leaps)

$$= \frac{1}{3} \text{ m.}$$

Hence, for 1 m gain he has to make 12 leaps.

Number of leaps required by the dog to gain 16 m = $12 \times 16 = 192 \text{ leaps.}$

? Example 15

A ship, 156 km away from the shore, springs a

leak which admits $\frac{7}{3}$ tonnes of water in 6.5 min.

A pump throws out 15 tonnes of water in 1 hr. If 68 tonnes would be suffice to sink the ship, find the average rate of sailing so as to just reach the shore.

✓ Solution

Net volume of water in the ship in 1 min

$$= \frac{7}{3} \times \frac{1}{6.5} - \frac{15}{60} = \frac{17}{156} \text{ tonnes.}$$

Time for 68 tonnes of water

$$= 68 \times \frac{156}{17} = 624 \text{ min.}$$

$$\text{Speed of the ship} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{156 \text{ km}}{624 \text{ min}} = \frac{1}{4} \text{ km/min} = 15 \text{ km/hr.}$$

? Example 16

A man rows 27 km downstream and 18 km upstream taking 3 hr each time. What is the velocity of the current?

✓ Solution

$$\text{Rate downstream} = \frac{27}{3} = 9 \text{ km/hr.}$$

$$\text{Rate upstream} = \frac{18}{3} = 6 \text{ km/hr.}$$

Velocity of the current = $0.5(9 - 6)$
= 1.5 km/hr.

? Example 17

A man can row upstream at 7 km/hr and downstream at 10 km/hr. Find his rate in still water and the rate of the current.

✓ Solution

Rate in still waters = $0.5(10 + 7)$
= 8.5 km/hr.

Rate of the current = $0.5(10 - 7)$
= 1.5 km/hr.

? Example 18

Buses take 12 hr to cover the distance of 120 km between A and B. A bus starts from A at 8 a.m. and another bus starts from B at 10 a.m. on the same day. When do the two buses meet?

✓ Solution

The distance between A and B is 120 km.

Speed of the buses = $\frac{120}{12} = 10$ km/hr.

By 10 a.m., the bus from A would have covered 20 km. Hence, the distance between the buses at 10 a.m. = $120 - 20 = 100$ km.

Relative speed of the buses = 20 km/hr.

Time taken to meet = $\frac{100}{20} = 5$ hr.

? Example 19

A and B can separately do a piece of work in 20 and 15 days respectively. They worked together for 6 days, after which B was replaced by C. If the work was finished in next 4 days, then in how many days could C alone do the work?

✓ Solution

(A + B)'s 6 days work = $6 \left(\frac{1}{20} + \frac{1}{15} \right) = \frac{7}{10}$.

(A + C)'s 4 days work = $\frac{3}{10}$;

(A + C)'s 1 day work = $\frac{3}{40}$.

A's 1 day work = $\frac{1}{20}$.

$\therefore$ C's 1 day work = $\left(\frac{3}{40} - \frac{1}{20} \right) = \frac{1}{40}$.

Hence, C alone can finish the work in 40 days.



My Doubts





Practice Exercise

1. A can do a piece of work in 7 days of 9 hr each, and B can do it in 6 days of 7 hr each. How long will they take to do it working together
 $\frac{42}{5}$ hr a day?
(a) 3 days (b) 4 days
(c) 4.5 days (d) None of these
2. A can do a piece of work in 80 days. He works for 10 days and then B alone finishes the remaining work in 42 days. The two together could complete the work in
(a) 24 days (b) 25 days
(c) 30 days (d) 29 days
3. A and B can do a piece of work in 45 and 40 days respectively. They begin together, but A leaves after some days and B completes the remaining work in 23 days. For how many days did A work?
(a) 6 days (b) 8 days
(c) 9 days (d) 12 days
4. A does half as much as work as B in three-fourths of the time. If together they take 18 days to complete the work, how much time will B take to do it?
(a) 30 days
(b) 35 days
(c) 40 days
(d) None of these
5. A can do a certain job in 12 days. B is 60% more efficient than A. B can do the work alone in
(a) 6 days (b) 6.25 days
(c) 7.5 days (d) 8 days
6. A and B can do a job in 25 days and 20 days respectively. A started the work and was joined by B after 10 days. The number of days taken to complete the work is
(a) 12.5 days (b) 14.22 days
(c) 15 days (d) 16.66 days
7. A is twice as good as B, and they together finish a job in 14 days. A will finish the work alone in
(a) 11 days (b) 21 days
(c) 28 days (d) 42 days
8. A and B can do a piece of work in 72 days; B and C in 120 days; A and C in 90 days. In how many days can A do it alone?
(a) 150 days
(b) 120 days
(c) 100 days
(d) 80 days
9. A and B can do a job in 12 days, and B and C can do it in 16 days. After A has been working for 5 days and B for 7 days, C finishes the work in 13 days. In how many days can C do the work alone?
(a) 16 days (b) 24 days
(c) 36 days (d) 48 days
10. Twelve men can do a job in 8 days. Six days after they start, 4 men join them. How many more days will it take to do the job?
(a) 2.5 days (b) 3.5 days
(c) 1.5 days (d) 4 days
11. Excluding the stoppages, the speed of a bus is 54 km/hr, and including the stoppages, it is 45 km/hr. For how many minutes does the bus stop per hour?
(a) 9 min (b) 10 min
(c) 12 min (d) 20 min
12. A job is done by 10 men in 20 days and 20 women in 15 days. How many days will it take for 5 men and 10 women to finish the work?
(a) $17\frac{1}{2}$ days (b) $17\frac{1}{7}$ days
(c) 17 days (d) $17\frac{1}{20}$ days
13. R and S can do a job in 8 and 12 days respectively. If they work on alternate days with R beginning, in how many days will the work be finished?
(a) $9\frac{1}{3}$ days (b) $9\frac{1}{2}$ days
(c) $9\frac{1}{24}$ days (d) $9\frac{1}{3}$ days

14. A, B and C can do a job in 11, 20 and 55 days respectively. How soon can the work be done if A is assisted by B and C on alternate days?
(a) 7 days (b) 8 days
(c) 9 days (d) 10 days
15. Machines A and B produce 8,000 clips in 4 hr and 6 hr respectively. If they work alternatively for 1 hr, A starting first, then 8,000 clips will be produced in
(a) 4.33 hr (b) 4.66 hr
(c) 5.33 hr (d) 5.66 hr
16. Father can work on a task as fast as his two sons working together. If one does the work in 3 hr and the other in 6 hr, in how many hours can the father do the work?
(a) 1 hr (b) 2 hr
(c) 3 hr (d) 4 hr
17. A sum of money is sufficient to pay A's wages for 21 days and B's wages for 28 days. It is then sufficient to pay the wages of both for
(a) 12 days (b) 14 days
(c) 12.25 days (d) 24.5 days
18. A does half as much as work as B, and C does half as much as work as A and B together in the same time. If C alone can do the work in 40 days, then all of them together will finish the work in
(a) 13.33 days
(b) 15 days
(c) 20 days
(d) 30 days
19. Pipes A, B and C take 20, 15 and 12 min to fill a cistern. Together they will fill the cistern in
(a) 5 min (b) 10 min
(c) 12 min (d) 15.66 min
20. Two pipes can fill a tank in 10 hr and 12 hr, while a third pipe can empty it in 20 hr. If all three pipes are opened, the cistern will be filled in
(a) 7 hr (b) 8 hr
(c) 7.5 hr (d) 8.5 hr
21. A cistern can be filled in 9 hr but it takes 10 hr due to a leak. In how much time can the leak empty the full cistern?
(a) 60 hr (b) 70 hr
(c) 80 hr (d) 90 hr
22. Taps A and B can fill a tank in 12 min and 15 min respectively. If both are opened and A is closed after 3 min, how long will it take for B to fill the tank?
(a) 7 min 45 sec (b) 7 min 15 sec
(c) 8 min 5 sec (d) 8 min 15 sec
23. If two pipes operate simultaneously, a tank will be full in 12 hr. If one pipe takes 10 hr less than the other, how much time does the other take to fill the tank?
(a) 25 hr (b) 28 hr
(c) 30 hr (d) 35 hr
24. A leak in the bottom of a tank can empty it in 6 hr. A pipe fills the tank at the rate of 4 L per min. When the tank is full, the inlet is opened but due to the leak the tank is emptied in 8 hr. What is the capacity of the tank?
(a) 5,260 L (b) 5,760 L
(c) 5,846 L (d) 6,970 L
25. A and B can fill a tank in 6 hr and 4 hr respectively. If they are opened on alternate hours and A is opened first, how many hours will it take to fill the tank?
(a) 4 hr (b) 5 hr
(c) 4.5 hr (d) 5.5 hr
26. A car can finish a journey in 10 hr at 48 km/hr. To cover the same distance in 8 hr, the speed must be increased by
(a) 6 km/hr
(b) 7.5 km/hr
(c) 12 km/hr
(d) 15 km/hr
27. Tarun can cover a certain distance in 1 hr 24 min by covering $\frac{2}{3}$ of the distance at 4 km/hr and the rest at 5 km/hr. The total distance is
(a) 5 km (b) 6 km
(c) 8 km (d) 9.2 km
28. Walking at $\frac{3}{4}$ of his normal speed, a man is late by 2.5 hr. The usual time is
(a) 7.5 hr (b) 3.5 hr
(c) 3.25 hr (d) $\frac{7}{8}$ hr



29. If a boy walks from his house at 4 km/hr, he reaches school 10 min early. If he walks at 3 km/hr, he reaches 10 min late. What is the distance from his house to the school?
(a) 6 km (b) 4.5 km
(c) 4 km (d) 3 km
30. If a train runs at 40 km/hr, it is late by 11 min, but if it runs at 50 km/hr, it is late by 5 min. The correct time to complete the journey is
(a) 13 min (b) 15 min
(c) 19 min (d) 21 min
31. Two trains start from opposite directions, 200 km apart, at the same time. They cross each other at a distance of 110 km from one of the stations. What is the ratio of their speeds?
(a) 11 : 20 (b) 9 : 20
(c) 11 : 9 (d) None of these
32. A thief steals a car at 2.30 p.m. and drives it at 60 km/hr. At 3 p.m. the owner sets off in another car at 75 km/hr. When will the owner overtake the thief?
(a) 4.30 p.m. (b) 4.45 p.m.
(c) 5 p.m. (d) 5.15 p.m.
33. The speeds of A and B are in the ratio 3 : 4. A takes 30 min more than B to reach the destination. How much time does A take to reach?
(a) 1.25 hr (b) 1.33 hr
(c) 2 hr (d) 2.5 hr
34. A man goes uphill at 24 km/hr and comes down at 36 km/hr. What is his average speed?
(a) 30 km/hr (b) 28.8 km/hr
(c) 32.6 km/hr (d) None of these
35. A plane travels 2,500 km, 1,200 km and 500 km at the rate of 500 km/hr, 400 km/hr and 250 km/hr respectively. The average speed is
(a) 405 km/hr (b) 410 km/hr
(c) 420 km/hr (d) 575 km/hr
36. A car has to cover 80 km in 10 hr. If it covers $\frac{1}{2}$ of the journey in $\frac{3}{5}$ of the time, what should be its speed for the remaining journey?
(a) 8 km/hr (b) 6.4 km/hr
(c) 10 km/hr (d) 20 km/hr
37. A is twice as fast as B, and B is twice as fast as C. The distance covered by C in 54 min will be covered by A in
(a) 216 min (b) 27 min
(c) 108 min (d) 13.5 min
38. The ratio of speeds of A and B is 3 : 4. If B covers a distance in 36 min, A will cover it in
(a) 27 min (b) 48 min
(c) $15\frac{3}{7}$ min (d) None of these
39. A train 110 m long is moving at 132 km/hr. How long will it take to cross a platform 165 m long?
(a) 5 sec (b) 7.5 sec
(c) 10 sec (d) 15 sec
40. A train travelling at the rate of 90 km/hr crosses a pole in 10 s. Its length is
(a) 250 m (b) 150 m
(c) 900 m (d) 100 m
41. A train 300 m long crossed a platform 900 m long in 1 min 12 sec. The speed of the train (in km/hr) is
(a) 45 (b) 50
(c) 54 (d) 60
42. A train crosses a pole in 15 sec and a platform 100 m long in 25 sec. Its length is
(a) 200 m (b) 150 m
(c) 50 m (d) Data insufficient
43. A train 110 m long is travelling at speed of 58 km/hr. What is the time in which it will pass a man walking in the same direction at a speed of 4 km/hr?
(a) 6 sec (b) 7.5 sec
(c) 7.33 sec (d) 7.33 min
44. A train 108 m long moving at the rate of 50 km/hr crosses another train of 112 m long coming from the opposite direction in 6 sec. The speed of the second train is
(a) 48 km/hr (b) 54 km/hr
(c) 66 km/hr (d) 82 km/hr
45. Two trains travel in opposite directions at 36 km/hr and 45 km/hr. A man sitting in the slower train passes the faster train in 8 sec. The length of the faster train is
(a) 80 m (b) 100 m
(c) 120 m (d) 180 m

46. A man sees a train passing over a bridge, 1 km long. The length of the train is half that of the bridge. If the train clears the bridge in 2 min, the speed of the train is
(a) 30 km/hr (b) 45 km/hr
(c) 50 km/hr (d) 60 km/hr
47. A man rows 13 km upstream and 28 km downstream in 5 hr each. The velocity of the stream is
(a) 1.5 km/hr (b) 2 km/hr
(c) 2.5 km/hr (d) 3 km/hr
48. If a man rows at 6 km/hr in still water and 4.5 km/hr against the current, then his rate of rowing along the current is
(a) 9.5 km/hr (b) 7.5 km/hr
(c) 7 km/hr (d) 5.25 km/hr
49. A river runs at 2 km/hr. If a man takes twice as long to row up the river as to row down, the speed of the man in still water is
(a) 6 km/hr (b) 4 km/hr
(c) 10 km/hr (d) 8 km/hr
50. A stream runs at 1 km/hr. A boat goes 35 km upstream and back again in 12 hr. The speed of the boat in still water is
(a) 6 km/hr (b) 7 km/hr
(c) 8 km/hr (d) 8.5 km/hr
51. Ajay completes one lap of a circular track of radius 21 m in 6 minute while Bijay completes it in 8 minutes. If they start together from the same point in the same direction after how much time (in minutes) will they meet for the first time?
(a) 24 (b) 16
(c) 12 (d) 20
52. In the above questions after how many minute they meet for the second time at the starting point?
(a) 24 (b) 36
(c) 48 (d) 40
53. Salma and Nagma are standing on a circular track diametrically opposite to each other. They start to run in anticlockwise direction. If the initial distance between them is 70 metres and speed of Salma is 5 m/s and that of Nagma is 6m/s, then how many times will they meet each other in first 6 minutes
(a) 1 (b) 2
(c) 3 (d) 4
54. In the above Question, if Nagma runs in clockwise direction then, after how many seconds they will meet for the third time?
(a) 70 (b) 40
(c) 50 (d) 90



My Doubts



Answers and Explanations

Chapter 1 : Number System

1	a	2	c	3	c	4	c	5	b	6	a	7	d	8	d	9	b	10	c
11	c	12	d	13	a	14	c	15	d	16	d	17	b	18	b	19	a	20	a
21	c	22	c	23	b	24	d	25	c	26	d	27	b	28	d	29	b	30	c
31	c	32	a	33	d	34	d	35	d	36	c	37	b	38	c	39	d	40	b
41	b	42	b	43	b	44	b	45	a										

Chapter 2 : Percentage

1	d	2	a	3	d	4	b	5	c	6	c	7	d	8	c	9	a	10	d
11	b	12	c	13	b	14	b	15	d	16	a	17	c	18	c	19	a	20	a
21	b	22	c	23	c	24	d	25	c	26	b	27	a	28	b	29	b	30	c
31	a	32	c	33	d	34	c	35	c	36	a	37	a	38	a	39	d	40	a
41	c	42	d	43	d	44	d	45	a	46	b								

Chapter 3 : Simple Interest & Compound Interest

1	a	2	d	3	c	4	a	5	b	6	c	7	c	8	d	9	b	10	a
11	b	12	a	13	c	14	c	15	d	16	b	17	c	18	a	19	c	20	b
21	d	22	c	23	c	24	b	25	b	26	c	27	b	28	a	29	b	30	b
31	b	32	d	33	c	34	a	35	b	36	c	37	a	38	c	39	a	40	a
41	a	42	b	43	c	44	b	45	a	46	b	47	c						

Chapter 4 : Profit, Loss & Discount

1	c	2	c	3	a	4	c	5	c	6	b	7	c	8	a	9	b	10	d
11	b	12	b	13	d	14	b	15	a	16	b	17	b	18	b	19	a	20	b
21	c	22	a	23	b	24	c	25	b	26	b	27	b	28	c	29	b	30	a
31	c	32	d	33	c	34	a	35	b	36	d	37	d	38	c	39	a		

Chapter 5 : Ratio & Proportion, Mixture Alligation, Partnership & Average

1	c	2	d	3	a	4	d	5	b	6	b	7	c	8	c	9	a	10	d
11	c	12	c	13	d	14	c	15	a	16	b	17	d	18	a	19	b	20	b
21	d	22	a	23	c	24	c	25	a	26	d	27	b	28	b	29	c	30	a
31	c	32	b	33	d	34	c	35	d	36	b	37	a	38	b	39	d	40	d
41	a	42	c	43	b	44	c	45	a	46	c	47	c	48	a	49	d	50	c
51	a	52	b	53	a	54	b	55	d	56	c								

Chapter 6 : Time, Speed, Distance and Time & Work

1	a	2	c	3	c	4	a	5	c	6	d	7	b	8	b	9	b	10	c
11	b	12	b	13	b	14	b	15	b	16	b	17	a	18	a	19	a	20	c
21	d	22	d	23	c	24	b	25	b	26	c	27	b	28	a	29	c	30	c
31	c	32	c	33	c	34	b	35	c	36	c	37	d	38	b	39	b	40	a
41	d	42	b	43	c	44	d	45	d	46	b	47	a	48	b	49	a	50	a
51	a	52	c	53	b	54	c												

Chapter 1 Number System

1. a $\text{LCM} \times \text{GCD} = \text{Product of numbers} = N_1 \times N_2$

$$\therefore N_2 = \frac{\text{Product}}{N_1} \Rightarrow N_2 = \frac{4200 \times 20}{100} = 840$$

2. c $y = \sqrt{2} + 1$

$$\Rightarrow y + \frac{1}{y} = \frac{y^2 + 1}{y} = \frac{(\sqrt{2} + 1)^2 + 1}{(\sqrt{2} + 1)}$$

$$= \frac{(4 + 2\sqrt{2})}{\sqrt{2} + 1} = \frac{2\sqrt{2}(1 + \sqrt{2})}{(\sqrt{2} + 1)} = 2\sqrt{2}$$

Short cut:

$$(\sqrt{2} + 1) + \frac{1}{\sqrt{2} + 1} = (\sqrt{2} + 1) + (\sqrt{2} - 1) = 2\sqrt{2}$$

3. c By observing, $\frac{4}{5} > \frac{3}{5} > \frac{2}{5} > \frac{1}{5}$.

Numerator is decreasing and denominator is same.

Now comparing $\frac{4}{5}$ and $\frac{7}{15}$.

$$\text{As, } \frac{7}{15} = \frac{7}{3 \times 5} \text{ and } \frac{7}{3} < 4$$

$\therefore \frac{4}{5}$ is the largest.

4. c $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$
 $\Rightarrow 1^2 + 2^2 + \dots + 10^2 = \frac{10(11)(21)}{6} = 385$

5. b $0! = 1$

6. a Whole number

7. d Change the fractions into decimals and then find

$$\frac{71}{84} = 0.845, \frac{5}{6} = 0.833, \frac{29}{35} = 0.828, \frac{57}{70} = 0.814.$$

Hence, the answer is all of these.

8. d $\text{LCM of } 2, 3, 4, 5, 6 \text{ and } 7 = 420$
 Smallest four-digit number divisible by LCM is
 $420 \times 3 = 1260$.

Short cut:

1260 is the only option divisible by 3.

9. b Difference between divisor and remainder
 $= 35 - 25 = 45 - 35 = 55 - 45 = 10$.
 $\text{LCM of } 35, 45 \text{ and } 55 = 5 \times 7 \times 9 \times 11 = 3465$.
 Required number $= 3465 - 10 = 3455$.

Short cut:

3455 is the only option from which when we subtract 35, we get 3420 which is divisible by 45.

10. c $x + y = 2(x - y)$
 $\Rightarrow x + y = 2(x - y)$
 $\Rightarrow x = 3y$
 Since, $xy = 48, \Rightarrow 3y^2 = 48$
 $\Rightarrow y^2 = 16$
 $\Rightarrow y = 4$
 Now, $y = 4, \therefore x = 12$

11. c $3 + \frac{3}{3 + \frac{1 \times 3}{10}} = 3 + \frac{3 \times 10}{33} = 3 + \frac{10}{11} = \frac{43}{11}$

12. d Units digit = Units digit of $(4 \times 1 \times 9 \times 8 = 4 \times 72) = 8$.
 Therefore, units digit of the number = 8.

13. a $\text{LCM of } 2, 3, 5 = 30$.
 $\therefore$ Number of coconuts $= 30 + 1 = 31$.

14. c Let the tens digit be x . Then units digit $= (x + 3)$.
 Sum of the digits $= x + (x + 3) = 2x + 3$
 Number $= 10x + (x + 3) = 11x + 3$
 $\frac{11x + 3}{2x + 3} = \frac{4}{1} \Rightarrow 11x + 3 = 4(2x + 3)$
 $\therefore x = 3$
 $\therefore$ Number $= (11x + 3) = 36$.

Short cut:

Pick up the option and check the conditions.

15. d $\therefore (6^5)^{3/5} = 6^3 = 216$.
 16. d $\frac{5}{6}$ is largest among four.

17. b $x = 6 - \sqrt{35}$
 $\therefore \frac{1}{x} = \frac{1}{6 - \sqrt{35}} \times \frac{6 + \sqrt{35}}{6 + \sqrt{35}} = 6 + \sqrt{35}$

18. b $2 + 4 + 6 + 8 + \dots + 100$ (50 even numbers)
 $= 50 \times (50 + 1) = 2550$.
 Sum of first n even numbers is $n(n + 1)$.

19. a $x + 2y = 4$ and $\frac{x}{y} = 2 \Rightarrow x = 2y$
 $\therefore 2y + 2y = 4$
 $\therefore y = 1$

20. a Numbers are 8, 9, 10.
 $\therefore$ Sum $= 8 + 9 + 10 = 27$.

21. c $18.18 \div 9 + 2.7 \text{ of } 3 = 18.18 \div 9 + 8.1$
 $= 2.02 + 8.1$
 $= 10.12$



22. c $8127 - 5422 + 1614 - 808 = 9741 - 6230 = 3511$

23. b $11^2 + 11^4 + 11^3 - 11 + (0.5)11^2$
 $= 121 + 11 - 11 + 0.5 \times 121$
 $= 121 + 60.5 = 181.5$

24. d The square root of 1296 is 36 and not 34.

25. c Let the numbers be x and $1365 + x$.
 Then, $1365 + x = 6x + 15$ or $x = 270$.

26. d $\frac{A}{2} = \frac{B}{3} = \frac{C}{4} = x \Rightarrow A = 2x, B = 3x \text{ and } C = 4x$
 $\Rightarrow A : B : C = 2 : 3 : 4$

Largest part = $\left(243 \times \frac{4}{9}\right) = 108$.

27. b Every such number must be divisible by LCM of 4, 5 and 6, i.e. 60.
 Such numbers are 240, 300, 360, 420, 480, 540.
 Hence, there are 6 numbers.

28. d Units digit is governed by product of individual units digit only.
 $7 \times 8 \times 7 \times 3 = 6$.

29. b $\therefore 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
 $\therefore 1 + 2 + 3 + \dots + 45 = \frac{45 \times 46}{2} = 1035$

30. c $\frac{451a603}{9}$.
 $\therefore$ Sum of all the digits must be divisible by 9,
 $\therefore a = 8$.

31. c Number formed by last 3 digits must be divisible by 8.

32. a Number should be divisible by 9 and 11 both.

33. d $111111 = 111 \times 1001 = (37 \times 3) \times (7 \times 11 \times 13)$
 Hence, number of such forms are divisible by all its factors.

34. d When 99547 is divided by 687, remainder is 619.
 $\therefore$ Nearest number = $99547 + 68 = 99615$.

35. d Out of the given answer choices only 99990 is divisible by 99. Also, in order to a number being divisible by 99, it should be divisible by both 9 and 11, which is true in case of option (d) only.

36. c Any number divisible by 111 should also be divisible by 3 and 37. Out of the given answer choices, both options (a) and (c) are divisible by 111 but, option (c) is the smallest.

37. b $3^{4n} - 4^{3n} = (3^4)^n - (4^3)^n = 81^n - 64^n$
 Since $a^n - b^n$ is always divisible by $(a - b)$
 $81^n - 64^n$ will be always divisible by $(81 - 64) = 17$.

Shortcut:
 Substitute $n = 1$.

38. c $8756 \times 99999 = 8756 \times [100000 - 1]$
 $= 875600000 - 8756 = 875591244$

39. d $1399 \times 1399 = (1400 - 1)^2$
 $= (1400)^2 + (1)^2 - 2(1400)(1) = 1960000 + 1 - 2800$
 $= 1957201$

40. b $397 \times 397 + 104 \times 104 + 2 \times 397 \times 104$
 $= (397 + 104)^2 = (501)^2 = (500 + 1)^2$
 $= (500)^2 + (1)^2 + 2(500)(1)$
 $= 250000 + 1 + 1000 = 251001$

41. b $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)}$
 $= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{(n+1)}$
 $= 1 - \frac{1}{(n+1)} = \frac{n}{n+1}$

As, numerator is always less than denominator, thus the expression will always be less than 1.

42. b $n = 4 \times q + 3$ ($q = \text{Quotient}$)
 $2n = 2 \times 4 \times q + 6$
 When $2n$ is divided by 4, quotient = $2q + 1$ and remainder = 2.

43. b LCM of 3, 6, 9, 12, 15 and 18 is 180.
 So the bells will toll together after every 180 s, i. e. 3 min.

In 32 min, they will toll together for $\frac{32}{3} = 10$ times.

44. b 3 has a cyclicity of 4. 33 when divided by 4 has remainder 1. Hence unit digit is 3^1 i.e. 3.

45. a $24^{37} = 4^{37} = (4^2)^{18} \cdot 4^1 = 4$ (unit digit)
 $37^{24} = 7^{24} = (7^4)^6 = (2401)^6 = 1$ (unit digit)
 Hence Units digit = $4 \times 1 = 4$.

Chapter 2 Percentage

1. d $18\frac{3}{4}\% \text{ of } 2000 = \frac{75}{4} \times \frac{2000}{100} = 375$

Alternative: $18\% \text{ of } 2000 + \frac{3}{4}\% \text{ of } 2000 = 375$

2. a $\frac{26}{48} \times 100 = \frac{13}{24} \times 100 = 54.16\%$

3. d $33\frac{1}{3}\% \text{ of } 972 = \frac{1}{3} \times 972 = 324$

4. b $\frac{37}{60} \times 100 = 61.66\%$

5. c Percentage increase = $\frac{\text{Final value} - \text{Initial value}}{\text{Initial value}} \times 100$
 $= \frac{7475 - 6500}{6500} = \frac{975}{6500} \times 100 = 15\%$

6. c 20% increase for 2 years

$$= 20 + 20 + \frac{20 \times 20}{100}$$

$$\left\{ \text{using successive \% change } a + b + \frac{ab}{100} \right\} = 44\%$$

$$\text{Now, population after 2 years} = 2500 \times 1.44 = 3600.$$

7. d 90% of 90% of 100 = 90% of 90 = 81.

8. c Let 37% of x = 990.86.

$$\text{Then, } \frac{37}{100} \times x = 990.86$$

$$\Rightarrow x = \frac{990.86 \times 100}{37} = \frac{99086}{37} = 2678.$$

$$\text{Now, } 19\% \text{ of } 2678 = \frac{19}{100} \times 2678 = 508.82$$

or 510 approximately.

9. a 1 hr = 60 min, $\therefore \frac{50}{60} \times 100 = 83.33\%$

10. d Loss percentage = $\frac{4}{20} = \frac{1}{5} = 20\%$

11. b A's income = 1.25 B's income.

$$\text{B's income} = \frac{1}{1.25} \text{ A's income} = 80\% \text{ of A's income.}$$

12. c 40% of 200 = 80

He obtained 72, i.e. he failed by 8 marks.

$$\text{Percentage by which he failed} = \frac{8}{200} \times 100 = 4\%.$$

13. b If x = 120% of y, then $y = \frac{100}{120}$ of x = 83.33% of x.

14. b If A is increased by 10%,
i.e. $A_1 = 1.1A$

$$\therefore A_1^2 = (1.1)^2 A^2 = 1.21A^2$$

$\therefore A^2$ increases by 21%.

15. d $A = \frac{1500}{9} \% \text{ of } \frac{9}{15} A$ i.e. 166.66%.

16. a Length becomes 0.9L.
Breadth becomes 0.9B.
Area = $0.9L \times 0.9B = 0.81LB$.
 $\therefore$ Area decreases by 19%.

Short cut:

Using successive % change

$$-10 - 10 + \frac{10 \times 10}{100} = -19\% (\text{decrease})$$

17. c Price of rice after 15% increase
 $= 1.15 \times 15 = ₹17.25$

After 30 paise reduction, price = ₹16.95

$$\text{Net increase} = \frac{16.95 - 15}{15} \times 100 = 13\%.$$

18. c 30% of 55% of 100 = 30% of 55 = 16.5

19. a Let the number be n.

Then 20% of n - 16% of n = 16

$$0.2n - 0.16n = 16 \text{ or } 0.04n = 16$$

$$\Rightarrow n = 400$$

20. a (i) $66\frac{1}{6}\%$, (ii) $\frac{3}{5} = 60\%$

(iii) 0.65 = 65%, (iv) $\frac{16}{25} = 64\%$

Hence option (a).

21. b After 20% increase, men = $2500 \times 1.2 = 3000$.

After 20% decrease, women = $2500 \times 0.8 = 2000$.

$$\text{So, women as a percentage of men} = \frac{2000}{3000} = 66.67\%.$$

22. c Net sales = Number of units $\times$ Price

$$1.28 S = y (\text{Number of units}) \times SP$$

$$y = \frac{1.28}{.8} = 1.6$$

$\therefore$ Number of units increase by 60%

23. c In a class of 300 students, boys = 200, girls = 100.

50% boys = 100, 48% girls = 48.

Total students who appeared = 148.

Hence, $300 - 148 = 152$ did not appear.

24. d Let he has ₹100. He invests ₹40 in shares. Out of ₹60, he invests ₹12 in property. Out of remaining ₹48, he loses ₹12 at casino. He is left with ₹36, i.e. 36%.

25. c Let the winner gets 'x' votes.

$\therefore$ The loser got $(x - 60)$ or $\frac{2}{3}x$ votes.

$\therefore x = 180$, i.e. the winner got 180 votes and the loser got 120 votes.

26. b $180 + 120 = 300$

27. a $\frac{180}{300} = \frac{6}{10} = \frac{3}{5} = 60\%$

28. b Let the original fraction = $\frac{x}{y}$.

$$\therefore \frac{x + 25\% \text{ of } x}{y - 20\% \text{ of } y} = \frac{5}{4}, \quad \frac{1.25x}{0.8y} = \frac{5}{4} \Rightarrow \frac{x}{y} = \frac{4}{5}$$

29. b Let 100 be the number. Then $43 - 28 = 15$. But difference is 75, i.e. five times 15. Therefore, actual number should be five times 100, i.e. 500.

$$\therefore 30\% \text{ of } 500 = \frac{30}{100} \times 500 = 150$$

30. c A student failed by 15%, i.e. he has got only 30%. So, 30% of total marks = 138.

$$\therefore \text{Total marks} = \frac{138}{30} \times 100 = 460.$$

31. a In 1 kg pack, ghee is 10%, i.e. 100 g. So, in 2 kg, ghee will be 200 g.

$$32. c \text{ Error on } 37\frac{1}{2} = \left(37\frac{1}{2} - 36 \right) = 1\frac{1}{2}.$$

$$\therefore \text{Error on } \frac{75}{2} = \frac{3}{2}.$$

$$\text{Error on } 100 = \left(\frac{3}{2} \times \frac{2}{75} \times 100 \right) = 4\%.$$

33. d 30% of petrol = ₹300. Hence, total sum = ₹1,000.

$$70\% = 700; \frac{1}{4} \text{ th of } 700 = ₹175$$

$$34. c \quad x\% \text{ of } a = y\% \text{ of } b \Rightarrow \frac{x}{100} a = \frac{y}{100} b$$

$$\Rightarrow b = \left(\frac{x}{100} \times \frac{100}{y} \right) a = \left(\frac{x}{y} \right) a$$

$$z\% \text{ of } b = z\% \text{ of } \left(\frac{x}{y} \right) a = \left(\frac{xz}{y \times 100} \right) a$$

$$= \left(\frac{xz}{y} \right) \% \text{ of } a$$

$$35. c \quad A = \frac{90}{100} B, B = \frac{125}{100} C \text{ and } C = \frac{80}{100} D$$

$$B = \frac{10}{9} A, C = \frac{4}{5} B \text{ and } D = \frac{5}{4} C \text{ then,}$$

$$B = \frac{10}{9} \times 360 = 400, C = \frac{4}{5} \times 400 = 320$$

$$\text{and } D = \frac{5}{4} \times 320 = 400$$

Percentage of marks obtained by

$$D = \left(\frac{400}{500} \times 100 \right) \% = 80\%$$

36. a $p = 6q$. So q is less than p by $5q$. Note that q has been compared with p , i.e. p is the base for comparison.

Required percentage

$$\left(\frac{5q}{p} \times 100 \right) \% = \left(\frac{5q}{6q} \times 100 \right) \% = 83\frac{1}{3}\%.$$

37. a Let the valid votes be x .

Then $52\% \text{ of } x - 48\% \text{ of } x = 98 \Rightarrow 4\% \text{ of } x = 98$.

$$\therefore \frac{4}{100} x = 98 \Rightarrow x = 98 \times 25 = 2450.$$

$$\therefore \text{Total votes polled} = (2450 + 68) = 2518.$$

38. a Let the original consumption = 100 units and the original price = ₹100 per unit.

$$\text{Original expenditure} = ₹(100 \times 100) = ₹10,000.$$

$$\text{New expenditure} = ₹(120 \times 75) = ₹9,000.$$

$$\text{Decrease in expenditure} = \left(\frac{10000}{10000} \times 100 \right) \% = 10\%.$$

Short Cut:

$$\text{Change in expenditure} = 20 - 25 - \frac{20 \times 25}{100} = -10\%$$

Thus, expenditure decreases by 10%

39. d Net growth on 1000 = $(32 - 11) = 21$.

$$\text{Net growth on } 100 = \left(\frac{21}{1000} \times 100 \right) = 2.1\%.$$

40. a Let length = l and breadth = b , and the required decrease in breadth be $x\%$.

$$\text{Then, } \frac{160}{100} l \times \frac{(100-x)}{100} \times b = lb$$

$$\Rightarrow 160(100-x) = 100 \times 100$$

$$\Rightarrow 100-x = \frac{10000}{160} = \frac{125}{2}$$

$$\Rightarrow x = \left(100 - \frac{125}{2} \right) = 37\frac{1}{2}\%$$

41. c In 50 L, 35% is water.

$$\text{i.e. } \frac{35}{100} \times 50 = 17.5 \text{ L}$$

So, quantity of milk = $50 - 17.5 = 32.5 \text{ L}$.

42. d If SP = 121, P = 10%

$$CP = \frac{121 \times 100}{110} = ₹110$$

If SP is reduced by 11, new SP = ₹(121 - 11) = ₹110.

Hence, no profit no loss.

43. d After giving a discount of 10%, the merchant makes a profit of 8%.

$$SP = MP - 10\% \text{ of } MP$$

$$SP = 0.9 MP$$

$$\text{Also } SP = 1.08 CP = 0.9 MP$$

$$\text{So } \frac{MP}{CP} = 1.2, \text{ i.e. } MP \text{ is } 20\% \text{ more than } CP.$$

44. d $M + 60 = 50\%$ of K . Clearly, M cannot be determined.

45. a Let A 's salary = x . Then B 's = $(2000 - x)$.
 5% of $A = 15\%$ of B , i.e.

$$\frac{5}{100}x = \frac{15}{100}(2000 - x) \Rightarrow x = ₹1500.$$

46. b Let original salary = ₹100.
 Now, final salary = 150% of $(50\%$ of 100)

$$= \left(\frac{150}{100} \times \frac{50}{100} \times 100 \right) = ₹75; \text{ Decrease} = 25\%$$

Chapter 3 Simple Interest & Compound Interest

1. a Total return = 8.5% of $2000 = ₹170$.

$$SI \text{ from } ₹500 = \frac{500 \times 7 \times 1}{100} = ₹35,$$

$$\text{and } SI \text{ from } ₹1500 = \frac{1500 \times r \times 1}{100} = 15r.$$

$$\text{We have, } 35 + 15r = 170$$

$$\Rightarrow 15r = 135 \Rightarrow r = 9\%.$$

2. d The family borrowed ₹325.
 The family paid = $16 \times 23 + 9 = ₹377$.
 Interest paid = $377 - 325 = ₹52$.

$$\text{So, percentage} = \frac{52}{325} \times 100 = 16\%.$$

$$3. c \quad \frac{500 \times R_1 \times 2}{100} - \frac{500 \times R_2 \times 2}{100} = 2.50$$

$$\Rightarrow 1000(R_1 - R_2) = 250$$

$$\Rightarrow R_1 - R_2 = \frac{250}{1000} = 0.25\%$$

$$4. a \quad \text{Interest} = \frac{3500 \times 5 \times 5}{100} = ₹875.$$

$$\text{Amount} = P + I = 3500 + 875 = ₹4375.$$

$$5. b \quad 2P = \frac{P \times 20 \times t}{100}, t = 10 \text{ years}$$

$$6. c \quad CI = P \left[1 + \frac{r}{100} \right]^3 - P$$

$$= 1000[1.1]^3 - 1000 = 1331 - 1000 = ₹331$$

$$7. c \quad CI \text{ for 3 years} = 700 \left[1 + \frac{10}{100} \right]^3 - 700 = ₹231.70$$

$$SI \text{ for 3 years} = \frac{700 \times 10 \times 3}{100} = ₹210$$

$$\therefore CI_3 - SI_3 = 21.7$$

Short cut:

$$CI_3 - SI_3 = 700(1.1^3 - 1.3) = ₹21.70$$

$$8. d \quad SI = \frac{P}{4} \therefore \frac{P}{4} = \frac{P \times r \times t}{100}$$

$$\text{Since } r = t,$$

$$\therefore r^2 = \frac{100}{4}, r = \frac{10}{2} = 5\%$$

$$9. b \quad CI = 1010 = P \left(1 + \frac{2}{100} \right)^2 - P$$

$$\Rightarrow 1010 = P[(1.02)^2 - 1] = 0.0404P$$

$$\Rightarrow P = ₹25,000$$

Alternative method:

$$CI = SI + SI + \frac{2}{100} SI$$

Where SI is simple interest for one year

$$\Rightarrow 1010 = \frac{202}{100} SI$$

$$\therefore SI = \frac{1010}{202} \times 100 = 500$$

$$\therefore P = \frac{500 \times 100}{2 \times 1} = ₹25000$$

$$10. a \quad CI = 203 = P \left(1 + \frac{3}{100} \right)^2 - P$$

$$\Rightarrow 203 = P[(1.03)^2 - 1]$$

$$\Rightarrow P = \frac{203}{0.0609} \approx ₹3,333$$

$$SI = \frac{P \times R \times T}{100} = \frac{3333 \times 3 \times 2}{100} \approx ₹200$$

Short cut:

SI must be less than CI .

$$11. b \quad 154 = \frac{P \times 5 \times 2}{100} \Rightarrow P = ₹1,540$$

12. a Present worth of ₹460

$$= \frac{460}{1 + \frac{2 \times 15}{100}} \approx ₹418.20$$

$$\text{Therefore, profit} = \frac{18.2}{400} \times 100 = 4.55\%$$



13. c March + April + May

$$23 + 30 + 21 = 74 \text{ days} = \frac{74}{365} \text{ year}$$

$$= \frac{1}{5} \text{ years (approximately)}$$

$$\therefore \text{Interest} = ₹ \left(1820 \times \frac{1}{5} \times \frac{15}{2 \times 100} \right) = ₹ 27.30$$

14. c Amount payable at the end of 1st year = ₹ 4,400;
Half of 4400 is paid. Remaining = ₹ 2,200.
Amount payable at the end of 2nd year
= ₹ 2,420; Half of which is paid.
Amount payable at the end of 3rd year
= 1210(1.1) = ₹ 1,331.

15. d Amount paid = 2200 + 1210 + 1331 = ₹ 4,741.
Therefore, interest paid = ₹ 741.

16. b $SI = \frac{P \times T \times R}{100}$

$$\Rightarrow 225 = 1125 \times \frac{R}{100} \times 4 \Rightarrow R = \frac{225 \times 100}{1125 \times 4}$$

$$\Rightarrow R = 5\%$$

17. c (i) $SI = \frac{5000 \times 3 \times 11}{100} = ₹ 1,650$

(ii) $CI = 5000 \left(1 + \frac{10}{100} \right)^3 - 5000 = ₹ 1,655$

Short cut:

Effective rate at SI and CI = 33% and 33.1% respectively.

18. a Let the sum be ₹ x and the original rate be R%.

$$\text{Then, } \frac{x \times (R+3) \times 2}{100} - \frac{x \times R \times 2}{100} = 72$$

$$\Rightarrow 2Rx + 6x - 2Rx = 7200 \Rightarrow x = ₹ 1,200.$$

19. c Let sum = x. Then SI = x.

$$\text{Rate} = \left(\frac{100 \times x}{x \times 8} \right) \% = 12.5\%.$$

20. b $y = \frac{x \times R \times T}{100} \Rightarrow RT = \frac{100y}{x}$

$$z = \frac{y \times R \times T}{100} \Rightarrow RT = \frac{100z}{y}$$

$$\therefore \frac{100y}{x} = \frac{100z}{y} \Rightarrow y^2 = xz$$

21. d Let the sum at 5% be ₹ x. Then,

$$\frac{x \times 5 \times 3}{100} + \frac{(1550 - x) \times 8 \times 3}{100} = 300 \Rightarrow x = ₹ 800$$

$$\frac{\text{Money at 5\%}}{\text{Money at 8\%}} = \frac{800}{(1550 - 800)} = \frac{800}{750} = \frac{16}{15}$$

22. c Let the total capital be x. Then

$$\left(\frac{x}{3} \times \frac{7}{100} \times 1 \right) + \left(\frac{x}{4} \times \frac{8}{100} \times 1 \right) + \left(\frac{5x}{12} \times \frac{10}{100} \times 1 \right) = 561$$

$$\Rightarrow \frac{7x}{300} + \frac{x}{50} + \frac{x}{24} = 561$$

$$\Rightarrow x = \left(\frac{561 \times 600}{51} \right) = ₹ 6,600$$

23. c $\left(\frac{2000 \times 8 \times 1}{100} \right) + \left(4000 \times \frac{15}{2} \times \frac{1}{100} \right) + \left(1400 \times \frac{17}{2} \times \frac{1}{100} \right)$

$$+ 2600 \times R \times \frac{1}{100} = \frac{813}{10000} \times 10000$$

$$\Rightarrow 160 + 300 + 119 + 26R = 813 \Rightarrow R = 9\%$$

24. b Let these parts be x, y and z.

Hence, by the simple interest formula,
4x = 8y, and 8y = 12z.

Hence, x : y : z = 6 : 3 : 2.

$$\text{The smallest part} = \frac{2189 \times 2}{11} = ₹ 398.$$

25. b $\text{Sum} = ₹ \left(\frac{100 \times 1200}{3 \times 5} \right) = ₹ 8,000$

$$\text{Amount} = ₹ \left[8000 \times \left(1 + \frac{5}{100} \right)^3 \right]$$

$$= ₹ \left(8000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} \right) = ₹ 9,261$$

$$\therefore CI = ₹ (9261 - 8000) = ₹ 1,261$$

26. c Let the sum be x. Then,

$$CI = x \left(1 + \frac{5}{100} \right)^2 - x = \left(\frac{441x}{400} - x \right) = \frac{41x}{400}$$

$$SI = \frac{x \times 10 \times 1}{100} = \frac{x}{10}$$

$$CI - SI = \frac{41x}{400} - \frac{x}{10} = \frac{x}{400}$$

$$\therefore \frac{x}{400} = 25 \Rightarrow x = ₹ 10,000$$

Short cut:

$$\text{Interest paid on principal for 6 month} = 25 \times \frac{100}{5} = ₹ 500$$

$$\text{Principal amount invested} = 500 \times \frac{100}{5} = ₹ 10,000$$

27. b $CI = ₹ \left[800 \times \left(1 + \frac{10}{100} \right)^2 - 800 \right] = ₹ 168$

$$SI = ₹ \left(\frac{800 \times 10 \times 2}{100} \right) = ₹160$$

$$\text{Gain} = CI - SI = ₹(168 - 160) = ₹8$$

28. a Interest on ₹4,624 for 1 year
= 4913 - 4624 = ₹289.

$$\therefore \text{Rate} = \frac{289}{4624} = 6\frac{1}{4}\%$$

Now if x = Sum,

$$x \left(1 + \frac{25}{4 \times 100} \right)^2 = 4624 \Rightarrow x \times \frac{17}{16} \times \frac{17}{16} = 4624$$

$$\Rightarrow x = \left(4624 \times \frac{16}{17} \times \frac{16}{17} \right) = ₹4,096$$

29. b Rate = 5% per half year.
Let time = 2n half years = n years

$$\text{Then, } 800 \left(1 + \frac{5}{100} \right)^{2n} = 926.10$$

$$\Rightarrow \left(\frac{21}{20} \right)^{2n} = \frac{926.10}{800} = \frac{9261}{8000} = \left(\frac{21}{20} \right)^3$$

$$\therefore 2n = 3 \Rightarrow n = \frac{3}{2} \text{ years.}$$

30. b $12000 \times \left(1 + \frac{R}{100} \right)^5 = 24000 \Rightarrow \left(1 + \frac{R}{100} \right)^5 = 2$

$$\therefore \left[\left(1 + \frac{R}{100} \right)^5 \right]^4 = 2^4 = 16 \Rightarrow \left(1 + \frac{R}{100} \right)^{20} = 16$$

$$\Rightarrow P \left(1 + \frac{R}{100} \right)^{20} = 16P$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^{20} = 16 \times 12000 = ₹1,92,000$$

31. b $P \left(1 + \frac{20}{100} \right)^n > 2P$ or $\left(\frac{6}{5} \right)^n > 2$

$$\text{Now } \left(\frac{6}{5} \times \frac{6}{5} \times \frac{6}{5} \times \frac{6}{5} \right) > 2.$$

So n = 4 years.

32. d Percentage increase = $\left(\frac{1}{8} \times 100 \right) \% = 12\frac{1}{2}\%$.

$$\text{Height after 2 years} = \left[64 \times \left(1 + \frac{25}{2 \times 100} \right)^2 \right] \text{ cm}$$

$$= \left(64 \times \frac{9}{8} \times \frac{9}{8} \right) \text{ cm} = 81 \text{ cm.}$$

33. c Let the sum be ₹x. Then

$$x \left(1 + \frac{R}{100} \right)^3 = 13380 \text{ and } x \left(1 + \frac{R}{100} \right)^6 = 20070.$$

$$\text{By dividing, we get } \left(1 + \frac{R}{100} \right)^3 = \frac{20070}{13380} = \frac{3}{2}.$$

$$\therefore x \times \frac{3}{2} = 13380 \Rightarrow x = \left(13380 \times \frac{2}{3} \right) = 8920$$

Hence, the sum is ₹8,920.

34. a Amount = ₹(30000 + 4347) = ₹34,347

$$\text{Let } 30000 \left(1 + \frac{7}{100} \right)^n = 34347$$

$$\Rightarrow \left(1 + \frac{7}{100} \right)^n = \frac{34347}{30000} = \frac{11449}{10000} = \left(\frac{107}{100} \right)^2$$

$\therefore n = 2$ years.

35. b $1200 \times \left(1 + \frac{R}{100} \right)^2 = 1348.32$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^2 = \frac{1348.32}{1200} = \frac{11236}{10000}$$

$$\therefore \left(1 + \frac{R}{100} \right)^2 = \left(\frac{106}{100} \right)^2 \Rightarrow 1 + \frac{R}{100} = 1.06 \Rightarrow R = 6\%$$

36. c Interest on ₹800 for one year = ₹(840 - 800) = ₹40

$$\therefore \text{Rate} = \left(\frac{100 \times 40}{800 \times 1} \right) \% = 5\%$$

37. a $P \left(1 + \frac{R}{100} \right)^3 = 8P \Rightarrow \left(1 + \frac{R}{100} \right)^3 = 8 = 2^3$

$$\therefore 1 + \frac{R}{100} = 2 \Rightarrow \frac{R}{100} = 1 \Rightarrow R = 100\%$$

38. c $A_{10} = A_{12}$

$$P_{10}[100 + 10(18 - 10)] = P_{12}[100 + 10(18 - 12)];$$

$$\frac{P_{10}}{P_{12}} = \frac{160}{180} = \frac{8}{9}$$

P_{10} = Sum of 10-year-old son

$$= \frac{8}{17} \times 85000 = ₹40,000.$$

39. a $\frac{9}{25} \times P = \frac{P \times R^2}{100} \Rightarrow R^2 = \frac{9 \times 100}{25} \Rightarrow R = \frac{3 \times 10}{5} = 6\%$

Hence, time period is 6 years.



$$40. a \quad SI = \frac{2000 \times 2 \times 10}{100} = ₹400$$

$$CI = 2000 \left\{ \left(1 + \frac{10}{100} \right)^2 - 1 \right\} = 2000 \times \frac{21}{100} = ₹420$$

$$\text{Hence difference} = 420 - 400 = ₹20.$$

Short cut:

Difference between S.I. and C.I.

$$= P \times \left(\frac{R}{100} \right)^2 = 2000 \times \frac{10}{100} \times \frac{10}{100} = ₹20$$

$$41. a \quad \frac{x \times 6 \times 5}{100} + \frac{(7000 - x)}{100} \times 5 \times 4 = 1600$$

$$30x + 140000 - 20x = 160000$$

$$\Rightarrow 10x = 20000$$

$$\Rightarrow x = ₹2,000$$

$$42. b \quad \text{Annual payment} = \frac{1200 + \frac{1200 \times 4 \times 4}{100}}{4} = ₹348.$$

$$43. c \quad T = \frac{9 \times 100 \times 4}{36 \times 25} = 4 \text{ years}$$

$$44. b \quad \frac{400 \times 8 \times 5}{100} = ₹160$$

$$45. a \quad SI \text{ for 1st year} = \frac{360}{3} = ₹120 = CI \text{ for 1st year.}$$

$$CI \text{ for 2 years} = ₹270 \Rightarrow 120 + CI \text{ for 2nd year.}$$

$$CI \text{ for 2nd year} = ₹120 + \text{Interest on ₹120 for 1 year i.e. } (150 - 120) = 30$$

$$\text{Rate} = \frac{30}{120} \times 100 = 25\%$$

$$46. b \quad r = \frac{9 \times 100}{1 \times 50} = 18\% \quad (\text{Since interest} = 10 - 1 = 9)$$

$$47. c \quad \frac{400 \times 3 \times r}{100} + \frac{600 \times 3 \times r}{100} = 90 \Rightarrow 30r = 90$$

$$\Rightarrow r = 3\%$$

Chapter 4 Profit, Loss and Discount

1. c Nothing is mentioned about the number of books.

2. c CP of 20 articles = SP of 15 articles.
SP of 15 = CP of 15 + CP of 5.
We know, SP = CP + Profit

$$\therefore \text{Percentage profit} = \frac{5}{15} \times 100 = 33.33\%$$

3. a Let CP = x.

$$\text{Then, SP} = \frac{4x}{3}.$$

$$\text{Gain} = \left(\frac{4x}{3} - x \right) = \frac{x}{3}.$$

$$\therefore \text{Gain percentage} = \left(\frac{x}{3} \times \frac{1}{x} \times 100 \right) = 33\frac{1}{3}\%.$$

4. c SP of 16 books = CP of 12 books.
SP of 16 = CP of 16 - CP of 4.

$$\therefore \text{Percentage Loss} = \frac{4}{16} \times 100 = 25\%.$$

5. c Loss = SP of 4 apples on selling 36 apples.

$$\therefore \text{SP of 40} = \text{CP of 36} = \text{CP of 40} - \text{CP of 4}$$

$$\therefore \text{Loss percentage} = \frac{4}{40} \times 100 = 10\%.$$

6. b Profit = $\frac{1}{4}$ of CP, SP = ₹375.

$$\text{Profit} = \text{SP} - \text{CP}$$

$$\frac{1}{4} \text{CP} = 375 - \text{CP} \Rightarrow 375 = \left(1 + \frac{1}{4} \right) \text{CP}$$

$$\text{CP} = 375 \times \frac{4}{5} = ₹300$$

7. c CP of one banana of first quality = ₹ $\frac{1}{3}$.

CP of one banana of second quality = ₹ $\frac{1}{2}$.

$$\text{Average CP} = \frac{\frac{1}{3} + \frac{1}{2}}{2} = ₹\frac{5}{12}.$$

$$\text{SP} = ? \quad P = 20\%$$

$$\text{SP} = \text{CP} \frac{(100 + \text{Gain percentage})}{100}.$$

$$\text{SP} = \frac{5}{12} \times \frac{120}{100} = ₹\frac{1}{2} \text{ per banana.}$$

$$\text{Price per dozen} = \frac{1}{2} \times 12 = ₹6.$$

8. a CP of one dozen oranges of first quality = ₹5.
CP of one dozen oranges of second quality = ₹2.

$$\text{Average CP} = \frac{5+2}{2} = ₹3.50 \text{ per dozen.}$$

$$\text{SP} = ₹5.50$$

$$\text{Profit per dozen} = ₹2.$$

$$\text{Total profit} = ₹50.$$

$$\therefore \text{Number of dozens} = \frac{50}{2} = 25.$$

9. b CP of first = ₹18 per kilogram.
CP of second = ₹20 per kilogram.
Suppose he mixes 5 kg of first and 3 kg of second (for 8 kg rice).
Total CP = $18 \times 5 + 20 \times 3 = 90 + 60 = ₹150$.
Total SP = $21 \times 8 = ₹168$.

$$\text{Profit} = \frac{18}{150} \times 100 = 18 \times \frac{2}{3} = 12\%$$

10. d Total CP = $35 \times 9.5 + 30 \times 10.5$
= $332.5 + 315 = ₹647.5$ (For 65 kg rice)

$$\text{SP} = \text{CP} \times \frac{(100 + \text{Gain percentage})}{100}$$

$$\text{SP} = \frac{647.5}{65} \times \frac{135}{100} \approx ₹13.50$$

11. b Let the ratio be $x : 1$ of ₹100 per liter and ₹50 per liter oil.
Total CP = ₹(100x + 50).
Total SP = ₹96(x + 1)

$$96(x + 1) = (100x + 50) \times \frac{120}{100}$$

$$96 \times 5(x + 1) = 6 \times 50(2x + 1)$$

$$8(x + 1) = 5(2x + 1)$$

$$\Rightarrow 8x + 8 = 10x + 5$$

$$\Rightarrow 2x = 3$$

$$\Rightarrow x = \frac{3}{2}$$

∴ Ratio is 3 : 2.

Short cut:

$$\text{CP of the mixture} = \frac{96}{1.2} = 80.$$

Using alligation,

$$\begin{array}{ccc} 100 & & 50 \\ & \searrow & \nearrow \\ & 80 & \\ & \nearrow & \searrow \\ 3 & & 2 \end{array}$$

12. b Let the quantity of rice be 2x and 3x kilograms.
Profit = 10%, SP = ₹22.
CP of 2x kg = ₹14 per kilogram.
Let CP of 3x kg = ₹y per kilogram.
Total CP = ₹(28x + 3xy)
Total SP = ₹22 × 5x = 110x.

$$\text{SP} = \text{CP} \times \frac{(100 + \text{Gain percentage})}{100}$$

$$110x = (28x + 3xy) \times \frac{110}{100}$$

$$\Rightarrow 100 = 28 + 3y; y = \frac{72}{3} = ₹24$$

Short cut:

CP of the mixture = 20.

Using alligation,

$$\begin{array}{ccc} 14 & & x \\ & \searrow & \nearrow \\ & 20 & \\ & \nearrow & \searrow \\ 2 & & 3 \end{array}$$

$$\Rightarrow \frac{x - 20}{20 - 1} = \frac{2}{3}$$

$$\Rightarrow x = 24$$

13. d Gain percentage = $\frac{100}{900} \times 100 = 11.11\%$.

14. b If he uses a weight of x grams, then profit percentage

$$= \frac{1000 - x}{x} \times 100, \text{ which is equal to } 6\frac{18}{47}\%$$

Therefore, x = 940 g.

15. a Number of toffees = $\frac{1}{\left(\frac{1}{20}\right)\left(\frac{1.2}{0.96}\right)} = 16$.

Short cut:

To make a gain, number of toffees must be less than 20. (from options)

16. b Let CP = 100. Therefore, SP = 110.
If SP = 220, Profit percentage = 120%.

17. b Let CP for A = ₹x.
Total CP after repairs = ₹(x + 110)
B's CP = (x + 110) × 1.2
C's CP = (x + 110) × 1.2 × 0.9
C's SP = (x + 110) × 1.2 × 0.9 × 1.1 = ₹1,188

$$\therefore x + 110 = \frac{1188 \times 1000}{12 \times 9 \times 11}$$

$$\Rightarrow x + 110 = 1000 \Rightarrow x = ₹890$$

18. b Let the cost of production be ₹a.
 $a \times 1.1 \times 1.15 \times 1.25 = ₹1,265$
 $\Rightarrow a = \frac{1265}{1.1 \times 1.15 \times 1.25} = \frac{1265 \times 100000}{11 \times 115 \times 125}$
 $\Rightarrow a = \frac{100000}{125} = ₹800$

19. a $\text{SP}_1 = \text{SP}_2 = ₹4,000$.
Gain₁ = 25%, loss₂ = ?

$$\text{CP}_1 = 4000 \times \frac{100}{125} = ₹3,200$$

$$\therefore \text{CP}_2 = ₹4000 + ₹800 = ₹4,800 \quad (\because \text{Total SP} = \text{Total CP})$$

$$\text{Therefore, loss percentage} = \frac{800}{4800} \times 100 = 16.66\%$$

20. b When SP of two articles is same, one is sold at a loss of x% and other at a gain of x%, then there is always an overall

$$\text{loss, by } \frac{x^2}{100}\% \text{ and which is } = \frac{20 \times 20}{100} = 4\%$$

$$\therefore \text{Total CP} = \frac{12000 \times 2}{0.96} = 25000$$

$$\therefore \text{Loss} = ₹(25000 - 24000) = ₹1000$$

21. c Let CP per kilogram be ₹1.
So total CP of 24 kg = ₹24.
Let he sell x kilograms apples at 20% gain and
(24 - x) kg at 5% loss.

$$SP_1 = \frac{120x}{100} = 1.2x \quad \dots (i)$$

$$SP_2 = 0.95(24 - x) = 22.8 - 0.95x \quad \dots (ii)$$

Overall profit = 10% on ₹24 = ₹2.4

But, Total SP - Total CP = ₹2.4

$$[1.2x + 22.8 - 0.95x] - 24 = 2.4$$

Solving for x, x = ₹14.4

∴ Amount sold at loss is 24 - x = 9.6 kg.

Short cut:

Using alligation,

$$\begin{array}{ccc} 20\% & & -5\% \\ & \searrow \quad \swarrow & \\ & 10\% & \\ & \swarrow \quad \searrow & \\ 3 & & 2 \end{array}$$

$$\text{Therefore, } \frac{2}{5} \times 24 = 9.6 \text{ kg.}$$

22. a CP = 40% of SP = 0.4 SP.

$$SP = \left(\frac{10}{4} \times 100\% \right) \text{ of CP} = 250\% \text{ CP.}$$

23. b Profit and loss are calculated on the cost price.

∴ Difference in percentages = 5 - (-2.5) = 7.5%
7.5% of C.P. = ₹6

$$\therefore \text{CP} = \frac{6 \times 100}{7.5} = ₹80$$

24. c Let CP = 100. SP = 105. New CP = 95. Profit = 10%.

Therefore, New SP = 104.5. Therefore, he is now selling at
₹0.5 less. He would sell at ₹1 less if CP = ₹200.

$$25. \text{ b } CP = \frac{355 + 425}{2} = ₹390$$

26. b Let CP = ₹x.

$$\text{Profit} = SP - CP = (900 - x)$$

$$\text{Loss} = CP - SP = (x - 450)$$

$$\therefore 900 - x = 2(x - 450)$$

$$900 - x = 2x - 900 \Rightarrow 1800 = 3x \Rightarrow x = ₹600 = \text{CP.}$$

Now to make a profit of 25%,

$$SP = 600 \times \frac{125}{100} = ₹750$$

27. b 30%, 20%, 10%

$$(1 \text{ and } 2) : -30 - 20 + \frac{600}{100} = -44\%$$

$$(1 \text{ and } 2) \text{ and } 3 : -44 - 10 + \frac{440}{100} = -49.6\%$$

The -ve sign here indicates discount percentage.

Short cut:

$$0.7 \times 0.8 \times 0.9 = 0.504 = 50.4\%.$$

$$\text{Therefore, } 100 - 50.4 = 49.6\%.$$

28. c 36% and 4% successive discounts equal to

$$-36 - 4 + \frac{144}{100} = -38.56\%$$

$$\text{Difference} = 40 - 38.56 = 1.44\%.$$

$$\therefore 1.44\% \text{ of } 500 = ₹7.20$$

$$29. \text{ b } MP = CP \frac{(100 + \text{Profit percentage})}{(100 - \text{Discount percentage})}$$

$$MP = CP \times \frac{133}{95} = 1.4 \text{ CP}$$

∴ MP is 40% above the CP.

$$30. \text{ a } \text{Net discount} = -20 - 10 + \frac{200}{100} = 28\%$$

$$SP = MP \frac{(100 - \text{Discount percentage})}{100}$$

$$108 = MP \times \frac{72}{100}, \text{ MP} = ₹150.$$

31. c Let MP = ₹100 = SP (Initially)

After a discount of 10%, SP would be ₹90.

But the shopkeeper wants to maintain the current price, i.e.

SP = ₹100.

When SP = ₹90, MP = ₹100.

$$\therefore \text{When SP} = ₹100, \text{ MP} = ₹100 \times \frac{100}{90}$$

$$= ₹111.11 \text{ or } 11.11\% \text{ increase.}$$

32. d The customer would get two successive discounts of 10% and 5%

$$\therefore \text{Net discount} = a + b + \frac{ab}{100}$$

$$= -10 - 5 + \frac{50}{100} = -14.5\%$$

33. c Total loss to the shopkeeper = (CP of article + Balance)
= (30 + 5) = ₹35

34. a If CP and SP increases by the same percentage, the profit remains same.

35. b In this case weight given is of no use. We have to calculate in percentage terms only.

Let total quantity of wheat = 100x

And percentage mark up = p%

$$\Rightarrow 40x \times \frac{80}{100} + 60x \left(\frac{100 + p}{100} \right) = 100x \times \frac{125}{100}$$

$$\Rightarrow 320 + 600 + 6p = 1250 \Rightarrow 6p = 330$$

$$\Rightarrow p = 55\%$$

Short cut:

Let r% be the percentage mark-up (or profit percentage).

Then using alligation,

$$\begin{array}{ccc} -20 & & r \\ & \searrow \quad \swarrow & \\ & 25 & \\ & \swarrow \quad \searrow & \\ 40 & & 60 \end{array}$$

$$\Rightarrow r = 55\%$$

36. d Let CP_1 per egg initially = ₹1 per egg
(Assuming she had 100 eggs.)
Due to loss of 10% eggs, CP of remaining 90 eggs increases.
 $CP_2 = \frac{100}{90} = ₹1.11$ per egg
To retain 10% profit,
 $SP_2 = 1.11 \times 1.1 = 1.221$ or a mark-up of 22.1%.
37. d If CP of tea is ₹1 per kilogram, then he is receiving ₹1,000 for something which is worth ₹900. But he gives a discount of 10% on ₹1000, i.e. sells at ₹900. Hence, no loss no profit.
38. c Let CP per litre milk be ₹x.
For 25 L, CP = ₹25x, SP = ₹1,250.
Profit = ₹5x = SP - CP
 $\Rightarrow 5x = 1250 - 25x \Rightarrow x = ₹41.66$ per litre.
Profit percentage = $\frac{50 - 41.66}{41.66} \times 100 = 20\%$.
39. a Let SP of 1 kg rice = ₹x.
SP of 100 kg rice = ₹100x.
CP of 100 kg rice = ₹1,100.
 $\therefore$ Loss = ₹20x = CP - SP = 1100 - 100x
 $\Rightarrow x = \frac{1100}{120} \approx ₹9.17$

Chapter 5 Ratio & Proportion, Mixture Alligation, Partnership & Average

1. c $\frac{A}{B} = \frac{3}{7}$. Let $A = 3x$, $B = 7x$.
 $\Rightarrow 3x + 7x = 45$, $x = \frac{45}{10} = 4.5$;
 $\therefore B = 7x = 31.5$
2. d Let the fraction be $\frac{x}{y}$; $\frac{x/y}{3/7} = \frac{1/27}{1/35} \Rightarrow \frac{7x}{3y} = \frac{35}{27}$
 $\frac{x}{y} = \frac{35}{27} \times \frac{3}{7} = \frac{5}{9}$
3. a If a, b, c are in continuous proportion. The mean proportion is b.
Therefore, $b^2 = ac$, $b^2 = 8 \times 72$, $b = \sqrt{576} = 24$.
4. d If a, b, c, d are in proportion, $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{3}{15} = \frac{27}{d}$
 $\Rightarrow d = 135$.
5. b Third proportional $c = \frac{b^2}{a} = \frac{30 \times 30}{20} = 45$.

6. b A's share = $P \left(\frac{x}{x+y+z} \right) = 3610 \times \frac{2000}{4750} = ₹1,520$,
where P = Profit and x, y and z are respective shares of A, B and C.
7. c Ratio of their profits = Ratio of their investments
 $= 4000 \times 12 : 8000 \times 9 : 12000 \times 2$
 $48 : 72 : 24$
 $2 : 3 : 1$
B's share = $\frac{3}{6} \times 5200 = ₹2,600$.
8. c $\frac{x}{y} = \frac{2}{7}$. Let x's share = 2a, y's share = 7a.
 $\frac{x's \text{ share}}{y's \text{ share} - x's \text{ share}} = \frac{2a}{5a} = \frac{2}{5}$.
9. a Let $\frac{A}{2} = \frac{B}{3} = \frac{C}{6} = k$
 $\Rightarrow A : B : C = 2 : 3 : 6$
B's part = $\frac{2}{11} \times 3960 = ₹1080$
10. d A's share = $\frac{A}{A + \frac{3}{2}A + 3A} \times P = \frac{2}{11}P$.
B's share = $\frac{B}{\frac{2}{3}B + B + 2B} \times P = \frac{3}{11}P$.
C's share = $P - \frac{(2+3)}{11}P = \frac{6}{11}P$.
 $\frac{A's}{B's - C's} = \frac{\frac{2}{11}}{\frac{6}{11} - \frac{3}{11}} = \frac{2}{3}$.
11. c Let the number be 5x and 3x.
Then, $\frac{5x-9}{3x-9} = \frac{23}{12}$
 $\Rightarrow 60x - 108 = 69x - 207$
 $\Rightarrow 9x = 99 \Rightarrow x = 11$
The first number is $11 \times 5 = 55$.
12. c B's money = $\frac{5}{4} \times 800 = ₹1000$.
C's money = $\frac{3}{2} \times 1000 = ₹1500$.
Therefore, A's + B's + C's money = ₹3,300.
13. d $4x = 3y = 2z$
Divide the three terms by their LCM.
LCM of 4, 3, 2 = 12;
 $\therefore x : y : z = 3 : 4 : 6$

14. c Let the parts be x , $(68 - x)$.

$$\frac{x}{7} = \frac{1}{10}(68 - x) = \frac{68}{10} - \frac{x}{10}$$

$$\frac{x}{7} + \frac{x}{10} = \frac{68}{10} \Rightarrow \frac{17x}{70} = \frac{68}{10}, x = 28.$$

15. a $A = \frac{2}{3}B$, $B = \frac{1}{4}C$

$$\frac{A}{B} = \frac{2}{3}, \frac{B}{C} = \frac{1}{4} = \frac{3}{12}$$

$$A : B : C = 2 : 3 : 12$$

$$B's \text{ share} = \frac{3}{17} \times 6800 = ₹1,200$$

16. b X's share $\Rightarrow 3x + 30$

$$Y's \text{ share} \Rightarrow 4x + 20$$

$$Z's \text{ share} \Rightarrow 5x + 50$$

$$\text{Sum is } 9700.$$

$$12x + 100 = 9700, 12x = 9600, x = 800.$$

$$Y's \text{ share} = 4x + 20 = 3200 + 20 = ₹3,220.$$

17. d Let B get ₹ x ,

$$A \text{ get } ₹x + 70,$$

$$C \text{ get } ₹x - 80.$$

$$x + x + 70 + x - 80 = 530$$

$$\Rightarrow 3x - 10 = 530, 3x = 540$$

$$\Rightarrow x = 180$$

$$\frac{A's \text{ share}}{C's \text{ share}} = \frac{180 + 70}{180 - 80} = \frac{250}{100} = \frac{5}{2}$$

18. a Ratio = $\frac{4}{9}$.

$$\text{Let antecedent} = 4x = 36, x = 9.$$

$$\text{Consequent} = 9x = 9 \times 9 = 81.$$

19. b $A = \frac{1}{2}B$, $B = 2C \Rightarrow \frac{A}{B} = \frac{1}{2}, \frac{B}{C} = \frac{2}{1}$.

$$A : B : C = 1 : 2 : 1$$

$$\text{Shares of A, B and C are } x, 2x \text{ and } x.$$

$$\frac{B}{A+B} = \frac{2x}{x+2x} = \frac{2x}{3x} = \frac{2}{3}$$

20. b Ratio of coins by value $\frac{3}{1} : \frac{4}{2} : \frac{12}{4} \Rightarrow 3 : 2 : 3$.

$$\therefore 3x + 2x + 3x = 600, x = \frac{600}{8} = 75.$$

$$\text{Value of 25-paise coins} = 75 \times 3 = ₹225.$$

$$\therefore \text{Number of coins} = 225 \times 4 = 900.$$

21. d $M : W = 2 : 1$. Total volume = 180 L.

$$\text{Milk} = 120 \text{ L, water} = 60 \text{ L.}$$

$$\text{Let 'x' litres of water is added to make the ratio } 1 : 2.$$

$$\therefore \frac{120}{60+x} = \frac{1}{2}$$

$$\Rightarrow 240 = 60 + x$$

$$\Rightarrow x = 180 \text{ L.}$$

22. a $\frac{M}{W} = \frac{3}{1}$ in 100 L mixture.

$$\text{milk} = 75 \text{ L, water} = 25 \text{ L.}$$

$$\text{After adding 200 L of water, water} = 225 \text{ L and milk} = 75 \text{ L.}$$

$$\text{Ratio} = \frac{75}{225} = 1 : 3.$$

23. c Initially number of boys = 250, number of girls = 250.

$$\text{New batch } \frac{1}{5} \text{ of the girls (50 girls) left the class and}$$

$$25 \text{ boys joined in.}$$

$$\text{Number of girls remaining} = 200, \text{ and number of boys} = 250 + 25 = 275.$$

$$\therefore \frac{\text{Boys}}{\text{Girls}} = \frac{275}{200} = \frac{11}{8}$$

24. c Original population is 15,000.

$$\text{Population after one year}$$

$$= 15000 + \frac{1}{10} \times 15000 = 15000 + 1500 = 16500.$$

$$\text{Population after two years}$$

$$= 16500 - \frac{1}{10} \times 16500 = 16500 - 1650 = 14850.$$

Short cut:

$$\text{Change in population} = 10 - 10 - \frac{10 \times 10}{100} = -1\%$$

$$\text{Population after 2 years} = 15,000 \times 0.99 = 14,850$$

25. a Let man's age be $7x$, son's age = $2x$.

$$4 \text{ according to question,}$$

$$\frac{7x+15}{2x+15} = \frac{2}{1}, 7x+15 = 4x+30, x = 5.$$

$$\text{Present age of father} = 35 \text{ years.}$$

$$\text{Present age of son} = 10 \text{ years.}$$

$$\text{Father's age when son was born}$$

$$= 35 - 10 = 25 \text{ years.}$$

26. d Let present age of son = x , and present age of father = y .

$$(y-4) = 6(x-4) \quad \dots (i)$$

$$(y+12) = 2(x+12) \quad \dots (ii)$$

$$\text{From (i) and (ii),}$$

$$y - 6x + 20 = 0$$

$$y - 2x - 12 = 0$$

$$4x = 32, x = 8 \text{ years, } y = 28 \text{ years.}$$

$$\text{Ratio of ages} = \frac{28}{8} = \frac{7}{2}$$

27. b Let $\frac{a}{3} = \frac{b}{4} = \frac{c}{7} = k$

$$a = 3k, b = 4k, c = 7k$$

$$\frac{a+b+c}{c} = \frac{3k+4k+7k}{7k} = \frac{14k}{7k} = 2$$

28. b Sum = 84
It must be divisible by the sum of the ratios.
Sum of the ratios are 12, 8, 21 and 12. Since, 8 does not divide 84.
Hence (b) is the correct option.

29. c 0.35 of x = 0.07 of y.

$$\therefore \frac{x}{y} = \frac{0.07}{0.35} = \frac{1}{5}$$

30. a $\frac{C_1}{I_1} = \frac{5}{8}, \frac{C_2}{I_2} = \frac{5}{3}$

$$\frac{\text{Copper}}{\text{Iron}} = \frac{\frac{5}{13} + \frac{5}{8}}{\frac{8}{13} + \frac{3}{8}} = \frac{40 + 65}{64 + 39} = \frac{105}{103}$$

Hence, ratio of iron : copper = 103 : 105.

31. c Quantity of milk in 80L mixture = $\frac{5}{8} \times 80 = 50\text{L}$

and in 16L mixture = $\frac{5}{8} \times 16 = 10\text{L}$

Quantity of water in 80L mixture = $\frac{3}{8} \times 80 = 30\text{L}$ and

in 16L mixture = $\frac{3}{8} \times 16 = 6\text{L}$

$\therefore$ Required ration = $\frac{(50-10)+16}{(30-6)} = \frac{56}{24}$, i.e. 7 : 3.

32. b Let copper = 13x, zinc = 7x.
13x + 7x = 500

$$x = \frac{500}{20} = 25 \text{ kg}$$

$\therefore$ Copper = 13 × 25 = 325 kg

33. d A's effective investment = 16000 × 6 = ₹96,000.
B's effective investment = 12000 × 8 = ₹96,000.
C's effective investment = 1000 × 12 = ₹12,000.
Profit sharing ratio = 96 : 96 : 12 i.e. 8 : 8 : 1.

34. c A gets 100 paise, B gets 65 paise and C gets 35 paise.

C's share = 560 = Sum × $\frac{35}{200}$. Sum = ₹3,200.

35. d Let the numbers be 2x and 3x.

Ratio of their squares = $\frac{4x^2}{9x^2} = \frac{4}{9}$.

Note: Ratio of squares is independent of difference of numbers.

36. b L : M : O = 5 : 7 : 3

Let the labour cost be 5x, material cost = 7x, overheads = 3x. Total cost = 15x.
Profits = 20% of 15x = 3x

$$\frac{\text{Material cost}}{\text{Profit}} = \frac{7x}{3x} = \frac{7}{3}$$

37. a $A = \frac{1}{2}(B + C)$
 $\Rightarrow 2A = B + C$... (i)

$B = \frac{1}{4}(A + C)$... (ii)
4B = A + C
Subtracting (ii) from (i),
2A - 4B = B - A $\Rightarrow 3A = 5B$

$\frac{3}{5}A = B$... (iii)

$C = 2A - B = 2A - \frac{3}{5}A = \frac{7}{5}A$

$\therefore A : B : C = 1 : \frac{3}{5} : \frac{7}{5} = 5 : 3 : 7$

A's share = $\frac{5}{15} \times 11250 = 750 \times 5$.

B's share = $\frac{3}{15} \times 11250 = 750 \times 3$.

A's share - B's share = 750 × 2 = ₹1500.

38. b $\frac{B}{G} = \frac{7}{5}$

Let number of boys = 7x.

Let number of girls = 5x.

7x + 5x = 72, x = 6

Number of boys = 42.

Number of girls = 30.

12 more girls should be admitted to make the number equal.

39. d Let the numbers be 2x, 3x, and 5x respectively.
Sum of their squares = 608 = 4x² + 9x² + 25x²
38x² = 608, x² = 16, x = ±4.
As the numbers are positive, the biggest number is 20.

40. d M : W = 2 : 5. Total volume = 28 L.

Milk = $\frac{2}{7} \times 28 = 8 \text{ L.}$

Water = $\frac{5}{7} \times 28 = 20 \text{ L.}$

2 L of milk is added and 5 L water is removed.

$$\left(\frac{M}{W}\right)_{\text{new}} = \frac{8+2}{20-5} = \frac{10}{15} = \frac{2}{3} \text{ i.e. } 2 : 3$$

41. a Let A's income = ₹3x.

B's income = ₹2x.

A's expenses = ₹5y.

B's expenses = ₹3y.

A's savings = 3x - 5y = 3000 ... (i)

B's savings = 2x - 3y = 3000 ... (ii)

Solving (i) and (ii), x = 6000

B's income = 2x = ₹12,000.

42. c Let A's, B's and C's investments be 8x, 7x and 5x respectively.

A's effective investment = (8x)5 + (4x)7
= 40x + 28x = 68x



B's effective investment = $7x \times 12 = 84x$.
C's effective investment = $5x \times 12 = 60x$.

$$\text{B's share} = \frac{84}{212} \times 26500 = ₹10,500.$$

43. b $\frac{B}{R} = \frac{5}{4}$. Let the price be $5x$ and $4x$
 $5x - 4x = 800$, $x = 800$.
Price of rickshaw = $800 \times 4 = ₹3,200$.

44. c $\frac{A}{B} = \frac{5}{2}$, $B = \frac{2}{5}A$
$$\frac{8A + 9B}{8A + 2B} = \frac{8A + \frac{18}{5}A}{8A + \frac{4}{5}A} = \frac{58}{44} = \frac{29}{22}$$

Short cut:

Let $A = 5$. Then $B = 2$.
Therefore, $(40 + 18) : (40 + 4) = 58 : 44 = 29 : 22$.

45. a 20% solution of sugar means one-fifth of sugar.
In 5 L of solution, 1 L is sugar and 4 L is water.
After adding 1 L of water,

$$\text{Percentage of sugar} = \frac{1}{6} \times 100 = 16.67\%.$$

46. c Let the original consumption be 1 unit costing ₹100.
New cost = ₹125.

$$\text{New consumption} = \left(\frac{1}{125} \times 100 \right) = \frac{4}{5} \text{ unit.}$$

$$\frac{\text{Reduction in consumption}}{\text{Original consumption}} = \frac{\left(1 - \frac{4}{5}\right)}{1} = \frac{1}{5}, \text{ i.e. } 1 : 5.$$

47. c Sum of present ages of all the three
= $25 \times 2 + 3 + 3 + 1 = 57$ years

$$\therefore \text{Average age of the family} = \frac{57}{3} = 19 \text{ years.}$$

48. a Runs scored in first 20 overs = $20 \times 4.2 = 84$
Runs to be scored to reach the target = $312 - 84 = 228$

$$\therefore \text{Required run rate} = \frac{228}{30} = 7.6.$$

49. d Sum of ages of
A and B = $60 \times 2 = 120$
C and D = $38 \times 2 = 76$
E and F = $10 \times 2 = 20$
 $\therefore A + B + C + D + E + F = 120 + 76 + 20 = 216$

$$\text{Hence, the average age of the family} = \frac{216}{6} = 36 \text{ years.}$$

50. c Total sales for 5 days
= $₹(6135 + 7227 + 6755 + 7430 + 6462) = ₹34009$.
 $\therefore$ Required sales on the sixth day
= $[(6600 \times 6) - 34009] = ₹(39600 - 34009)$
= ₹5591.

51. a Average of 50 numbers = 0.
 $\therefore$ Sum of 50 numbers $(0 \times 50) = 0$.
It is quite possible that almost 49 of these numbers can be positive and if their sum is 'a' then 50th number will be $(-a)$, making the sum total of 50 numbers zero.

52. b Increase in weight due to replacement of the person weighing 56 kg by a new person
= $2 \times 10 = 20$ kg
 $\therefore$ The weight of the new person = $(56 + 20)$ kg
= 76 kg.

53. a Let the monthly incomes of Ram, Rahim and Salim are 'x', 'y' and 'z' respectively, then
 $x + y = (4500 \times 2) = 9000$... (i)
 $y + z = (5500 \times 2) = 11000$... (ii)
 $x + z = (5100 \times 2) = 10200$... (iii)
Adding (i), (ii) and (iii), we get; $2(x + y + z) = 30200$ or
 $x + y + z = 15100$... (iv)
Subtracting (ii) from (iv), we get $x = 4100$.
 $\therefore$ Ram's monthly income = ₹4,100.

54. b Since the month begins with a Sunday, so there will be five Sundays in the month.
Required average
= $\frac{720 \times 5 + 450 \times 25}{30} = \frac{3600 + 11250}{30} = 495$.

55. d Runs made in first 20 overs = $24 \times 4.5 = 108$
Total run to be scored = $50 \times 6 = 300$
 $\therefore$ Runs needed to be scored = $300 - 108 = 192$.

56. c Let the number of students in the class = x
Increase in the total marks = $0.3x$
 $\therefore 0.3x = (83 - 38)$

$$\Rightarrow x = \frac{45}{0.3} = 150$$

Alternative method:

$$\text{Total number of students} = \frac{\text{Increase in marks}}{\text{Increase in average}}$$

$$= \frac{45}{0.3} = 150.$$

Chapter 6 Time, Speed, Distance and Time & Work

1. a A takes $7 \times 9 = 63$ hr.
 $\therefore$ In 1 hr A does $\frac{1}{63}$ of work.
B takes $6 \times 7 = 42$ hr.
 $\therefore$ In 1 hr B does $\frac{1}{42}$ of work.
 $(A + B)$'s work in 1 hr = $\frac{1}{63} + \frac{1}{42}$.
 $\therefore (A + B)$ in $\frac{42}{5}$ hr = $\left(\frac{1}{63} + \frac{1}{42} \right) \times \frac{42}{5}$

$$= \frac{105}{63 \times 42} \times \frac{42}{5} = \frac{1}{3}$$

$$\therefore \text{Number of days} = \frac{1}{\left(\frac{1}{3}\right)} = 3 \text{ days.}$$

2. c A's work in one day = $\frac{1}{80}$.

$$\therefore 10 \text{ days work} = \frac{10}{80} = \frac{1}{8}.$$

$$\text{Work to be done by B} = \left(1 - \frac{1}{8}\right) = \frac{7}{8}.$$

$$x \times 42 = \frac{7}{8}, x = \frac{7}{42 \times 8} = \frac{7}{336}.$$

$$\text{Number of days required by B} = \frac{336}{7} = 48 \text{ days.}$$

$$A + B = \frac{1}{80} + \frac{1}{48} = \frac{128}{80 \times 48}$$

$$\therefore \text{Number of days} = \frac{80 \times 48}{128} = 30 \text{ days.}$$

3. c B's work lasts 23 days.

$$\therefore \text{Work done in 23 days by B} = \frac{23}{40}.$$

$$\text{Remaining work} = \left(1 - \frac{23}{40}\right) = \frac{17}{40}.$$

$$(A + B) \text{ in one day} = \left(\frac{1}{45} + \frac{1}{40}\right)^{\text{th}} \text{ of work.}$$

$$\text{Number of days required to finish } \frac{17}{40} \text{ of work}$$

$$= \frac{17}{40} \times \frac{(40 \times 45)}{(45 + 40)} = \frac{17 \times 40 \times 45}{40 \times 85} = 9 \text{ days.}$$

4. a Work done A : B = 1 : 2.

$$\text{Time taken A : B} = 3 : 4.$$

$$\therefore \text{A works 1 unit in 3 hr.}$$

$$\therefore \text{In 1 hr} = \frac{1}{3} \text{ unit of work.}$$

$$\therefore \text{B works 2 units in 4 hr.}$$

$$\therefore \text{In 1 hr} = \frac{2}{4} \text{ unit of work.}$$

$$\therefore \text{Work done per hour} = A : B = \frac{1}{3} : \frac{1}{2} = 2 : 3.$$

$$\therefore A + B \text{ can do 5 units in 1 day.}$$

$$\therefore A + B \text{ can do } 5 \times 18 \text{ units in 18 days.}$$

$$\therefore \text{Total work} = 90 \text{ units.}$$

$$\text{Time taken by B to complete the work} = \frac{90}{3} = 30 \text{ days.}$$

5. c A : B work done in same time = 10 : 16 = 5 : 8.
(As B is 60% more efficient)
A can finish work in 12 days.
 $\therefore$ Units of work = $12 \times 5 = 60$.

$$\therefore \text{B can finish work in } \frac{60}{8} \text{ days} = 7.5 \text{ days.}$$

6. d A's 10 days' work = $\frac{1}{25} \times 10 = \frac{2}{5}$.

$$\text{Remaining work} = \left(1 - \frac{2}{5}\right) = \frac{3}{5}$$

$$\left(\frac{1}{25} + \frac{1}{20}\right) \text{ work is done by (A + B) in 1 day}$$

$$\frac{3}{5} \text{ work is done by them in} = \frac{100}{9} \times \frac{3}{5} = 6.66 \text{ days.}$$

$$\text{Number of days} = 10 + 6.66 = 16.66 \text{ days}$$

7. b Let A can finish a job in x days and B can do it in 2x days

$$\Rightarrow \frac{1}{2x} + \frac{1}{x} = \frac{1}{14}$$

$$\therefore x = 21 \text{ days.}$$

8. b A + B = 72 days, B + C = 120 days, A + C = 90 days.
Assume units of work = 360 = [LCM(72, 120, 90)].

$$(A + B)'s \text{ work in 1 day} = \frac{360}{72} = 5 \text{ units.}$$

$$(B + C)'s \text{ units of work in 1 day} = \frac{360}{120} = 3 \text{ units.}$$

$$(A + C)'s \text{ of work in 1 day} = \frac{360}{90} = 4 \text{ units.}$$

$$\therefore 2(A + B + C) \text{ units in 1 day} = 5 + 3 + 4 = 12.$$

$$(A + B + C)'s \text{ work in 1 day} = \frac{12}{2} = 6.$$

$$\therefore A's \text{ work in 1 day} = 6 - 3 = 3 \text{ units.}$$

$$\therefore A \text{ will finish 360 units in } \frac{360}{3} = 120 \text{ days.}$$

9. b Assume a units of work = LCM of (12, 16) = 48.

$$(A + B)'s \text{ work in 1 day} = \frac{48}{12} = 4 \text{ units.}$$

$$(B + C)'s \text{ work in 1 day} = \frac{48}{16} = 3 \text{ units.}$$

$$(A + B) \text{ worked for 5 days} = 4 \times 5 = 20 \text{ units done.}$$

$$(B + C) \text{ worked for 2 days} = 3 \times 2 = 6 \text{ units done.}$$

$$\text{Remaining work} = 48 - 26 = 22 \text{ units.}$$

$$C \text{ finishes in } (13 - 2) \text{ days} = 11 \text{ days.}$$

$$\therefore \text{Units of work in 1 day by C} = \frac{22}{11} = 2.$$

$$\therefore \text{Number of days required for C} = \frac{48}{2} = 24 \text{ days.}$$

10. c Total work = $12 \times 8 = 96$.
 12 men finish 6 days of work = $12 \times 6 = 72$.
 $\therefore$ Work left = $96 - 72 = 24$.
 Now number of men = $12 + 4 = 16$.
 $\therefore$ Time = $\frac{24}{16} = 1\frac{1}{2} = 1.5$ days.
11. b Due to stoppage, it covers 9 km less.
 Time taken to cover 9 km = $\left(\frac{9}{54} \times 60\right)$ min = 10 min.
12. b 10 men in 20 days = 20 women in 15 days.
 $\therefore$ 5 men in 40 days and 10 women in 30 days.
 (5 men + 10 women) in $\frac{1}{\left(\frac{1}{40} + \frac{1}{30}\right)}$ days
 $= \frac{1200}{70}$ days = $17\frac{1}{7}$ days.
13. b Assume the units of work done = 24 = [LCM(12, 8)].
 Units of work done by R in a day = $\frac{24}{8} = 3$.
 Units of work done by S in a day = $\frac{24}{12} = 2$.
 In 2 days units of work done = 5.
 In 8 days units of work done = 20.
 In 9 days units of work done = 23.
 On 10th day, work to be done = $24 - 23 = 1$ unit.
 $\therefore$ Time required by S = $\frac{1}{2}$ days
 $\therefore$ Total time = $9\frac{1}{2}$ days.
14. b Units of work done = 220 = LCM (11, 20, 55).
 Units of work in a day by A, B and C = 20, 11, 4.
 1st day = (A + B) = 31,
 2nd day = (A + C) = 24,
 work done in 2 days = 55.
 work done in 8 days = 220.
15. b Assume units of work = 12 [LCM(4,6)]
 A's 1 hour work = $\frac{12}{4} = 3$ units
 B's 1 hour work = $\frac{12}{6} = 2$ units
 $\therefore$ (A + B)'s 2 hour work = $3 + 2 = 5$ units
 $\therefore$ In 4 hours 10 unit of work will be completed.
 Next 2 unit of work will take $\frac{2}{3}$ hr = 0.66 hours.
 $\therefore$ Total time = $4 + 0.66 = 4.66$ hours.
16. b Assume the units of work = 18.
 Father in 1 hr = 9 units. (6 + 3)
 $\therefore$ Time taken = 2 hr.

17. a Assume total wages = 21×28 .
 A's wages for 1 day = $21 \times \frac{28}{21} = 28$.
 B's wages for 1 day = $28 \times \frac{21}{28} = 21$.
 A + B's wages of 1 day = 49.
 A + B's wages will last for $21 \times \frac{28}{49} = 12$ days.
18. a A : B = 1 : 2
 C : (A + B) = 1.5 : 3
 If C does 1.5 units of work per day, A does 1 and B does 2.
 Total units done by C = $40 \times 1.5 = 60$.
 $\therefore$ Number of units done by (A + B + C) in 1 day = 4.5, Hence
 all three will take $\frac{60}{4.5} = 13.33$ days.
19. a Units to fill up = LCM(20, 15, 12) = 60.
 Units filled up by A, B, C in 1 min = 3, 4, 5
 Time taken by (A + B + C) to fill up = $\frac{60}{12} = 5$ min.
20. c Say, capacity in units = LCM(10, 12, 20) = 60.
 Rate of A, B, C per hour = $\frac{60}{10}, \frac{60}{12}, \frac{60}{20} = 6, 5$ and 3 units.
 $\therefore$ A + B = 11 units, C = 3 units.
 $\therefore$ Per hour intake = $11 - 3 = 8$ units.
 Time to fill up = $\frac{60}{8} = 7.5$ hr.
21. d Capacity in units = LCM(9, 10) = 90.
 Filling rate = 10 units/hr (i.e. 90/9).
 Similarly, filling rate with leak = 9 units/hr (i.e. 90/10).
 $\therefore$ Resultant outflow = $10 - 9 = 1$ unit/hr.
 Total outflow = 90 units.
 So time required = $\frac{90}{1} = 90$ hr.
22. d Say, capacity of tank = LCM(12, 15) = 60 units.
 In 1 min total intake in units = $\frac{60}{12} + \frac{60}{15} = 9$.
 In 3 minutes total intake in units = $9 \times 3 = 27$.
 Units to be filled = $60 - 27 = 33$.
 Time taken by B = $\frac{33}{4} = 8.25$ (8 min 15 sec).
23. c Let two pipes A and B are there
 $\therefore \frac{1}{A} + \frac{1}{B} = \frac{1}{12}$, say A takes x hours to fill. Then B will take x
 - 10 hr to fill the tank.
 $\frac{1}{x} + \frac{1}{x-10} = \frac{1}{12} \Rightarrow \frac{x-10+x}{x(x-10)} = \frac{1}{12}$
 $\Rightarrow \frac{2x-10}{x(x-10)} = \frac{1}{12}$

$$\begin{aligned} \Rightarrow 24x - 120 &= x^2 - 10x \\ \Rightarrow x^2 - 34x + 120 &= 0 \\ \Rightarrow x^2 - 30x - 4x + 120 &= 0 \\ \Rightarrow x(x - 30) - 4(x - 30) &= 0 \\ \Rightarrow (x - 4)(x - 30) &= 0 \\ \Rightarrow x &= 4, 30 \\ \therefore x &= 30 \text{ hr} \quad (\because x \text{ cannot be } 4) \end{aligned}$$

24. b 4 L per minute = 240 L per hour.
Let's assume that capacity of tank = LCM(6, 8) = 24 units.

$$\text{When inlet is also opened, net outflow} = \frac{24}{8}$$

= 3 units per hour.

$\therefore 4 - 3 = 1$ unit per hour of water contributed by the pipe.

$\therefore 1 \text{ unit} = 240 \text{ L.}$

24 units = $24 \times 240 = 5,760 \text{ L.}$

25. b Say capacity of tank = 12 units.

$$\text{A can fill} = \frac{12}{6} = 2 \text{ units per hour.}$$

$$\text{B can fill} = \frac{12}{4} = 3 \text{ units per hour.}$$

A + B alternately in 2 hr = $2 + 3 = 5$ units.

A + B alternately in 4 hr = $5 \times 2 = 10$ units.

In the 5th hour, A is opened.

$\therefore$ A fills the remaining 2 units.

$\therefore$ Total time = 5 hr.

26. c (a) Distance = Speed $\times$ Time = $48 \times 10 = 480 \text{ km.}$
(b) To cover the same distance in 8 hr.

$$\text{Speed} = \frac{d}{t} = \frac{480}{8} = 60 \text{ km/hr.}$$

$\therefore$ Speed must be increased by $60 - 48 = 12 \text{ km/hr.}$

27. b Total time = $60 + 24 = \frac{84}{60} \text{ hr.}$

$$\text{At } 4 \text{ km/hr, time } t_1 = \frac{2/3d}{4} = \frac{1}{6}d \text{ hr.}$$

$$\text{At } 5 \text{ km/hr, time } t_2 = \frac{1/3d}{5} = \frac{1}{15}d \text{ hr.}$$

$$\text{Total time} = \frac{d}{6} + \frac{d}{15} = \frac{84}{60} \Rightarrow d = 6 \text{ km.}$$

28. a Let his usual time be t hours and his usual speed be $s \text{ km/hr.}$

$$\text{Distance } d = st = \frac{3}{4}s \times (t + 2.5)$$

$$\Rightarrow 4t = 3t + 7.5 \Rightarrow t = 7.5 \text{ hr.}$$

29. c Let his normal speed be $s \text{ km/hr.}$
Let his normal time be t hours.

$$d = st \Rightarrow 4\left(t - \frac{1}{6}\right) = 3\left(t + \frac{1}{6}\right)$$

$$4t - \frac{4}{6} = 3t + \frac{3}{6}, t = \frac{7}{6} \text{ hr.}$$

$$\text{Distance} = 4 \times \left(\frac{7}{6} - \frac{1}{6}\right) = 4 \text{ km.}$$

30. c Let the normal speed be $s \text{ km/hr}$ and normal time be t hours.

$$d = st \Rightarrow 40\left(t + \frac{11}{60}\right) = 50\left(t + \frac{5}{60}\right)$$

$$4t + \frac{44}{60} = 5t + \frac{25}{60}$$

$$t = \frac{19}{60} \text{ hr} = 19 \text{ min}$$

31. c By the time the trains cross each other, let one cover 110 km, second cover 90 km.

Ratio of speeds = Ratio of the distances covered

$$= \frac{110}{90} = 11:9$$

32. c Thief starts at 2.30 p.m.

Owner starts at 3 p.m.

$$\text{In } 30 \text{ min} \left(\text{or } \frac{1}{2} \text{ hr}\right), \text{ the thief covers } 60 \times \frac{1}{2}$$

= 30 km

At that time

distance between them = 30 km,

Relative speed = $75 - 60 = 15 \text{ km/hr.}$

$$\therefore \text{Owner will overtake the thief in } \frac{30}{15} = 2 \text{ hr,}$$

i.e. at 5 p.m.

33. c Let speed of A = $3x \text{ km/hr}$ and speed of B = $4x \text{ km/hr.}$

$$\text{Time taken by A} = t_B + \frac{1}{2} \text{ hr.}$$

$$\text{Time taken by B} = t_B \text{ hr}$$

$$d \Rightarrow s \times t \Rightarrow 3x \times \left(t_B + \frac{1}{2}\right) = 4x \times t_B$$

$$\Rightarrow 3t_B = \frac{3}{2} = 4t_B \Rightarrow t_B = \frac{3}{2}, t_A = \frac{1}{2} + \frac{3}{2} = 2 \text{ hr.}$$

Short cut:

Speed is inversely proportional to time taken.

34. b Let the distance be d .

$$\text{Uphill travelling time} = \frac{d}{24}$$

$$\text{Downhill travelling time} = \frac{d}{36}$$

$$\therefore \text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$= \frac{2d}{\frac{d}{24} + \frac{d}{36}} = \frac{2 \times 24 \times 36}{24 + 36} = \frac{144}{5} = 28.8 \text{ km/hr}$$

35. c Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{2500 + 1200 + 500}{\frac{2500}{500} + \frac{1200}{400} + \frac{500}{250}} = \frac{4200}{10} = 420 \text{ km/hr}$$

36. c $d = s \times t$

$$\frac{1}{2} \text{ journey} (= 40 \text{ km}) \text{ in } \frac{3}{5} \text{ of time} (= 6 \text{ hr})$$

Total distance = 80 km.

Total time = 10 hr.

$$80 = 40 + S_2 \times 4,$$

$$\therefore S_2 = 10 \text{ km/hr}$$

37. d $A = 2B, B = 2C$. C takes 54 min. So, B would cover the same

$$\text{distance in half of the time} = \frac{54}{2} = 27 \text{ min.}$$

$$\text{So, time taken by A} = \frac{27}{2} \text{ min} = 13.5 \text{ min.}$$

38. b $\frac{S_A}{S_B} = \frac{3}{4}$

Speed of A = $3x$ km/min. Speed of B = $4x$ km/min.
Distance is same.

$$d = S_A \times t_A = S_B \times t_B$$

$$3x \times t_A = 4x \times 36;$$

$$t_A = 48 \text{ min}$$

39. b To cross the platform the train has to cover
 $165 + 110 = 275 \text{ m.}$

$$132 \text{ km/hr} = 132 \times \frac{5}{18} = \frac{110}{3} \text{ m/sec}$$

$$t = \frac{275}{110} \times 3 = 7.5 \text{ sec}$$

40. a $t = 10 \text{ sec.}$

$$\text{Speed} = 90 \text{ km/hr} = 90 \times \frac{5}{18} = 25 \text{ m/s.}$$

$$\text{Length} = s \times t = 25 \times 10 = 250 \text{ m.}$$

41. d Total length to be covered
= Length of train + Length of platform
= $900 + 300 = 1200 \text{ m}$
Total time = $60 + 12 = 72 \text{ sec}$

$$\text{Speed} = \frac{1200}{72} = \frac{50}{3} \text{ m/s} = \frac{50}{3} \times \frac{18}{5} = 60 \text{ km/hr.}$$

42. b Let length of the train be l m and its speed be x m/s.
It crosses a pole in 15 sec.

$$\therefore l = 15x \quad \dots (i)$$

It crosses a platform of 100 m long in 25 sec.

$$\therefore \frac{100+l}{x} = 25 \quad \dots (ii)$$

Solving (i) and (ii),

$$100 + 15x = 25x$$

$$\therefore x = \frac{100}{10} = 10 \text{ m/s} \Rightarrow l = 150 \text{ m}$$

43. c Length of the train = 110 m.

$$\text{Speed of the train} = 58 \times \frac{5}{18} \text{ m/s} = 16.11 \text{ m/sec.}$$

$$\text{Speed of man} = 4 \text{ km/hr} = 4 \times \frac{5}{18} = 1.11 \text{ m/sec.}$$

$$\text{Time} = \frac{110 \text{ m}}{16.11 - 1.11} = \frac{110}{15} = 7.33 \text{ sec.}$$

44. d Length of the first train = 108 m. Its speed = 50 km/hr.
Length of the second train = 112 m.

Let its speed = x km/hr.

Time to cross = 6 s

$$\Rightarrow 6 = \frac{108 + 112}{(50 + x) \times \frac{5}{18}} = \frac{220 \times 18}{5(50 + x)}$$

$$250 + 5x = 660 \Rightarrow 5x = 410$$

$$x = 82 \text{ km/hr}$$

45. d Length of the faster train

$$= (36 + 45) \times \frac{5}{18} \times 8 \text{ m} = 81 \times \frac{5}{18} \times 8 = 180 \text{ m.}$$

46. b Length of the bridge = 1 km.
Length of the train = 0.5 km.

$$\text{Time to clear the bridge} = 2 \text{ min} = \frac{2}{60} \text{ hr.}$$

$$\text{Speed} = \frac{1 + 0.5}{2/60} = \frac{1.5}{2} \times 60 = 45 \text{ km/hr.}$$

47. a Let speed of the boat be x km/hr and speed of the stream be y km/hr.

$$\text{Speed upstream} = \frac{13}{5} = x - y \quad \dots (i)$$

$$\text{Speed downstream} = \frac{28}{5} = x + y \quad \dots (iii)$$

$$\text{Solving for (y)} \Rightarrow y = 1.5 \text{ km/hr.}$$

48. b Speed of the boat in still water be ' x ' = 6 km/hr.
Let speed of the stream be ' y '.

$$x - y = 4.5 \text{ km/hr}$$

$$\therefore y = 1.5 \text{ km/hr}$$

$$\therefore \text{Rate along the stream} = (x + y) = 1.5 + 6 = 7.5 \text{ km/hr.}$$

49. a Let x be the speed of the man in still water.

Speed of the river = 2 km/hr = ' y '.

Speed upstream = $x - y$ km/hr.

Speed downstream = $x + y$ km/hr.

$$(x - y)2t = (x + y)t$$

$$2(x - 2) = x + 2 \Rightarrow x = 6 \text{ km/hr.}$$

50. a Speed of the stream = 1 km/hr.

Let speed of the boat in still water = x km/hr.

Total time = 12 hr.

$$12 = \frac{35}{x-1} + \frac{35}{x+1} = 35 \left[\frac{1}{x-1} + \frac{1}{x+1} \right] = 35 \left[\frac{2x}{x^2-1} \right]$$

$$12x^2 - 70x - 12 = 0$$

$$\Rightarrow 6x^2 - 35x - 6 = 0$$

$$\Rightarrow 6x^2 - 36x + x - 6 = 0$$

$$\Rightarrow 6x(x-6) + 1(x-6) = 0$$

$$\Rightarrow (x-6)(6x+1) = 0$$

$$\Rightarrow x = 6 \text{ km/hr}$$

51. a Length of the track $2\pi r = 2 \times \frac{22}{7} \times 21 = 132$ metres

$\therefore$ Speed of Ajay is $\frac{132}{6}$ m/minute and that of Bijay is

$$\frac{132}{8} \text{ m/minute}$$

$\therefore$ They are running in same direction,

$\therefore$ Relative Speed = $\frac{132}{6} - \frac{132}{8} = \frac{132}{24}$ m/minute

$\therefore$ Time taken to meet first time = $\frac{132}{\frac{132}{24}} = 24$ minutes

52. c Time taken to meet first time at starting point = LCM of 6 and 8 = 24 minutes

So they will meet second time after $2 \times 24 = 48$ minutes

53. b The length of the track = 220 metres

Two situations will arise in this, for meeting first time it will

take $\frac{1}{2} \left(\frac{220}{6-5} \right)$ seconds = 110 seconds and after first meet

it will take 220 second for each consecutive meet, Because for first meeting, they will have to fill the gap of half of the length of the track.

So in 6 minutes (i.e. 360 seconds) they will meet two times.

54. c They are running in opposite directions

Relative speed = $6 + 5 = 11$ m/s

For meeting third time the distance they will have to cover

= $2\frac{1}{2}$ times of length of the track

= $\frac{5}{2} \times 220 = 550$ metres

(half for first and two for next two meetings)

Time taken = $\frac{550}{11} = 50$ seconds



My Doubts

SAMPLE

CL Media

